

**INTEGRATED MANAGEMENT OF DRY ROOT ROT OF
CHICKPEA AND MOLECULAR CHARACTERIZATION
OF POTENTIAL BIOCONTROL AGENTS**

By

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Certificate

This is to certify that **Mr. G. NAGARJUNA REDDY** has satisfactorily prosecuted the course of research and that the thesis entitled “**INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHICKPEA AND MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS**” submitted is the result of original research work and is of sufficiently high standard to warrant its presentation to the examination. I also certify that the thesis or part there of has not been previously submitted by him for a degree of any university.

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No part of the thesis has been submitted for any other degree or diploma. The published part has been fully acknowledged. All assistance and help received during the course of the investigations have been duly acknowledged by the author of the thesis.

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LIST OF ABBREVIATIONS AND SYMBOLS

%	-	Per cent
@	-	At the rate of
°C	-	degree celsius
µg	-	Micro gram
µl	-	microlitre
µm	-	micrometre
a.i	-	Active ingredient
CD	-	Critical difference
cfu	-	Colony forming units
cm	-	centimetre
DAI	-	Days after inoculation
<i>et al.</i>	-	co-workers
Fig.	-	Figure
g	-	gram (s)
h	-	Hour(s)
ha	-	Hectare
<i>i.e.,</i>	-	That is
IDM	-	Integrated disease management
ITS-PCR	-	Internal Transcribed Spacer – Polymerase Chain Reaction
kg	-	kilogram
mg	-	milligram
mha	-	million hectares
ml	-	millilitre
mm	-	millimetre
mM	-	millimolar
N	-	Normality
ng	-	nangogram
No.	-	Number
PDA	-	Potato dextrose agar
PDI	-	Per cent disease incidence
pH	-	Power of hydrogen ion concentration
ppm	-	parts per million
psi	-	Pounds per square inch
RAPD	-	Random Amplified Polymorphic DNA
RFLP	-	Restriction Fragment Length Polymorphism
SEm	-	Standard error of mean
Sp. or Spp.	-	Species (singular or plural form)
<i>viz.</i>	-	Namely
WP	-	Wettable powder

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Declaration

*I, Mr. G. NAGARJUNA REDDY hereby declare that the thesis entitled “**INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHICKPEA AND MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS**” submitted to Acharya N.G. Ranga Agricultural University, Hyderabad for the degree of **MASTER OF SCIENCE IN AGRICULTURE** is the result of original research work done by me. I also declare that the material contained in this thesis has not been published earlier.*

Date :

(G. NAGARJUNA REDDY)

Place :

ABSTRACT

Title of thesis : **INTEGRATED MANAGEMENT OF DRY
ROOT ROT OF CHICKPEA AND
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OF POTENTIAL BIOCONTROL AGENTS**

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Chickpea (*Cicer arietinum* L.) is one of the major grain legume pulse crop of India covering 40% of area under pulse crops. Chickpea is affected by *Rhizoctonia bataticola* (Taub) Butler. causing dry root rot an important disease with yield losses ranging from 10 to 100 per cent.

A roving survey was conducted in Kadapa and Kurnool districts of Rayalaseema region, Andhra Pradesh for the incidence of dry root rot of chickpea. *In vitro* antagonism of microflora isolated from rhizosphere and root habitats against *Rhizoctonia bataticola* and their compatibility with different fungicides was studied. *In vitro* evaluation of fungicides against pathogen, integrated management of *Rhizoctonia bataticola* and molecular

characterization of potential biocontrol agents by using RAPD and 16S rDNA analysis was also carried.

A roving survey was conducted on dry root rot incidence in six major chickpea growing mandals of Kadapa and Kurnool districts, A.P. and the incidence was ranged from 6.22 to 13.50 per cent with lowest and highest incidence in Rajupalem and Sanjamala mandals, respectively.

The Pathogen was isolated from infected plant showing typical dry root rot symptoms viz., withering and drying of the plants, presence of dark tap root showing signs of rotting and devoid of its lateral and finer roots, purified and identified as *Rhizoctonia bataticola*.

A total of 23 antagonistic microflora (8 fungi and 15 bacteria) were obtained from rhizosphere soil and root endophytes from chickpea. Among 8 fungal isolates, *Trichoderma* isolate-3 (T₃) inhibited the growth of *Rhizoctonia bataticola* to the extent of 57.83 per cent. Among the 15 bacterial isolates REB-8 inhibited the growth of *Rhizoctonia bataticola* to the extent of 76.47 per cent followed by RB-1 (74.11%) and REB-9 (71.76%).

In vitro efficacy of four fungicides viz., thiram, copper oxychloride, captan and carbendazim was evaluated against *Rhizoctonia bataticola* using poisoned food technique at different concentrations. Carbendazim was found to be effective as it completely inhibited the mycelial growth even at lower concentration. Thiram was found to be next best fungicide.

In vitro compatibility of four fungicides used against *R. bataticola* were tested on potential antagonist REB-8 at different concentrations by using spectrophotometric method. The carbendazim showed high compatibility followed by thiram and copper oxychloride, whereas captan was found to be less compatible with REB-8.

The efficacy of potential biocontrol agent (REB-8) and compatible fungicide (carbendazim) was tested in pot culture against dry root rot of chickpea. The results revealed that treatment T6 (soil application with potential biocontrol agent + soil drenching with fungicide) was superior in reducing per cent disease incidence and increasing plant growth parameters, like root length, shoot length, dry weight of shoot and root when compared to other treatments.

The RAPD banding profiles with random primers viz., OPA-11, OPA-12, OPA-14, OPA-18 and OPD-3 reflected the genetic diversity among the antagonistic bacterial isolates with formation of two main clusters. Amplified 16S rDNA with universal primers 63F and 1387R produced approximately 1300 bp fragments as expected. 16S rDNA-RFLP results with *Taq* I enzyme showed no polymorphism among isolates under the study. The 1300bp amplified product of 16S rDNA from potential biocontrol agent i.e., REB-8 was cloned into the vector and sent for sequencing to MWG technologies, Bangalore.

CHAPTER – I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is one of the major grain legume pulse crops of India and other semi-arid regions of the world. It is cultivated in different regions of North India viz., Punjab, Uttar Pradesh, Haryana and parts of Central and South India.

Among the major pulse crops, chickpea contributes nearly 30.20 per cent and 25.80 per cent of total pulse area and total pulse production respectively. In India, it is grown over an area of 7.10 m.ha with an annual production of 5.75 m.t. In Andhra Pradesh, it is grown in an area of 6.30 lakh hectares with an annual production of 9.12 lakh tonnes. In Rayalaseema region alone it is being cultivated in 3.73 lakh ha with 2.71 lakh tonnes production (Directorate of Economics and Statistics, Hyderabad, 2007-08).

Several factors are responsible for low productivity, among which diseases like blight, root rot, wilt etc., are very important. Among of these, dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler, is a major problem. The disease occurs mostly when there is a moisture stress. It has been estimated that root rot may result in 10 to 100 per cent crop loss (Singh *et al.*, 1990).

Dry root rot generally appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The

symptoms include drooping of petioles and leaflets which are confined to top of the plants. The tap root is dark, shows signs of rotting and is devoid of most of its lateral and finer roots. The most important diagnostic symptom is shredding of bark and which comes out in the form of flakes (Haware, 1990). The disease occurred in severe form during *rabi* 2008 and 2009 in Kadapa and Kurnool districts of Rayalaseema region of Andhra Pradesh. This initiates to work on this disease to generate basic information on percentage of disease incidence, to search for an alternate approach other than use of fungicides for the management of the disease.

Biological control of plant pathogens is a distinct possibility for the future and can be successfully exploited in the modern agriculture, especially within the frame work of integrated disease management (IDM) system which is needed to hold disease below economic threshold level without damaging the agro ecosystem (Papavizas, 1985). So far, the biocontrol agents were isolated and characterized mainly from rhizosphere. The information on root endophytic biocontrol agents is scanty and as such it is worthwhile to isolate microflora having antagonistic activity from that new habitat of root.

Integrated disease management is gaining importance which involves blending of compatible systems of control measures for effective management of disease from profitability to food and environmental safety (Jacksen and Backmon, 1993).

Recently, molecular techniques gained importance in characterization and diagnosis of microbial population. Hence, molecular characterization of the potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD), 16S rDNA and 16S rDNA-RFLP which helps in identification of antagonists are of immense use. These molecular techniques will also help in developing SCAR markers for the diagnosis of potential biocontrol agents in future.

Hence, in the present investigation, an attempt was made to explore the feasibility of using biocontrol agents along with fungicides for the management of dry root rot of chickpea incited by *Rhizoctonia bataticola* with the following objectives.

1. Survey for incidence of dry root rot of chickpea in Kadapa and Kurnool districts of Rayalaseema region, Andhra Pradesh.
2. Isolation and identification of pathogen from infected plants.
3. To evaluate the efficacy of fungicides against causal agent of dry root rot under *in vitro*.
4. To isolate potential fungicidal compatible biocontrol agents from rhizosphere and root endophytes under *in vitro*.
5. Integrated disease management of dry root rot of chickpea under greenhouse conditions.
6. Molecular characterization of the potential biocontrol agents.

CHAPTER – II

REVIEW OF LITERATURE

A brief review of literature pertaining to the aspects under investigation is presented in this chapter.

Wherever, the literature on particular aspects of *Rhizoctonia bataticola* on chickpea was scanty, it was amply supplemented with and supported by relevant literature from other pathogen and other crops.

2.1 SURVEY

Taya *et al.* (1988) studied the influence of soil type on the severity of dry root rot caused by *R. bataticola* of chickpea and observed more severe incidence of the disease in sandy soils than clay soils.

Pandey and Singh (1990) reported *Fusarium oxysporum* and *R. bataticola* on greengram in Allahabad. These pathogens were found associated with the crop in an average incidence of 19.2 and 5.0 per cent respectively.

Sahu and Jena (1997) surveyed and studied the seed microflora of 10 cultivars of greengram and isolated 20 fungal taxa belonging to 13 genera of which *Macrophomina phaseolina* was the dominant species associated with all the cultivars tested.

Kratisharma and Tribhuwan Singh (2000) observed twenty four per cent of mungbean seed samples collected from 11 districts of Rajasthan during 1996-97 showing 0.5 to 38 per cent *Rhizoctonia bataticola* infection.

Singh and Agarwal (2002) conducted a survey in the vindhya plateaux zone (Bhopal, Raisen, Sagar, Seshore and Vidish) of Madhya Pradesh to evaluate the prevalence of dry root rot in chickpea and reported that the incidence of *R. bataticola* was ranged from 8 to 20 per cent.

Prajapati *et al.* (2003) reported that the incidence of dry root rot of chickpea caused by *R. bataticola* was highest in bold- seeded cultivars.

Singh and Sirohi (2003) evaluated the incidence of chickpea diseases in the Himachal Pradesh and reported that the incidence of dry root rot was highest in Una (4.86%) and Sirmour (3.04%) districts.

Gurha and Trivedi (2008) conducted a survey in all the chickpea growing districts of Karnataka and reported that *R. bataticola* was found as the predominant pathogen which infected 60.0 per cent plants in the fields of Gulberga.

2.2 THE PATHOGEN.

The pathogen occurs in the sclerotial form with the taxonomic nomenclature as *Rhizoctonia bataticola* (Taub) Butler. The pycnidial stage of this pathogen is *M. phaseolina* (Tassi) Goid. Mc Rae (1929) reported the

occurrence of *Macrophomina phaseolina* on wilted plants of mung bean and urdbean and established the genetic similarity between *M. phaseolina* and *R. bataticola*. Different plant parts of host were harboring the pathogen and isolated the pathogen from various plant parts viz., root, stem, leaf, pod and seed and the pathogen also differed in their cultural and morphological characters (Jain *et al.*, 1973).

2.2.1 Morphology

Philip *et al.* (1969) noticed the differences in morphology of *Macrophomina phaseolina* isolates from various parts of host plants. They include the pycnidial development *in vitro* on mung bean roots infected with *Macrophomina*, as black, globose or depressed and 150-200 µm in diameter. Pycnidiospores were oval or elliptical, hyaline, non-septate, thin walled with 10-24 x 6-10 µm in size. The pycnidiospores develop into *R. bataticola* in culture at room temperature. Occurrence of numerous small black sclerotia of this pathogen under *in vitro* conditions was also reported by Jain *et al.*, (1973). According to him, sclerotial morphology of *Macrophomina phaseolina* of urdbean was found to be different when isolated from different plant parts. The soil and seed isolate developed small sclerotia which were more pathogenic than leaf isolate which developed larger sclerotia.

2.2.2 DISTRIBUTION AND ECONOMIC IMPORTANCE OF *R. bataticola*

Rhizoctonia bataticola is a soil borne pathogen with a wide host range (Dhingra and Sinclair, 1978) and causes charcoal rot (seedling phase), root rot and stem blight in more than 500 species of plants (Sinclair, 1982).

The pathogen has been found to occur through out the world particularly in tropical and sub-tropical regions, namely India (Butler, 1918), Uganda (Hansford, 1943), United states (Hoffmaster *et al.*, 1943), Australia (Anonymous, 1965), Phillipines (Yang, 1977) and Pakistan (Shahzed and Ghaffer, 1986).

The disease development is influenced by dry soil conditions, especially at flowering causing the plants to sudden drying (Singh and Mehrotra, 1982). The incidence of dry root rot of chickpea has been reported to be as high as 24.70 per cent in Madhya Pradesh by Sharma *et al.* (1983).

Kataria and Grover (1977) and Tyagi *et al.* (1988) reported a yield loss of 10.8 per cent and 24.1 per cent due to *R. bataticola* in mungbean from the states of Haryana and Rajasthan respectively.

Ahmed and Mohammad (1986) reported a yield loss of 50 to 71 per cent due to *R. bataticola* in chickpea growing areas of Bihar.

Pineda and Avila (1993) reported a yield loss of 37 to 79 per cent due to *Macrophomina phaseolina* in sunflower under rainfed conditions. Plant

losses upto 77 per cent have been reported due to *Rhizoctonia bataticola* in soybean (Muthuswamy and Mariappan, 1991).

Hwang *et al.* (2003) studied the impact of seedling blight and root rot caused by *Rhizoctonia solani* Ag-4 on nodulation and seed yield of chickpea and reported that emergence and dry matter production declined and root rot severity increased with increasing inoculum concentrations under green house conditions.

2.3 SYMPTOMATOLOGY

Nene *et al.* (1978) observed continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of dry root rot of chickpea.

Baldev *et al.* (1988) reported symptoms of dry root rot of chickpea which consisted of straw colored leaves and stems. Tap root of infected plants was observed to be dry and devoid of lateral and finer roots. Roots were turned dark and showed signs of rotting. The dead root was observed brittle towards the tip and showed shredding of bark.

Haware (1990) reported that dry root rot of chickpea appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The symptoms induced were drooping of petioles and leaflets which were confined to top of the plants. Shredding of bark in form of flakes was observed.

Singh *et al.* (1990) observed that *R. bataticola* inoculated roots of chickpea, upon microscopic examination, showed disintegration of cortical tissue and plugging of xylem vessels with mycelial and sclerotial bodies of fungus.

Rangaswamy (1996) reported field symptoms which include yellowing of plants with drooping of leaves. Due to decay of roots, plants can be easily uprooted. Root portion appears brownish from outside. The stem and root below the region shows rotting with frequently pinkish white mycelial growth. Dried plants scattered through out the field are indicative of root rot incidence.

Singh (1999) reported that dry root rot of chickpea occurred from flowering to podding stage. Infected plants were suddenly dried in the field. The tap roots turned dark brittle and devoid of lateral roots.

Khalid and Ilyas (2000) considered the presence of root lesions and sclerotium plugging in xylem vessels of roots and collar region as symptoms of dry root rot of chickpea for screening of germplasm against this disease.

Singh and Agarwal (2002) observed the withering and drying of chickpea plants in the field due to the infestation of *Rhizoctonia bataticola*.

2.4 ISOLATION OF THE PATHOGEN

Rhizoctonia bataticola can be isolated from different plant parts viz., diseased seeds and seedlings (Rajeevpant and Mukhopadhyay, 2001; Bagri *et al.*, 2004; Vinod Kumar *et al.*, 2007), roots (Sajeena *et al.*, 2004), stem (Pan and Bhagat, 2008) and leaves (Sarkar and Pradhan, 1999).

2.4.1 Maintenance of the pathogen

Potato dextrose agar (PDA) was found to be the best supporting medium for *R. bataticola* (Vinod Kumar, 2007; Konde *et al.*, 2008). The pathogen *R. bataticola* was stored in an Agar Papa dextrose culture medium (ADP, Difco) in a bioclimatic chamber at 4°C (Valiente *et al.*, 2008).

2.5 PATHOGENECITY TESTS

Vishwadar and Sarabhay (1993) studied the variation in isolates of *R. bataticola* isolated from soybean plants collected from 11 locations at IARI. About 44 isolates were isolated from different parts of the plants and it was observed that the isolates differed with reference to colour of the colony, mycelial growth, shape and size of sclerotia. Morphological and cultural variations in isolates of *R. bataticola* isolated from different crops were also reported by other workers.

Prameela Devi and Singh (1998) reported 10 per cent inoculum density causing 71 per cent seedling mortality in greengram and blackgram.

Kataria *et al.* (2007) assessed nine isolates of *R. bataticola*, the incitant of dry root rot of chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea in both sterilized and unsterilized soil. The root rot incidence varied from 26 to 100 per cent in sterilized soil and 20 to 80 per cent in an unsterilized soil.

Jaiman and Jain (2008) conducted pathogenicity test for *M. phaseolina* in cluster bean seeds and observed that seed inoculation with *M. phaseolina* resulted in pre-(33%) and post emergence mortality and less of seed germination (59%) and vigour index (856) in comparison to control (93%).

2.6 ANTAGONISTS

2.6.1 Isolation of antagonistic mycoflora and bacteria from rhizosphere soil and root endophytes

Singh and Mehrotra (1980) isolated the *Bacillus* and actinomycetes from chickpea rhizosphere and were found to be antagonistic to *R. bataticola* under *in vitro* conditions.

Nautiyal (1997) isolated 478 chickpea rhizosphere competent bacteria for suppression of chickpea pathogenic fungi *Fusarium oxysporum* f. sp. *ciceri*, *Rhizoctonia bataticola* and *Pythium* sp. and found 386 strains that effectively colonize chickpea roots.

Parmer and Dadarwal (1997) isolated *Pseudomonas* and *Bacillus* sp. from rhizosphere and rhizoplane of healthy chickpea plants.

Jayalakshmi *et al.* (2003) isolated *Trichoderma harzianum* from rhizosphere of healthy pigeonpea plants.

Sendhilvel (2005) isolated five different isolates of *Pseudomonas fluorescens* from cowpea rhizosphere region and screened against the *Macrophomina phaseolina* the causal organism of cowpea dry root rot.

Ramesh and Korikanthimath (2006) isolated biocontrol agents like *Trichoderma viride*, *T. harzianum*, *Pseudomonas fluorescens* and *Bacillus subtilis* from rhizosphere of various crops and tested their efficacy against the *Macrophomina phaseolina* causing root rot of groundnut.

Zeidan (2006) isolated root endophytes from peanut healthy roots and found that *Bacillus subtilis* (No: 1) abundantly colonized peanut root than *P. fluorescens* and effectively controls the root and pod rot diseases.

Mendes *et al.* (2007) isolated endophytic bacteria from roots and stems of healthy sugarcane and found that *Burkholderia* isolates produced the antifungal metabolite pyrrolnitrin.

Tiwari and Thrimurthy (2007) isolated 21 isolates of *Pseudomonas fluorescens* from rhizosphere of rice, maize, wheat, chickpea, mung, urd and soyabean from Raipur and Bastar regions and revealed that seven isolates

were found to be effective against *R. solani*, the incitant of rice sheath blight.

2.6.2 *In vitro* evaluation of the efficacy of antagonists against *R. bataticola*

Parakhia and Vaishnav (1986) reported the growth of *R. bataticola*, causal organism of dry root rot of chickpea was inhibited by *T. harzianum* in dual culture and the antagonist over grew the pathogen. The growth of *Macrophomina phaseolina* was significantly inhibited by *Bacillus subtilis* under *in vitro* conditions (Jharia and Khare, 1986).

Selvarajan and Jeyarajan (1996) screened three *Trichoderma* spp. under *in vitro* against chickpea root rot causing organisms viz., *Fusarium solani* and *Macrophomina phaseolina*. They observed that *T. viride*, *T. harzianum* and *T. hamatum* formed inhibition zones against both the chickpea root rot pathogens. The antagonist reduced sclerotial size and germination of *Macrophomina phaseolina*.

Khot *et al.* (1996) reported that five rhizobacteria isolated from rhizosphere of chickpea plants inhibited the growth of *F. o. sp. ciceri* and *Rhizoctonia bataticola* under *in vitro*.

Jayasree *et al.* (2000) obtained ten fluorescent *Pseudomonas* strains from rhizosphere of blackgram and studied their efficacy against *Macrophomina phaseolina* under *in vitro*. Among these strains, pf₁ was

found to be more effective in inhibiting the mycelial growth of *Macrophomina phaseolina* of blackgram.

Patel and Anahosur (2001) observed that *T. harzianum* formed inhibition zone against *Fusarium* sp. and *Macrophomina phaseolina*.

Sindhan *et al* (2002) reported the antagonistic activity of *T. viride*, *T. harzianum*, *A. flavus*, *A. niger*, *Penicillium* sp., *B. subtilis* and *Pseudomonas fluorescens* against *Rhizoctonia bataticola* under *in vitro* separately using dual culture method. Among the antagonists, *Pseudomonas fluorescens* exhibited maximum antagonistic activity causing 65 and 90.5 per cent inhibition in mycelial growth and sclerotia production respectively.

Ananthan *et al* (2003) studied the antagonistic effect of four *Trichoderma* sp. in dual culture technique. All the *Trichoderma* spp. significantly inhibited the growth of *M. phaseolina* to an extent of 16.11% as compared to control.

Suriachandraselvan (2004) tested the efficacy of *Trichoderma* sp. against *Macrophomina phaseolina* causal organism of sunflower charcoal rot and reported that *T. viride* and *T. harzianum* significantly inhibited the mycelial growth of pathogen (73-74%).

Sendhilvel (2005) reported that *Pseudomonas fluorescens* isolate SVPF2 was found to be effective in inhibiting the mycelial growth of *M. phaseolina* incitant of cowpea root rot.

Paramasivan *et al.* (2007) tested the various fungal and bacterial antagonists against *Macrophomina phaseolina* causal organism of dry root rot of Coleus and revealed that *T. viride* isolate-4 and *Pseudomonas fluorescens* were effective in inhibiting mycelial growth of pathogen.

Rajeena and Ahmad (2007) evaluated the efficacy of *Pseudomonas fluorescens* against *Aspergillus flavus*, *A. niger* and *M. phaseolina* pathogens of okra seedlings and reported that *Pseudomonas fluorescens* showed complete inhibition of *A. flavus* and *M. phaseolina* under *in vitro*.

Vinod Kumar *et al.* (2007) tested the efficacy of *Pseudomonas fluorescens* isolates against *M. phaseolina* incitant of charcoal rot of chickpea under *in vitro* and reported that the isolate pf 4-99 was found to be effective in inhibiting the mycelial growth of pathogen.

Konde *et al.* (2008) tested the five antagonist's of soybean and observed that *Trichoderma viride* (96.39%) was superior in inhibiting the growth of *R. bataticola*.

Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* causal organism of chickpea dry root rot.

Khan and Gangopadhyay (2008) studied the efficacy of *Pseudomonas fluorescens* strains against *R. bataticola* causal organism of chickpea dry root rot and reported that *P. fluorescens* strains PFBC-25 and 26 were effective in reducing the growth of the pathogen.

Pan and Bhagat (2008) isolated ten isolates of *Trichoderma* and evaluated their efficacy against *M. phaseolina* incitant of stem rot of jute. The results revealed that *Trichoderma* isolate-6 was most effective in parasitizing growth of the pathogen.

Sharma *et al.* (2009) isolated eight endophytic *Trichoderma* isolates from leaves of *Ficus religiosa* and evaluated their efficacy against soil borne pathogens viz., *R. solani* and *S. rolfsii*. The results revealed that *T. viride* I and II were effective in inhibiting the mycelial growth of *R. solani* and *S. rolfsii*.

2.7 IN VITRO EVALUATION OF FUNGICIDES AGAINST PATHOGEN

Pershney *et al.* (1992) studied the effect of fungicides viz., thiram, captan, mancozeb, PCNB, carboxin, carbendazim and tridemorph on the growth of *Rhizoctonia bataticola* causing charcoal rot of sorghum. The fungus was found to be highly sensitive to carbendazim and thiobendazole even at 600 ppm concentration.

Singh and Kaiser (1995) evaluated some systemic and non-systemic fungicides under *in vitro* against charcoal rot pathogen of maize. Complete inhibition of growth of the pathogen was observed at a very low concentration of 30 ppm with carbendazim followed by topsin (40 ppm), agrizime (100 ppm), rhizolex (120 ppm), derosol (20 ppm), foltaf (500 ppm) and thiram (1000 pm).

The effect of six fungicides on growth of *M. phaseolina* causing root rot disease in greengram was studied under *in vitro*. Carbendazim (0.1%) completely inhibited the growth of the pathogen. The fungicides copper oxychloride (0.25%), thiram (0.2%), thiophanate methyl (0.1%), captan (0.2%) and methoxy ethyl chloride (0.05%) also suppressed the pathogen by 95.89, 91.11, 89.67, 78.56 and 77.44 per cent respectively (Ebenezar and Wesely, 2000).

Rajeevpant and Mukhopadhyay (2001) evaluated the fungicides against *Macrophomina phaseolina* causal organism of soybean dry root rot and revealed that vitavax was found highly inhibitory to *Macrophomina phaseolina*.

Chattopadhyay and Kalpana (2002) reported that carbendazim could cause 100 per cent inhibition of mycelial growth of *Macrophomina phaseolina* even at 25 ppm concentration.

Konde *et al.* (2008) evaluated the fungicides against *R. bataticola* causing root rot of soybean and revealed that combination of carbendazim + thiram (0.1 + 0.2 % concentration), penconozol (0.1%) and thiophanate-M (0.1%) were significantly effective in completely (100%) inhibiting the radial growth of *R. bataticola*.

Khan and Gangopadhyay (2008) evaluated the fungicides against dry root rot of chickpea and reported that captan and carbendazim were highly inhibitory to *M. phaseolina*.

Paul *et al.* (2008) evaluated the fungicides against the root rot of pea caused by *F. solani*, *Rhizoctonia solani* and *S. sclerotiorum* and results revealed that carbendazim was most effective fungicide against all the pathogens.

2.8 IN VITRO COMPATIBILITY OF POTENTIAL ANTAGONISTS WITH FUNGICIDES

Combination of fungicides and biocontrol agents for controlling soil borne pathogens has been successfully used by many workers (Papavizas and Lumsden, 1980 and Poddar *et al.*, 2004).

Vyas (1987) reported that *Trichoderma* sp. and *B. subtilis* showed high degree of tolerance to thiram and carbendazim.

Vidhyasekharan *et al.* (1995) reported that thiram and carbendazim were not inhibitory to *P. fluorescens* under *in vitro* conditions.

Rajeevpanth and Mukhopadhyay (2001) observed that vitavax was compatible with *G. virens* and *T. harzianum* for management of seed and seedling rot of soybean caused by *M. phaseolina*.

Girija and Umamaheswaran (2003) reported the compatibility of *T.virens* with carbendazim under *in vitro* at three concentrations (100,150 and 1000 ppm) and observed that the antagonist *T. virens* was compatible with carbendazim at 100 ppm concentration.

Gupta (2004) reported that carbendazim completely inhibited the mycelial growth of *Trichoderma harzianum* at concentrations 1, 10, 100, and 1000 ppm concentrations.

Tiwari and Singh (2004) evaluated the *in vitro* efficacy of different fungicides against *T. harzianum* @ 1500 ppm and reported that the mycelial growth of *T. harzianum* was completely inhibited by carbendazim and hexaconazole @ 1500 ppm and the inhibition with copper oxychloride and mancozeb was 90 and 41 per cent.

Gupta *et al.* (2005) reported that carbendazim was incompatible with *Trichoderma viride* TV2 isolate while with carboxin was compatible for integrated treatment.

Naseema Beevi *et al.* (2005) tested the *in vitro* compatibility of *T.harzianum* with mancozeb, carbendazim and copper oxychloride and found that carbendazim at 0.1 per cent completely inhibited the mycelial growth while mancozeb and copper oxychloride showed compatibility with the antagonist at 0.2 and 0.1 per cent respectively.

Khan and Gangopadhyay (2008) tested the compatibility of *Pseudomonas fluorescens* with the fungicides and revealed that carboxin, and carbendazim were least toxic to *P. fluorescens* strain PFBC-25 whereas captan was most inhibitory.

2.9 MASS MULTIPLICATION OF *Rhizoctonia bataticola*

The pathogen *Rhizoctonia bataticola* was mass multiplied on sand maize medium (Sajeen *et al.*, 2004; Kataria *et al.*, 2007). Sterilized sorghum grains were used for mass culturing of *R. bataticola* (Rajeswari *et al.*, 1999; Durai, 2004).

2.10 MASS MULTIPLICATION OF ANTAGONISTS

Wheat bran has been used as best substrate for mass multiplication of *Trichoderma viride* (Dubey and Patel, 2002; Patibanda *et al.*, 2002; Upadhyay *et al.*, 2004 and Gaur *et al.*, 2005).

Roopali Sharma *et al.* (2002) reported that FYM + Jhingora (Barnyard millet) in 3:1 w/w was more suitable for mass multiplication of *Trichoderma harzianum*.

Umamaheswari *et al.* (2002) multiplied *P. fluorescens* and *Bacillus subtilis* in King's B and nutrient broth medium (Padmodaya and Reddy, 1998 and Gogoi *et al.*, 2002).

Zaidi and Singh (2004) reported that cow dung and poultry manure supported good growth of *Trichoderma harzianum*. They also tested three substrates for multiplication of *Pseudomonas fluorescens* (PBAD-27). The results revealed that the population of *P. fluorescens* was more in cow dung (42.6×10^{12}) and least in press mud (1.6×10^6).

Pre-boiled and sterilized sorghum grains supplemented with anhydrous dextrose was used for mass multiplication of *G. virens* and *T. harzianum* (Rajeevpant and Mukhopadhyay, 2001).

Diby Paul *et al.* (2005) multiplied *P. fluorescens* in nutrient broth at 28°C for 48 hrs.

Muthukumar and Bhaskaran (2007) multiplied *P. fluorescens* in king's B broth at room temperature ($28 \pm 2^\circ\text{C}$) for three days (Bora and Deka, 2008).

2.11 IN VIVO EVALUATION OF EFFICACY OF FUNGICIDES AGAINST *Rhizoctonia bataticola*

Raut and Somani (1987) reported that seed treatment with captan (0.2%) and thiram (0.2%) were found effective against chickpea dry root rot.

Pall *et al.* (1990) reported that seed treatment with carbendazim (0.1%) was found effective against mungbean dry root rot.

Rajpurohit (1997) evaluated eight fungicides against stem and root rot of sesame caused by *Macrophomina phaseolina*. Seed treatment with carbendazim, carboxin, captan and thiram were effective in reducing the disease.

Sharma and Tripathi (2001) reported the efficacy of opus (0.2%), carbendazim (0.2%), propiconazole (0.1%), hexaconazole (0.1%) and propineb (0.25%) against *M. phaseolina*. Seed treatment and foliar sprays of propiconazole at 15 days interval resulted in maximum reduction in disease severity (30-32%) and increase in grain yield (950-1012 Kg/ha) and 1000 grain weight (35g).

Seed and soil application of thiophanate methyl resulted in less disease incidence (12.2%) as compared to soil treatment (16.55%) or seed treatment (16.2%) alone in management of sunflower charcoal rot caused by *Macrophomina phaseolina* in pot culture (Padmalatha, 2002).

Prajapati *et al.* (2002) used carbendazim as seed treatment against root rot of chickpea and reported that carbendazim was effective in reducing the disease incidence.

Gupta (1995) evaluated twenty fungicides including two seed treatments, six sprays and twelve combinations against stem blight of cowpea caused by *M. phaseolina*. Seed treatment with carbendazim +

thiram followed by one spray of mancozeb was found economical and superior in controlling the disease.

Ramesh and Koriakanthimath (2006) studied the efficacy of fungicides on root rot incidence of groundnut and revealed that seed treatment with carbendazim showed low disease incidence (6.7%).

Vijay Mohan *et al.* (2006) used carbendazim (0.2%) and Etaconazole (0.1%) as seed treatment, soil drenching and seed treatment plus soil drenching against chickpea root rot incidence and recorded lowest disease incidences of 15.60 and 18.2 per cent and highest grain yield 19.20 and 18.90 q/ha respectively.

Kaushal (2008) reported that seed treatment with thiram and carbendazim at 2g/kg was effective in control of chickpea dry root rot.

2.12 IN VIVO EVALUATION OF EFFICACY OF ANTAGONISTS AGAINST *Rhizoctonia bataticola*

Kheri and Chandra (1991) studied the efficacy of *Trichoderma viride* in controlling the pathogenic activity of *R. bataticola* causing dry root rot of chickpea. The antagonist applied as seed coating reduced mortality of chickpea from 8-9 per cent in unsterilized soils under greenhouse conditions and the biocontrol efficacy of the antagonist increased in sterilized soils.

Bacterization of chickpea and soybean seeds with siderophore producing fluorescent pseudomonads resulted in increased seed germination, growth and yield of the plants (Dileep kumar and Dubey, 1992).

Raghuchander *et al.* (1993) studied the effect of different isolates of *T. viride* in the control of root rot of mungbean caused *M. phaseolina*. Application of biocontrol agents in soil resulted in reduction of root rot incidence to the extent of 16 per cent as compared to control.

Jeyarajan *et al.* (1994) developed a talc based formulation of *T. harzianum* which reduced the root rot incidence in urdbean, chickpea, peanut and gingelly by 66, 50, 77 and 67 per cent and increased yield of 20, 13, 12 and 12 per cent respectively.

Seed coating with *Bacillus subtilis* inhibited dry root rot incidence by 52 per cent in chickpea (Saxena and Saxena, 1995).

Khot *et al.* (1996) reported that seed inoculation with *Pseudomonas fluorescens* resulted in significant control of root rot due to *M. phaseolina* in chickpea under field conditions.

Nautiyal (1997) reported that chickpea seed bacterization with *Pseudomonas fluorescens* NBRI1303 increased the germination of seedlings by 25 per cent and reduced the number of diseased plants to 45 per cent.

Rajendra singh *et al.* (1998) reported that *Pseudomonas fluorescens* was found effective in reducing the incidence of dry root rot of chickpea caused by *Rhizoctonia bataticola*.

Singh *et al.* (1998) reported that *T. viride* and *T. harzianum* used as seed treatment, soil application and soil application plus seed treatment were found effective in reducing the incidence of dry root rot by *M. phaseolina* in chickpea.

Rangeshwaran *et al.* (2003) studied the effect of seed treatment with two bacterial antagonists viz., *Pseudomonas putida* and *P. fluorescens* in control of root rot of chickpea caused by *Rhizoctonia* sp. and results indicated that seed treatment with both the antagonists significantly reduced root rot incidence of chickpea when compared to control. Among the two bacterial antagonists *P. fluorescens* was found to be highly effective in reducing the incidence by 6.4 per cent compared to *P. putida*. Seed treatment with these two biocontrol agents also resulted in significant increase in yield.

Meena *et al.* (2001) reported the reduction of groundnut root rot incidence significantly in field experiments by seed treatment with powder formulation of *Pseudomonas fluorescens*.

Prasad *et al.* (2002) evaluated two antagonistic fungi *Trichoderma harzianum* (PDBCH 10) and *T. viride* (PDBCV) against wilt and root rot of

chickpea under pot culture. These results revealed that soil application of *Trichoderma harzianum* showed only 4.9 and 1.2 per cent root rot incidence at 30 and 60 days respectively.

Mondal and Bhattacharya (2003) studied the efficacy of bacterial isolates of *Bacillus* sp. in controlling the pathogenic activity of *Macrophomina phaseolina* causing dry root rot of chickpea and reported that the two bacterial isolates (S12 and S17) when applied as both seed and soil drench reduced the root rot incidence of chickpea under pot culture studies.

Tewari and Mukhopadhyay (2003) reported that application of carboxy methyl cellulose (CMC) with *Gliocladium virens* powder (10^9 spores/g) in combination with vitavax showed maximum protection (81.90%) to chickpea crop against root rot and collar rot pathogen.

Gaur *et al.* (2005) revealed that soil application of wheat bran based *T. harzianum* (@15g/3kg of soil) and FYM (100g) mixture applied at sowing was found to be significantly superior to check in reducing root rot of chickpea by 76 per cent and enhancing overall germination (35%) over control.

Khan and Gangopadhyay (2008) reported that seed treatment with *Pseudomonas fluorescens* isolates PFBC-25 and PFBC-26 significantly reduced the root rot incidence of chickpea.

2.13 INTEGRATED MANAGEMENT OF DRY ROOT ROT

Vyas and Khare (1986) obtained good control of soybean root rot by combined application of *T. harzianum* and carbendazim.

Vyas (1987) confirmed that the disease could be reduced by combination of thiram + carbendazim (2:1) followed by seed treatment with fungicide tolerant strain of *Trichoderma* sp. or *Bacillus subtilis* in *Macrophomina* sick soils. Simultaneous application of *T. viride* or *T. harzianum* with carbendazim treatment was effective in reducing dry root rot of soybean (Vyas, 1994).

Sankar and Jeyarajan (1995) conducted a field experiment to manage root rot of sesame caused by *Macrophomina phaseolina* by seed treatment with antagonists *T. harzianum* and *T. viride* that significantly reduced the root rot incidence to 10.1 and 12.8 per cent respectively, compared to 60 per cent in control and were superior to carbendazim seed treatment.

Rajeswari (1997) reported that dry root rot of mungbean caused by *Macrophomina phaseolina* was effectively controlled by integration of *Trichoderma harzianum* as soil application (5g/kg) + seed treatment (10^8 cfu/ml) with sub-lethal doses of carbendazim (0.02%) significantly reduced the dry root rot incidence (95.3 %) over soil treatment (91.5%) and seed treatment alone.

Patel and Anahosur (2001) enumerated rhizosphere microflora of chickpea under irrigated conditions and found that carbendazim as seed treatment and soil drench coupled with soil application of *Trichoderma harzianum* reduced maximum fungal population of *Fusarium* sp. and *Macrophomina phaseolina*. FYM also reduced pathogen population and stimulated all other microflora more than other treatment there by showing antagonism against *Fusarium* sp. and *M. phaseolina*.

Venkateswar Rao (1998) reported that an integration of soil solarization (six weeks), *Trichoderma viride* (native) and mancozeb (0.3%) effectively controlled the stem and root rot of sesame to the tune of 82.2 per cent caused by *Macrophomina phaseolina* under field conditions.

Jayasree *et al.* (2000) investigated that the *Pseudomonas fluorescens* strain pf₁ effectively inhibited the mycelia growth of *Macrophomina phaseolina*, the pathogen causing dry root rot in blackgram and sesame. Among various methods of pf₁ application, seed treatment + soil application recorded maximum yield in blackgram (1238 kg/ha) and sesame (1200 kg/ha) followed by seed treatment + soil drenching of carbendazim, which recorded grain yield to the extent of 1205 kg/ha for blackgram and 820 kg/ha for sesame.

Prasanthi *et al.* (2000) evaluated fungal and bacterial antagonists as seed and soil application against safflower root rot caused by *R. bataticola*.

Both seed treatment and soil drenching with antagonists increased safflower seedling percentage survival being seed treatment being more effective than soil drenching with highest survival rate 83.33% with *T. viride* and 86.66 % with *Pseudomonas fluorescens*.

Gaikwad *et al.* (2002) studied the about management of charcoal rot of sorghum using three biocontrol agents viz., *T. viride*, *T. harzianum* and *P. fluorescens* and the fungicides thiram and carbendazim. They have found that seed treatment with biocontrol agents and fungicides were highly effective against *M. phaseolina*.

Sindhan *et al.* (2002) reported satisfactory disease control when *Pseudomonas fluorescens* was used as seed treatment along with carbendazim against *Macrophomina phaseolina* in chickpea.

Singh and Sinha (2007) evaluated field efficacy of foliar application of different formulations of *P. fluorescens* (pfr₁ and pfr₅) at three doses viz., 2, 4 and 8 g/l against sheath blight of rice. A higher dose (8g/l) was found highly effective in reducing disease severity (57%), incidence (31%) and increasing grain yield (32%) and 1000 grain weight (13%).

Vinod kumar *et al.* (2007) tested the efficacy of *P. fluorescens* isolates against charcoal rot of chickpea both in green house as well as field conditions. The observations revealed that isolate pf4-99 was effective in reducing the charcoal rot disease and also increased seed yields.

Khan and Gangopadhyay (2008) evaluated the combined seed treatment of *Pseudomonas fluorescens* and fungicides against the dry root rot of chickpea and revealed that *P. fluorescens* isolate PFBC-26 was effective against the pathogen when used in combination with carbendazim.

Many of the workers reported that soil application of *Trichoderma* sp. and *P. fluorescens* effectively controlled the diseases caused by soil borne plant pathogens and also a synergistic effect in the growth of plants was observed (Suriachandraselvan, 1997; Manoranjitham *et al.*, 2000; Bharati *et al.*, 2004 and Poddar *et al.*, 2004).

2.14 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

Misbah *et al.* (2005) identified the *Acinetobacter* of clinical isolates by amplification of 16S r DNA genome region consisting of approximately 1500 nucleotides using three primer pairs viz 27F, 780R; 529F, 1099R; 925F and 1491R.

Ramesh Kumar *et al.* (2002) studied the genetic variability among the isolates of *Pseudomonas* by RAPD with random primers, and the primer pgs3 produced several bands, including a unique band with size of 800 bp.

Megha *et al.* (2007) studied the diversity of fifteen isolates of fluorescent pseudomonads using RAPD - PCR with eight random primers viz., OPC-9, OPD-2, OPD-3, OPO-6, OPO-09, OPO-13, 15 and 16. The

PCR amplicons of fluorescent pseudomonads obtained from eight random primers produced 127 polymorphic bands. The minimum bands (9) were produced by the primer OPD-02 and maximum number of bands (25) were produced by OPO-16.

CHAPTER - III

MATERIALS AND METHODS

3.1 LOCATION OF WORK

The laboratory experiments pertaining to the research work were conducted during the year 2009-10 in the Department of Plant Pathology, S.V. Agricultural College, Tirupati, Chittoor District, Andhra Pradesh.

3.2.1 Glassware

Glassware made of corning or borosil were used throughout the present investigation. They were Petri plates (90 mm dia), test tubes, conical flasks (100, 250 and 500 ml), beakers (100, 500 and 1000 ml), pipettes (1, 2, 5 and 10 ml) and measuring cylinders (10, 50, 100 and 500 ml). The glassware was first washed with detergent followed by thorough cleaning with tap water before placing them in cleaning solution for 24 hrs and finally rinsed with distilled water for 3-4 times and were air dried before use.

The composition of cleaning solution

Potassium dichromate	:	60 g
Concentrated sulphuric acid	:	60 ml
Distilled water	:	1000 ml

3.2.2 Chemicals

Chemicals used in the present study were of analytical reagent (AR) and guaranteed reagent (GR) grades of standard mark. The pH of the media was adjusted using either 0.1N HCl or 0.1 N NaOH.

3.2.3 Equipments

Hot air oven and autoclave were used for sterilization of glassware and media respectively. Incubators were used for incubating test materials at different temperatures. The cultures were stored in refrigerator. Compound microscope (10x, 40x magnifications) was used for observing the fungi. Weighments were done on a single pan electronic balance with a sensitivity of 0.001g. The sclerotia and mycelium of *Rhizoctonia bataticola* were observed by using microscope with built in camera attachment (Motic Camera, USA). Other tools which were used in the present investigation for various purposes include camel brush, inoculation needle, inoculation loop, cork borer, slides, cover slips, plastic pots etc.

3.2.4 For molecular study

DNA amplifications in RAPD and 16S rDNA were carried out in Gradient PCR master cycler (Eppendorff) and DNA banding patterns were documented in gel documentation system. Horizontal gel electrophoresis for gel running and UV transilluminator for observing the bands were used. Centrifugation was carried out in refrigerated eppendorf centrifuge. DNA

samples were stored at -20°C in deep freezer. Sterile pestle and mortar, liquid nitrogen, eppendorf tubes (1.5 ml), PCR tubes (0.2 ml), tips (10, 50, 200 and 1000 µl) and micro pipettes were used.

3.2.5 Culture media used

i. Potato Dextrose Agar (PDA) medium (Ainsworth, 1961)

Peeled potato slices	: 200 g
Dextrose	: 20 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 6.5

ii. Nutrient Agar (NA) medium (Tuite, 1969)

Peptone	: 5 g
Beef extract	: 3 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 7.0

iii. Nutrient Broth (NB)

Peptone	: 5 g
Beef extract	: 3 g
Distilled water	: 1000 ml
pH	: 7.0

iii. Rose Bengal Agar Medium (Martin, 1950)

Glucose	: 10 g
Peptone	: 2 g
K ₂ HPO ₄	: 1 g
MgSO ₄ .7H ₂ O	: 0.5 g
Rose Bengal	: 0.3 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 6.0

iv. Potato Dextrose Broth (PDB)

Peeled potato slices	: 200 g
Dextrose	: 20 g
Distilled water	: 1000 ml
pH	: 7.0

3.2.6 Sterilization

Glassware used for present investigation were kept in sterilization tins or wrapped in aluminum foil and were sterilized in hot air oven at 160°C for 90 minutes.

Surface of laminar air flow (LAF) was sterilized by wiping with cotton swab dipped in alcohol. Inoculation loop, cork borer and scalpel were sterilized by dipping in alcohol and heating to red hot.

The culture media and distilled water were sterilized in an autoclave at 15 p.s.i. for 20 minutes.

For pot culture experiment, the soil was sterilized in autoclave at 20 p.s.i. for 30 minutes for 2 consecutive days.

3.2.7 Laboratory techniques

The general laboratory techniques described by Dhingra and Sinclair (1995), Rangaswami and Mahadevan (1999), Nene and Thapliyal (1993), Aneja (1993) were followed for preparation of media, sterilization, isolation and maintenance of fungal cultures with slight modifications wherever necessary. The total genomic DNA from the isolates was extracted using bacterial genomic DNA kit M/S Medox, chennai. RAPD was performed as per the procedure given by Williams *et al.* (1990). The 16S rDNA was amplified as per the procedure given by Misbah *et al.* (2005).The amplified

product was cloned in TA-cloning kit supplied by M/s Fermentas, Bangalore and were sent to M/s MWG. Bangalore for sequencing.

3.2.8 Source of seed

Chickpea seeds of variety JG-11 popularly cultivated in Kadapa and Kurnool districts of Andhra Pradesh were obtained from Regional Agricultural Research Station, Nandyal and used for pot culture studies.

3.3 SURVEY

Roving survey was conducted in major chickpea growing mandals of Kadapa and Kurnool districts of Andhra Pradesh during *rabi* 2009-10 to study the incidence of dry root rot.

3.4. ISOLATION OF PATHOGEN

The pathogen was isolated from dry root rot infected chickpea plants by using tissue segment method (Rangaswami and Mahadevan, 1999).

Small pieces of tissue about 3 mm from infected collar region along with some healthy tissue were cut with sterile scalpel. Then the pieces were surface sterilized with 0.1 per cent HgCl_2 for 30 sec. followed by three washings in sterile distilled water to eliminate mercury ions on the bits of tissue. These bits were transferred to PDA plated Petri plates. Plates were incubated at $28 \pm 2^\circ\text{C}$ and observed periodically for growth of the fungus.

The culture was purified by single hyphal tip method and maintained on PDA by periodical transfer throughout the present investigation.

3.4.1 Identification of pathogen

The pathogen was identified based on its mycelial and sclerotial characters (Barnett and Hunter, 1972).

3.4.2 Pathogenicity test

Pathogenicity test was proved by soil infestation method. The pathogen was mass multiplied on sterilized sorghum grains in 250 ml conical flasks. The flasks were autoclaved at 15 p.s.i for 20 min. Then the flasks were inoculated with 4 discs of 5.0 mm diameter mycelial growth of three days old culture of *Rhizoctonia bataticola* grown on PDA plate. The flasks were incubated at $28 \pm 2^{\circ}\text{C}$ for seven days. Then the inoculum was mixed with sterilized soil @ 100 g kg^{-1} soil as the pots (22.5 cm) were filled. The seeds of chickpea were sown simultaneously with pathogen inoculation @ 10 seeds per pot and an uninoculated control was maintained. The plants were observed for root rot symptoms. Each treatment replicated three times.

3.5 *IN VITRO* EVALUATION OF EFFICACY OF FUNGICIDES AGAINST THE PATHOGEN

In vitro efficacy of fungicides against the pathogen was evaluated by poisoned food technique (Nene and Thapliyal, 1993). The list of fungicides used in the present studies is given below:

Sl. No.	Trade name	Common name	Active ingredient	Concentration (%)
1.	Thiride	Thiram	75% WP	0.25
2.	Captaf	Captan	50% WP	0.25
3.	Blitox	Copper oxychloride	50% WP	0.25
4.	Bavistin	Carbendazim	50% WP	0.10

To 50 ml of sterilized distilled water, required quantity of fungicide was added and mixed thoroughly. This solution was added to 50 ml of sterilized cool molten double strength PDA medium, mixed thoroughly and poured into Petri plates. 6 mm discs of four days old culture of pathogen were inoculated at the centre of Petri plates and then incubated at $28 \pm 2^{\circ}\text{C}$. Three replications were maintained for each fungicide. Medium without fungicide was kept as control. Per cent inhibition of the growth of the fungus over the control was calculated using the formula:

$$I = \frac{C - T}{C} \times 100$$

where,

I = Per cent reduction in growth of pathogen

C = Radial growth (mm) in control

T = Radial growth (mm) in treatment.

3.6 ISOLATION OF POTENTIAL FUNGICIDE TOLERANT BIOCONTROL AGENTS

3.6.1 Isolation of native antagonistic mycoflora and bacteria from rhizosphere.

Antagonistic mycoflora and bacteria were isolated by following serial dilution technique (Jhonson and Curl, 1977). Composite soil sample was collected from rhizosphere of healthy plants. The soil was dried under shade and then used for serial dilution. To get 10^{-1} dilution, 10g of this soil was dissolved in 90 ml of sterile distilled water from that 1ml of soil suspension was taken and added to 9 ml of sterile distilled water to get 10^{-2} dilution. This was repeated until a dilution of 10^{-4} for isolation of fungi and 10^{-6} for bacteria.

Antagonistic mycoflora were isolated on Rose Bengal Agar medium. One ml of final dilution of soil suspension was poured into sterilized Petri plates, then the melted and cooled media was poured. Plates were rotated gently to get uniform distribution of soil suspension in the medium. The plates were incubated at $28 \pm 2^\circ\text{C}$ and observed at frequent intervals for the

development of colonies. Three days old colonies of mycoflora were picked up and purified by single spore method, whereas one day old colonies of bacteria were picked up and purified by streak plate method.

3.6.2 Isolation of root endophytes

During the survey, the healthy chickpea plants were also collected for the isolation of root endophytes. For isolation of endophytes, 5 g of root was surface sterilized for 5 min. with 70.0 per cent ethanol and homogenized in 20 ml of sterilized phosphate buffer (0.2M Na₂HPO₄+0.2M NaH₂PO₄) using mortar and pestle. Appropriate dilutions (10⁻⁴ for fungi and 10⁻⁶ for bacteria) of these suspensions were plated on PDA and NA for the isolation of fungi and bacteria respectively. The plates were incubated for 72 hr at 28 ± 2°C (Kishore *et al.*, 2005).

3.6.3 Identification of potential biocontrol agents

Dual culture technique (Morton and Stroube, 1953) was used to identify the potential antagonists from rhizosphere and root endophytes.

3.6.3.1 Dual culture techniques

The antagonistic activity of microflora of rhizosphere and root endophytes against *Rhizoctonia bataticola* was determined by dual culture technique under *in vitro* conditions.

Mycelial discs measuring 6 mm diameter from four day old cultures of both fungal antagonist and the test pathogen were placed at equidistant on sterile Petri plate containing PDA medium. One day old cultures of bacteria were streaked on opposite side of the pathogen on PDA medium. The Petri plates were then incubated at $28 \pm 2^{\circ}\text{C}$. Three replications were maintained in each treatment. Suitable controls were kept without antagonist. Growth of antagonists, pathogen and zone of inhibition were measured after recording full growth of the pathogen in control plate. Per cent inhibition of mycelial growth of test pathogen was calculated by the formula:

$$I = \frac{C - T}{C} \times 100$$

where,

I = Per cent reduction in growth of test pathogen

C = Radial growth (mm) in control

T = Radial growth (mm) in treatments.

3.6.4 Identification of compatibility of fungicides with potential biocontrol agent under *in vitro*

Native potential biocontrol agents were tested for their compatibility with the fungicides viz., copper oxychloride (0.25%), carbendazim (0.1%), captan (0.25%) and thiram (0.25%). Fungal isolates were tested for their compatibility by poisoned food technique (Nene and Thapliyal, 1993) and

spectrophotometric method (Kishore *et al.* 2005) for bacterial isolates under *in vitro*.

3.6.4.1 Poisoned food technique

To 50 ml of sterilized distilled water, required quantity of fungicide was added and mixed thoroughly. This solution was added to 50 ml of sterilized cool molten double strength PDA medium, mixed thoroughly and poured into Petri plates. 6 mm discs of four days old culture of potential fungal antagonists were inoculated at the centre of Petri plates and then incubated at $28 \pm 2^{\circ}\text{C}$. Three replications were maintained for each fungicide. Medium without fungicide was kept as control. Per cent inhibition of the growth of the fungus over the control was calculated using the formula:

$$I = \frac{C - T}{C} \times 100$$

where,

I = Per cent reduction in growth of pathogen

C = Radial growth (mm) in control

T = Radial growth (mm) in treatment.

3.6.4.2 Spectrophotometric method

Five hundred microlitres of antagonistic bacterial cultures grown in Nutrient Broth (NB) for 16 hours at $28 \pm 2^{\circ}\text{C}$ and 150 rpm were added to 50

ml of NB in 250 ml flasks containing different fungicides. Inoculated flasks were incubated at $28 \pm 2^{\circ}\text{C}$ and 150 rpm on orbital shaker. Bacterial growth was determined in systronic spectrophotometer at 600 nm after 24 hours of incubation. Each treatment was replicated thrice.

3.7 INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHIKPEA

3.7.1 Pot culture studies

The potential biocontrol agent and compatible fungicides was evaluated under glass house conditions against the pathogen on popularly cultivated chickpea variety JG-11 by imposing the following treatments. In all the treatments the drought was imposed at 50 days after sowing (DAS). Three replications were maintained for each treatment.

SL. No.	Treatment No.	Treatment
1.	T ₁	Seed treatment with potential biocontrol agent @ 4g/Kg.
2.	T ₂	Seed treatment with fungicide @ 2g/Kg
3.	T ₃	Soil application with potential biocontrol agent
4.	T ₄	Soil drenching with fungicide @ 2.5g/l
5.	T ₅	T ₁ + T ₂
6.	T ₆	T ₃ + T ₄
7.	T ₇	Inoculated control
8.	T ₈	Uninoculated control

Design : Completely Randomized Design

Replications : 3

3.7.2 Mass multiplication of *Rhizoctonia bataticola*

The test pathogen *R. bataticola* was mass multiplied on sterilized sorghum seeds for pot culture studies. For this, 100g of sorghum seeds were washed thoroughly in tap water and soaked in water overnight in 250 ml conical flask with addition of 20 ml of 4% dextrose. After removing the water, the flasks were autoclaved for 20 min at 15 p.s.i and inoculated with 2-3 discs of 4 days old culture of test pathogen. After seven days the inoculum was mixed with sterilized soil in pots @100g/kg.

3.7.3 Preparation of talc based formulation of potential bacterial isolate

The talc based formulation of potential biocontrol agent was prepared by following the method as described by Vidhyasekharan and Muthamilan (1995).

A loopful of potential antagonistic bacteria was inoculated into Nutrient broth and incubated in a rotary shaker at 150 rpm min⁻¹ for 48 hours at room temperature (28 ± 2°C). One kg of talc powder (montmorillonite) was taken in a metal tray under aseptic conditions and pH was adjusted to 7.0 by adding CaCO₃ at the rate of 15 g kg⁻¹. 10 g of carboxymethyl cellulose (CMC) was added to 1 kg of talc powder, mixed well and the mixture was autoclaved for 30 min. at 121°C for 2 successive days. 400 ml of the bacterial suspension containing 1x10⁸cfu/ml was mixed

with carrier cellulose mixture under aseptic conditions. After drying to 35% moisture content overnight under aseptic conditions, the mixture was packed in polypropylene bags and sealed.

3.7.4 Seed treatment

Chickpea seeds were treated with talc based formulation of potential biocontrol agent @ 4g per kg of seed and the seeds were used for sowing. For treatment with fungicide, the chickpea seeds were treated with compatible and effective fungicide @ 2.5 g/kg of seeds and sown in the pathogen infested soil in the pots. For treatment with both potential biocontrol agent and fungicide, at first seeds were treated with biocontrol agent followed by compatible fungicide.

3.7.5 Soil application

The potential biocontrol agent was multiplied in Farm Yard Manure (FYM) and 100 g of FYM/pot was applied before sowing. For application of fungicide to the soil, soil was drenched with fungicide @ 2.5 g/l.

3.7.6 Observations

i. Percentage of disease incidence (PDI)

$$PDI = \frac{\text{Number of diseased plants}}{\text{Total number of seeds planted}} \times 100$$

ii. Shoot and root length

The average shoot and root length of the plants was recorded.

iii. Dry weight of shoots and roots

The samples were allowed to dry under room temperature and the dry weights of shoots and roots were recorded.

3.8 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

The genetic variability among the isolates of bacterial biocontrol agents was studied by Random Amplified Polymorphic DNA (RAPD), 16S rDNA and 16S rDNA Restriction fragment length polymorphism (16S rDNA-RFLP).

3.8.1 Bacterial cultures

The antagonistic bacterial isolates having different degree of antagonistic activity against *Rhizoctonia bataticola* were selected for molecular characterization. Bacterial isolates were grown on nutrient broth at $28 \pm 2^\circ\text{C}$ for overnight for DNA extraction.

3.8.2 Buffers used for electrophoresis

1. Composition of 10 x TBE buffer

Tris base	:	54.0 g
Boric acid	:	27.5 g
EDTA	:	4.65 g
Distilled water	:	500 ml
pH	:	8.0

Preparation

Each chemical was dissolved in separate beakers using distilled water and all were mixed finally. The pH was adjusted to 8.0 by using 0.1 HCl or NaOH and volume was made upto 500 ml and sterilized by autoclaving at 15 lbs for 15 minutes.

2. Composition of loading dye (10x)

Glycerol	:	5ml
10 x TBE	:	1ml
Bromo1phenol blue (saturated)	:	1ml
Xylene cyanol (10%)	:	1ml
Double distilled water	:	10ml

Preparation

Contents were mixed well and divided into 1ml aliquot, sterilized and stored at -20°C for further use.

3.8.3 DNA extraction

The DNA from potential bacterial antagonists was extracted by using bacterial genomic DNA isolation kit from M/s Medox, Chennai. The quantity and quality of DNA was verified on 1% agarose gel and by Nanodrop.

3.8.4 Preparation of gels

Gel plates (13 x 14 cm) were washed thoroughly with cleaning solution followed by distilled water and dried. The two open sides of the plates were sealed with cellophane tape. Gel solution was prepared by mixing 1.0g of agarose in 100 ml of 1 x TBE buffer (1.0% gel) in a conical flask and boiled in an oven until a clear solution was obtained and 4 µl of ethidium bromide (10mg µl⁻¹) was added. The solution was poured onto the sealed plate, inserted the suitable comb and allowed to polymerize.

3.8.5 Loading and running of gels

The inserted comb was gently removed from the gel after polymerization. The gel plate was placed in horizontal apparatus and fixed with 1 x TBE buffer. The samples were loaded in the wells with help of

micro pipettes. After loading, the electrophoretic unit was connected to power pack with a regulated electric power supply of 100V. At the end of run, the gel was carefully removed and analyzed.

3.8.6 Qualitative and quantitative verification of DNA from different isolates of potential biocontrol agents

DNA samples (5 µl) from each isolate mixed with 4 µl of 1 x loading dye were loaded on the wells of the 1% agarose gel along with 5 µl of DNA marker in order to verify the quality and quantity of DNA. Alternatively the quality and quantity of DNA was also verified by Nanodrop.

3.8.7 RAPD profiles through polymerized chain reaction (PCR)

Five different random primers belong to operon “A” and “D” series viz. OPA-11, 12, 14, 18, and OPD-2 (operon technologies Inc.,) were used to detect polymorphism among the isolates under the study. The experiment was repeated thrice and results were reproducible. The primer sequences used in RAPD technique are given below.

Sl. No.	Operon	Sequence
1	OPA-11	5 ¹ (CAA TCG CCGT)3 ¹
2	OPA-12	5 ¹ (TCG GCG ATAG)3 ¹
3	OPA-14	5 ¹ (TCT GTG CTGG)3 ¹
4	OPA-18	5 ¹ (AGG TGA CCGT)3 ¹
5	OPD-3	5 ¹ (GTC GCC GTCA)3 ¹

3.8.7.1 Standardization of RAPD technique

The RAPD technique has been standardized and the following conditions were used for the amplification of DNA from different isolates.

Master mix for RAPD has been prepared as given below

1.	Assay buffer (10x)	2.5µl
2.	MgCl ₂ (25Mm)	2.0 µl
3.	dNTPS (20 mM)	1.0 µl
4.	Primer (10 p mol)	1.0 µl
5.	Taq polymerase (3u/ µl)	0.4 µl
6.	DNA sample (100 ng)	2.0 µl
7.	Sterile double distilled water	16.1 µl
	Total volume	25 µl

Conditions used for RAPD amplification

Stage-I: Initial denaturation at 94°C for 4 min.

Stage-II: Denaturation at 94°C for 1 min.

Annellation at 37°C for 3 min and

Extension at 72°C for 2 min.

Number of cycles: 40

Stage-III: Final extension at 72°C for 10 min.

Amplified PCR products were subjected to 1.0 per cent agarose gel electrophoresis with 1.0 x TBE as running buffer. The banding pattern was visualized under UV trans-illuminator with ethidium bromide (10 mg ml⁻¹) staining. The DNA banding profiles were documented in the gel documentation system (Alpha Innotech) and compared with 1 kb DNA ladder.

3.8.7.2 Scoring and data analysis

Each amplified band was considered as RAPD marker and recorded for all samples. Data were entered using a matrix in which all observed bands or characters were listed. The RAPD pattern of each isolate was evaluated by assigning character state 1 to all the bands that could be reproducible and detected in the gel and '0' for the absence of band.

The data matrix thus generated was used to calculate Jaccard's similarity coefficient for each pair wise comparison. The coefficients were calculated insilica using the following formula.

$$\text{Similarity coefficient} = \frac{a}{n}$$

where,

a = Number of matching band for each pair of comparisons

n = Total number of bands observed in two samples.

The similarity coefficients were subjected to unweighted pair-group method on arithmetic average (UPGMA) cluster analysis to group the isolate based on their over all similarities statistical package for social sciences (SPSS) was used for the cluster analysis and subsequent dendrogram preparation.

3.8.8 Amplification of 16S rDNA from antagonistic bacterial isolates

The 16S rDNA sequence has been selected for identification and to detect polymorphism among potential biocontrol agents. The 16S rDNA from potential antagonistic isolates has been amplified by using 63F and 1387R primers. The primer sequences are given below.

63F – 5¹ CAG GCC TAA CAC ATG CAA GTC – 3¹

1387R – 5¹ GGG CGG (AT) GT GTA CAA GGC – 3¹

As a part of this, PCR technique has been standardized and the following components were used for the amplification of 16S rDNA.

1.	Assay buffer (10x)	2.5 µl
2.	MgCl ₂ (25mM)	2.0 µl
3.	dNTPs (20mM)	1.0 µl
4.	Forward primer (63F) (10p mol)	2.0 µl
5.	Reverse primer (1387R) (10p mol)	2.0 µl
6.	Taq polymerase (3U/ µl)	0.4 µl
7.	DNA sample (100 ng)	2.0 µl
8.	Sterile double distilled water	13.1 µl
	Total volume	25.0 µl

The 16S rDNA amplification was carried under following conditions.

Stage-I: Initial denaturation at 94°C for 4 minutes.

Stage-II: Denaturation of 94°C for 1.0 min

Annealing at 55.4°C for 1 min and

Extension at 72°C for 1.5 min.

Number of cycles: 35

Stage-III: Final extension at 72°C for 5 min.

The amplified products were visualized in 1.0% agarose gel.

3.8.9 16S rDNA-RFLP of antagonistic bacterial isolates

Amplified PCR products of 16S rDNA region for the isolates under the study were digested with restriction enzyme i.e. *Taq-I*. Digestion was carried out with 20 µl reaction mixture which contains 5 µl of 16S rDNA product (150 ng), 1 µl of enzyme (10U/ µl), 2 µl of 10 x enzyme buffer and 12 µl of sterile PCR water. The digestion was carried out overnight at 65°C for *Taq-I* in a water bath. Restricted fragments were analyzed on 3.0 per cent agarose gel and observed under UV transilluminator with ethidium bromide staining. The banding patterns were documented through gel documentation system (Alfa Innotech). The size of the restricted fragments was estimated by comparison with known DNA marker.

3.8.10 Cloning, Sequencing and identification of potential bacterial antagonist

The 16S rDNA from potential fungicide tolerant biocontrol agent was amplified as per the procedure given in section 3.8.8. The amplified 16S rDNA fragment was cloned using TA-cloning Kit from Fermentas, Bangalore was used. The following steps were carried as per the procedure given in the kit:

Ligation:

Component	Volume
Vector pTZ57/RT, (0.18 pmol ends)	3 µl
5 X ligation buffer	6 µl
PCR product (0.54 pmol ends)	10 µl
Water, nuclease free	10 µl
T4 DNA ligase	1 µl
Total	30µl

Incubated the ligation mixture at room temperature (22°C) for 1 hour followed by 4°C for overnight as per the procedure given by the company.

Transformation of *E.coli* (DH5α) cells

10 µl of ligation mixture was added to 100 µl of *E.coli* DH5α competent cells and incubated on ice for 30min. The cells were subjected to heat shock at 42°C for 90 sec. To this, 800 µl of LB broth was added and

incubated for 1 hr at 37°C at 100 rpm. Later the *E.coli* cells were plated on LB agar medium containing ampicillin (100g/ml), X-gal (20mg/ml) and IPTG (4µl/plate). The plates are incubated overnight at 37°C for the development of colonies. The plates were observed for blue and white colonies for non-transformed and transformed cells respectively.

Plasmid isolation

Transformed white colonies were further subjected to plasmid DNA isolation as per the kit supplied by M/s Chromous biotech, Bangalore.

Restriction digestion

Representative 4-6 white colonies were analyzed for the presence of the DNA insert using the restriction enzymes *EcoR I* and *Hind III*.

Colony PCR

The above colonies were also subjected to colony PCR for analyzing the presence of insert.

The PCR conditions and PCR mixture are the same as used in 16S rDNA amplification but instead of using the DNA, an individual colony was used for PCR amplification.

The banding pattern of both vector and insert and its size was observed to confirm the desired recombinant plasmid.

Sequencing

The recombinant plasmid was sent for sequencing to M/S MWG Biotech, Bangalore.

3.9 STATISTICAL ANALYSIS

Wherever necessary, the data was statistically analyzed (Gomez and Gomez, 1984). Completely Randomized Design (CRD) was used for total growth, pot culture experiment, dual culture technique, poisoned food technique and two way CRD was used for spectrophotometric method and the treatments were compared at $P \leq 0.05$.

CHAPTER-IV

RESULTS

The results of the experiments conducted in the present investigation on “Integrated Management of Dry root rot of Chickpea and Molecular Characterization of Potential Biocontrol Agents” are presented below.

4.1 SURVEY

A preliminary field survey was conducted to know the incidence of dry root rot caused by *Rhizoctonia bataticola* in major chickpea growing mandals of Kadapa and Kurnool districts of Rayalaseema region of Andhra Pradesh (Fig.1a and 1b). Three mandals viz., Kovelakuntla, Sanjamala, Dornipadu of Kurnool district and other three mandals viz., Jammalamadugu, Peddamandium, Rajupalem of Kadapa district were surveyed. In each mandal three villages were taken into account for survey (Table 1 and Fig. 2). In each village 2 to 3 fields were selected, in each field one m² area were chosen as representative of the whole field in 10 spots randomly and counted the number of diseased and healthy plants in that area. Soil samples were collected from the rhizosphere of healthy plants for the isolation of antagonistic mycoflora in the diseased fields. Simultaneously, diseased and healthy plants were also collected for the isolation of the pathogen and antagonistic root endophytes respectively.

The disease incidence in Kovelakuntla, Sanjamala and Dornipadu mandals of Kurnool district were 9.55, 13.50 and 7.71 per cent respectively, while in Jammalamadugu, Peddamandium, Rajupalem mandals of Kadapa district were 7.55, 7.68 and 6.22 per cent respectively.

The highest average per cent disease incidence was recorded in Sanjamala mandal (13.5%), while the least average per cent disease incidence (6.22%) was recorded in Rajupalem mandal. Among villages in different mandals, the lowest per cent disease incidence (5.42%) was observed in Arkatavemula village of Rajupalem mandal. While the highest per cent disease incidence (15.31%) was recorded in Reddipalli village of Sanjamala mandal.

4.2 THE PATHOGEN

4.2.1 Isolation of pathogen

The dry root rot affected chickpea plants were identified in the field based on key symptoms like withering and drying of plants (Plate 1). When such plants were pulled out showed blackening of tap root and devoid of lateral and finer roots (Plate 2a) and shredding of bark and coming out in the form of flakes (Plate 2b). Such infected plants were collected from Reddipalli village of Sanjamal mandal of Kurnool district for pathogen isolation where the disease incidence was high.

The pathogen was isolated from root rot affected plants using tissue segment method on PDA. The fungus was further purified by single hyphal tip method on PDA.

4.2.2 Identification of pathogen

The fungus produced radial hyaline colonies, which later become carbonaceous brown to black (Plate 3). Mycelium was septate and dark brown in colour. Typical right angled branching of mycelium (Plate 4) was observed. Sclerotia (Plate 5) were black, varied from spherical to irregular in shape and measured to 80 to 85µm in diameter. Pycnidial production was not observed in culture plates.

The colony characters and morphological characters of mycelium and sclerotia were in agreement with earlier reports. Thus, the fungus under present investigation was identified as *Rhizoctonia bataticola*.

4.2.3 Pathogenicity test

Soil inoculation method was used to prove the pathogenicity of *R. bataticola*. About 10 seeds of chickpea were sown in each pot with a diameter of 22.5 cm (sterilized) inoculated with pathogen @100g Kg⁻¹ soil. Pre-emergence rot of seedlings was observed and the survived plants showed stunted growth followed by wilting and drying of leaves and stems (Plate 6). When the infected plants were pulled out the tap root was blackened with devoid of lateral and finer roots.

On re-isolation, the characters of the pathogen showed similarity with the original pathogen isolated from the field (section 4.2.2), thus Koch's postulates was fulfilled.

4.3 IN VITRO EVALUATION OF EFFICACY OF FUNGICIDES AGAINST *R. bataticola*

Efficacy of four commonly used fungicides viz., captan, thiram, copper oxychloride were evaluated against *R. bataticola* at different concentrations viz., 1000, 1500, 2000 and 2500 ppm, whereas carbendazim was evaluated at concentrations viz., 500, 750, 1000 and 1500 ppm by using poison food technique. The data revealed that all the fungicides at all concentrations reduced mycelial growth (Table 2) of *R. bataticola* when compared to control.

It is evident from the data (Table 2 and Fig. 3), among all the fungicides that were tested captan and thiram showed 100 per cent inhibition (Plate 7 and 8) of mycelial growth at the concentrations of 2000 and 2500 ppm. However, the carbendazim showed 100 per cent inhibition even at 1000 and 1500 ppm (Plate 9). Whereas the fungicide copper oxychloride failed to give 100 per cent inhibition (Plate 10) even at 2500 ppm (88.37%).

Among the fungicides evaluated, overall per cent inhibition of *R. bataticola* mycelial growth was maximum with carbendazim followed by thiram and captan. Copper oxychloride was found to be least effective on *Rhizoctonia bataticola* which inhibited mycelial growth by 63.95 per cent.

4.4 POTENTIAL FUNGICIDAL TOLERANT BIOCONTROL AGENTS

4.3.1 Isolation and identification of native antagonistic mycoflora and bacteria from rhizosphere soil and roots of chickpea against *R. bataticola*

Antagonistic microflora from rhizosphere soil and roots of healthy chickpea plants were isolated as per procedure given in section 3.6.1 and 3.6.2. The mycoflora were isolated on Rose Bengal Agar (RBA) medium (Plate 11) and bacteria on Nutrient Agar (NA) medium (Plate 12). The fungal antagonists were purified by single spore method and were maintained on PDA medium. While bacteria were purified by streak plate method and maintained on Nutrient agar medium. A total of 8 fungi and 5 bacteria were obtained from rhizosphere soil; whereas 10 bacteria were obtained from root as root endophytes (Table 3).

Based on colony and morphological characters, mycoflora and bacteria were identified. Eight fungi viz., *Fusarium* sp. (Plate 9), *Aspergillus flavus*, *A. niger* (Plate 14) and *Trichoderma* isolates-1, 2, 3, 4 and 5 were isolated (Plate 15). Among the 15 bacterial isolates, 5 bacteria isolated from rhizosphere soil were designated as RB-1 to RB-5 followed by root endophytic bacteria as REB-1 to REB-10 (Plate 16).

4.4.2. *In vitro* evaluation of the antagonistic activity of microflora against *Rhizoctonia bataticola* in dual culture.

The antagonistic effect of native microflora was assessed based on their ability to inhibit the pathogen growth in dual culture technique. The effect of these native antagonists on the mycelial growth of the pathogen was calculated and expressed as per cent inhibition.

4.4.2.1 *In vitro* evaluation of efficacy of antagonistic mycoflora and bacteria against *R. bataticola* in dual culture technique.

Efficacy of antagonistic mycoflora and bacteria against *Rhizoctonia bataticola* was evaluated by using dual culture technique. All the native antagonists showed significant reduction in mycelial growth of *R. bataticola* when compared to control. The data pertaining to per cent inhibition of mycelial growth of *R. bataticola* due to antagonistic mycoflora and bacteria are presented in Table 4 and Table 5 respectively.

Among the eight mycoflora tested (Plate 17), *Trichoderma* isolates reduced mycelial growth of *Rhizoctonia bataticola* more significantly than others. The data revealed that *Trichoderma* isolate-3(T₃) showed maximum inhibition of growth of *Rhizoctonia bataticola* (57.83%) followed by *Trichoderma* isolate-4 (54.21%) and *Trichoderma* isolate-2 (51.80%) (Plate 18). However both were found to be on a par with each other in inhibiting the pathogen.

Overall per cent inhibition of mycelial growth of *R. bataticola* was maximum incase of *Trichoderma* isolate-3(T₃) (Fig. 2) and minimum incase of *Fusarium* sp.(37.50%)(Plate 17).

A total of 15 bacterial antagonists of chickpea were tested for their efficacy under *in vitro* against *R. bataticola*. The results are presented in Table 5 and Plate 19.

Among the 15 bacterial antagonists tested the isolate REB-8 showed the maximum inhibition (76.47%) of growth of *R. bataticola* followed by RB-1 (74.11%) and REB-9 (71.76%). However, both were found to be on a par with each other in inhibiting the pathogen. The isolate RB-4 was recorded least (29.42%) per cent inhibition (Fig.4).

4.4.3 *In vitro* evaluation of the compatibility of potential antagonist with different fungicides.

The potential root endophytic bacteria REB-8 was selected for fungicidal compatibility studies since it has shown maximum inhibition of *Rhizoctonia bataticola* growth in dual culture studies when compared to all other antagonists. Spectrophotometric method was used to evaluate the compatibility of REB-8 with different fungicides.

4.4.3.1 Compatibility of antagonistic bacterial root endophyte (REB-8) with different fungicides.

Higher OD values at 600 nm indicate high compatibility of the antagonist with that specific fungicide.

It is evident from the data (Table 6 and Fig 6) that the isolate REB-8 was more compatible with carbendazim (0.985) followed by thiram (0.932) and copper oxychloride (0.840). The less compatibility was recorded in case of captan (0.820) compared to control (1.207).

4.5 INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHICKPEA

The Chickpea variety JG-11 was used for pot culture studies. The different treatments were imposed as indicated in the section 3.7.1 and the results are presented in Table 7.

4.5.3 Mass multiplication of *Rhizoctonia bataticola*

The pathogen was mass multiplied on sorghum seeds (Plate 20) and added to soil @ 100g Kg⁻¹ at the time of sowing.

4.5.4 Preparation of Talc based formulation of potential fungicidal tolerant antagonist (REB-8)

Talc based formulation of potential biocontrol agent (REB-8) was prepared by following the procedure given in section 3.7.3 (Plate 21 and 22). The population estimation was done at the time of application and it was 7.6×10^7 cfu/g (Plate 23).

4.5.5 Mass multiplication of potential antagonistic bacteria

The potential biocontrol agent (REB-8) was multiplied in Farm Yard Manure (FYM) and 100 g of FYM/ pot was applied before sowing.

4.5.6 Observation

The data on per cent incidence of dry root rot and plant growth parameters viz., plant height, root length, shoot dry weight and root dry weight of chickpea in each of the treatment were recorded and presented in table 7 and plate 24.

4.5.3.1 Per cent disease incidence

From the data (Table 7 and Fig 7) it is evident that all the treatments were significantly superior over control in reducing the per cent disease incidence. Maximum reduction was observed in treatment T₆ (soil application with potential BCA + soil drenching with fungicide) in which PDI of 6.67 per cent was recorded when compared to treatment T₇ inoculated control (83.10%).

Treatment T₁ recorded 42.41 per cent and was on par with T₂ and T₃ treatments.

4.5.3.2 .Effect of different treatments on plant growth parameters

a) Plant height

Maximum plant height (25.60cm) was recorded in treatment T₆ (soil application with potential BCA + drenching with fungicide). It is evident from Fig. 8 that least plant height was recorded in inoculated control (T₇).

It is evident from Fig 8 that treatment T₆ stimulated the plant growth and development (25.6 cm) when compared to inoculated control (16.00 cm). The treatment T₆ and T₅ were on par with each other.

b) Root length

Maximum root length was recorded in treatment T₆ (9.15cm) followed by treatment T₅ (8.86 cm). It is evident from Fig. 9 that least root length (4.50 cm) was recorded in inoculated control (T₇).

c) Dry weight of shoot and root

The maximum shoot weight was recorded in treatment T₆ (0.28g) followed by treatment T₅ (0.230 g). It is evident from Fig. 10 that least shoot weight (0.09g) was recorded in inoculated control.

Maximum root weight (0.09g) was recorded in treatment T₆ and least (0.05 g) was recorded in inoculated control.

From the above results it is evident that dry weights of both shoot and root were maximum in treatment T₆ (Fig 10). Thus overall, the efficacy of treatment T₆ (soil application with potential BCA + soil drenching with fungicide) was found to be superior which recorded least PDI, maximum shoot and root dry weight when compared to other treatments.

4.6 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

The genetic variability among the isolates of biocontrol agents was studied by using molecular techniques like RAPD and rDNA analysis.

4.6.1 Qualitative and Quantitative verification of DNA from different isolates of biocontrol agents

Six isolates having different degrees of antagonistic activity i.e. RB-1, RB-2, RB-3, RB-4, REB-8, and REB-9 were selected for molecular characterization.

The genomic DNA from the isolates under the investigation was extracted as per the procedure given in section 3.8.3. The quantity and quality of DNA was analyzed by running 2 µl of each sample in 1% agarose gel. Alternatively the quantity and quality of DNA was analyzed by nanodrop. These results indicated that the ratio between 260/280 was more than 1.7. The agarose gel analysis and nanodrop results indicated the good quality of DNA. The DNA concentration was adjusted to 100 ng/µl for the RAPD and 16S rDNA analysis.

4.6.2 Characterization of potential biocontrol agents by RAPD

In the present investigation, the genetic variability among the six isolates of bacterial biocontrol agents in the study was analyzed by RAPD.

Five random primers viz., OPA-11, OPA-12, OPA-14, OPA-18, and OPD-3 generated reproducible polymorphism among the isolates of biocontrol agents (Plate 25a and 25b). Amplified products with all the primers have shown polymorphic and distinguishable banding pattern indicating genetic diversity among all the isolates. A total of 48 reproducible and scorable polymorphic bands ranging approximately as low as 500 bp to as high as 2500 bp were generated with 5 primers among the six isolates.

The primer OPA-11 yielded a specific band approximately 1100 bp in case of RB-2 and RB-3 and absent in all other isolates. The 1300 and 2500 bp bands were specifically amplified in case of isolate RB-3 and RB-4.

The primer OPA-12 amplified unique bands of approximately 1100 bp and 1700 bp in case of RB-4 and REB-9 and 500 bp in case of RB-4.

The primer OPA-14 yielded a specific band approximately 900 and 1000 bp in case RB-2.

The primer OPA-18 amplified unique band of approximately 1700 bp in case of isolate RB-3 and RB-4. Whereas specific band of approximately 800 bp has been amplified in case of isolate RB-2 and REB-8.

The primer OPD-3 amplified unique band of approximately 800 bp and 1500 bp in case of isolate RB-4 and REB-9. Approximately 1100 bp band is very specific to isolate REB-8 which is highly potential antagonistic organism.

Primer survey was carried out using 4 primers from OPA series and 1 primer from OPD series of Operon technology, MWG-Biotech AG, Bangalore. All the 5 primers used for amplification of DNA for 6 bacterial isolates, gave reproducible and scorable bands with high percentage of polymorphism.

PCR amplification with 5 primers was done twice before scoring for presence and absence of bands. Number of amplification products obtained was specific to each primer and it was ranged from 2-10. All the primers used in the present analysis showed 10% polymorphism as all the bands obtained were polymorphic with size ranging from 500 bp to 2.5 kb.

Relationship among the isolates was evaluated by cluster analysis of data based on similarity matrix. The dendrogram (Fig. 11) was generated using UPGMA package based on Ward's squared Equalidean distance method. Based on the results obtained all the 6 isolates were grouped into two main clusters.

Cluster I contains three isolates viz., RB-1, RB-2 and REB-8, of which the first two and last one form two separate clusters i.e., cluster Ia (RB-1 and RB-2) and cluster Ib (REB-8). Cluster II contains 3 isolates viz., RB-3, RB-4 and REB-9.

Jaccard's Similarity co-efficient among 6 isolates were calculated to establish the genetic relationships. The similarity index values ranged from 0.00 to 100 per cent indicating the presence of a high range of variability at nucleic acid level among the 6 antagonistic bacterial isolates under the study.

4.6.3 Characterization of bacterial isolates by 16S rDNA

The structure of rDNA cluster and the expected amplified products with 63F and 1387R primers are shown in Fig. 12. .

16S rDNA specific target primers viz., 63F and 1387R were used for PCR amplification of 16S region of rDNA cluster of all isolates. Both primers produced amplified product size of approximately 1300 bp in all the isolates under the study as expected (Plate 26).

4.6.4. Characterization of Isolates by 16S rDNA – RFLP.

The 16S region of rDNA amplified with specific primers (63F and 1387R) yielded the single band of approximately 1300 bp. The band was further subjected to restriction analysis with endonuclease in order to observe the polymorphism among the antagonistic bacterial isolates. The amplicon of 1300 bp was digested with restriction enzyme *Taq* I and the restriction fragment were separated in 3.0 per cent agarose gel (Plate 27). The results further confirmed that there was no polymorphism in restriction banding pattern of 1300 bp of 16S rDNA among the isolates under the study.

4.6.5 Identification of potential bacterial antagonists based on analysis of 16S rDNA.

The 16S rDNA sequence has been selected for the identification of the potential biocontrol agents (Fig. 12).

4.6.5.1 Cloning and sequencing of 16S rDNA for the identification of REB-8 isolate

The potential bacterial antagonist, REB-8 which performed better in dual culture and pot culture studies was selected for further characterization. The 16S rDNA region of REB-8 was amplified with the primer 63F and 1387R. The 1300 bp amplified product of 16S rDNA region was cloned into the vector pTZ57/RT using TA cloning kit supplied by M/s Fermentas, Bangalore and sent for sequencing to MWG technologies, Bangalore. The recombinant clones were analysed for the presence of 1300 bp insert by restriction analysis and colony PCR (Plate 28). This positive recombinant clone was sent for sequencing to M/s MWG Technologies, Bangalore and awaiting for sequence data for the identification of REB-8 based on 16S rDNA sequence.

Fig.2 : Disease incidence of dry root rot of chickpea in different mandals of Kadapa and Kurnool districts of Andhra Pradesh

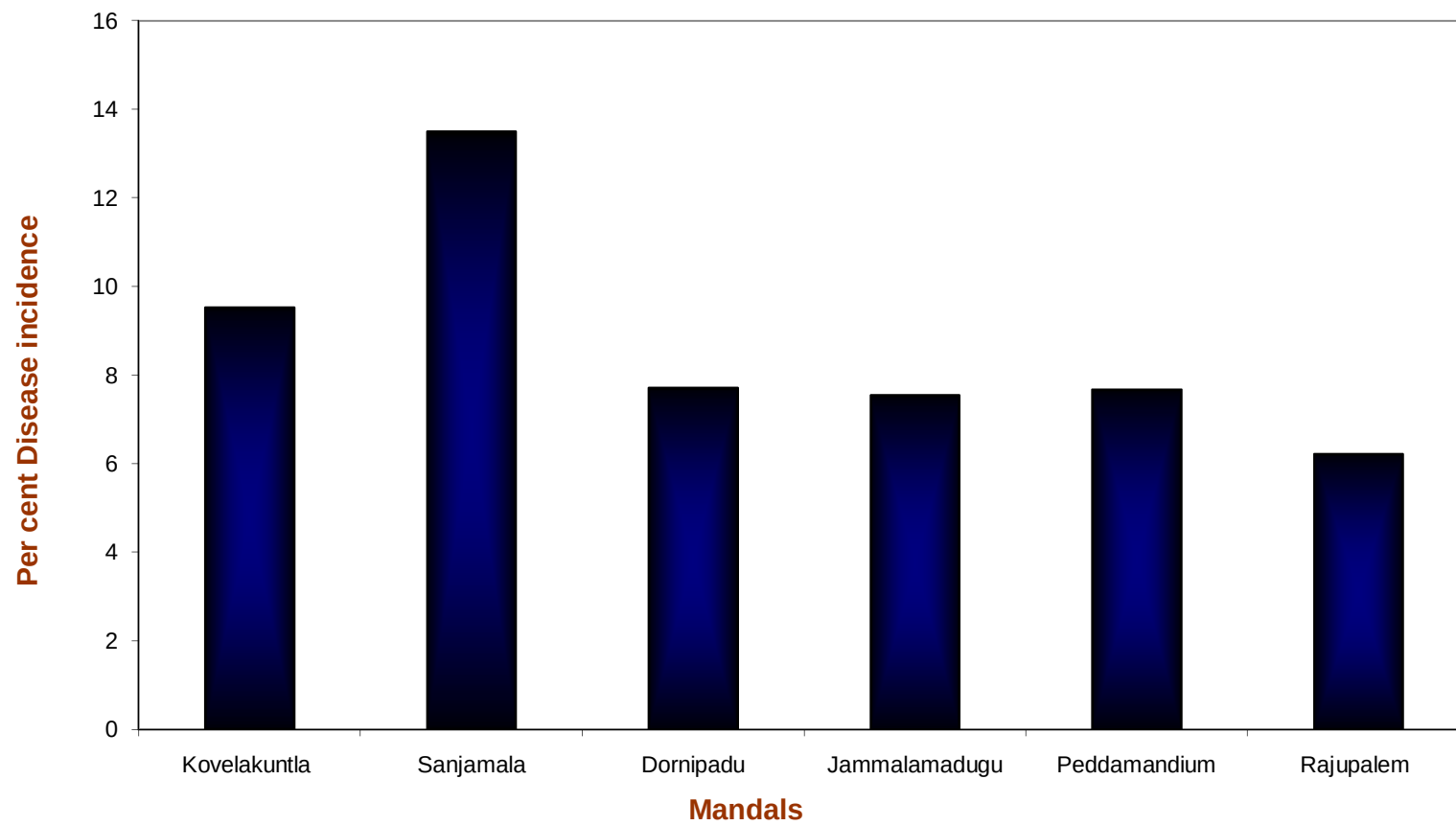


Fig. 3: *In vitro* evaluation of efficacy of different fungicides on mycelial growth of *R. bataticola* in poisoned food technique

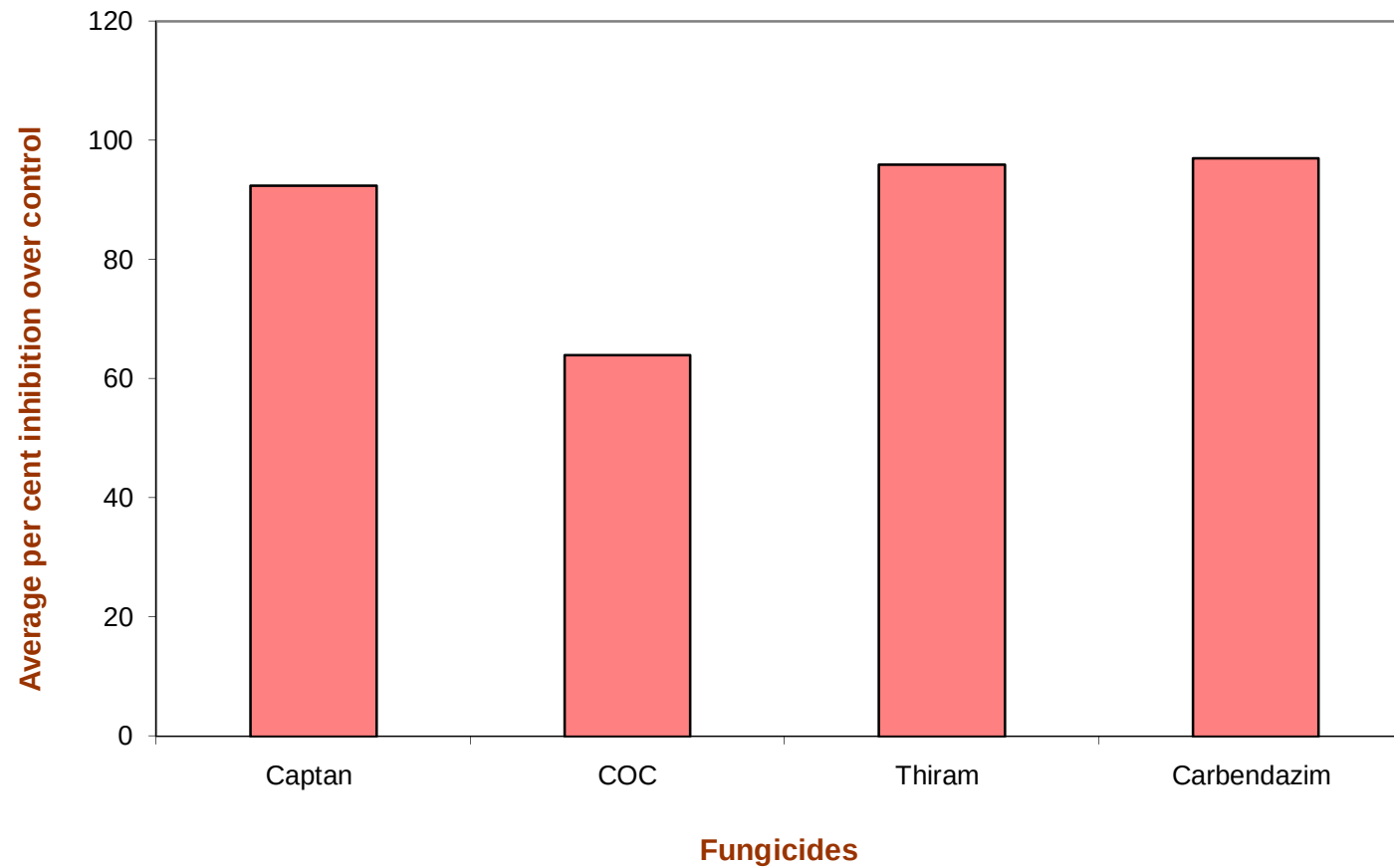


Fig. 4: *In vitro* evaluation of efficacy of antagonistic mycoflora against *R. bataticola* by dual culture technique

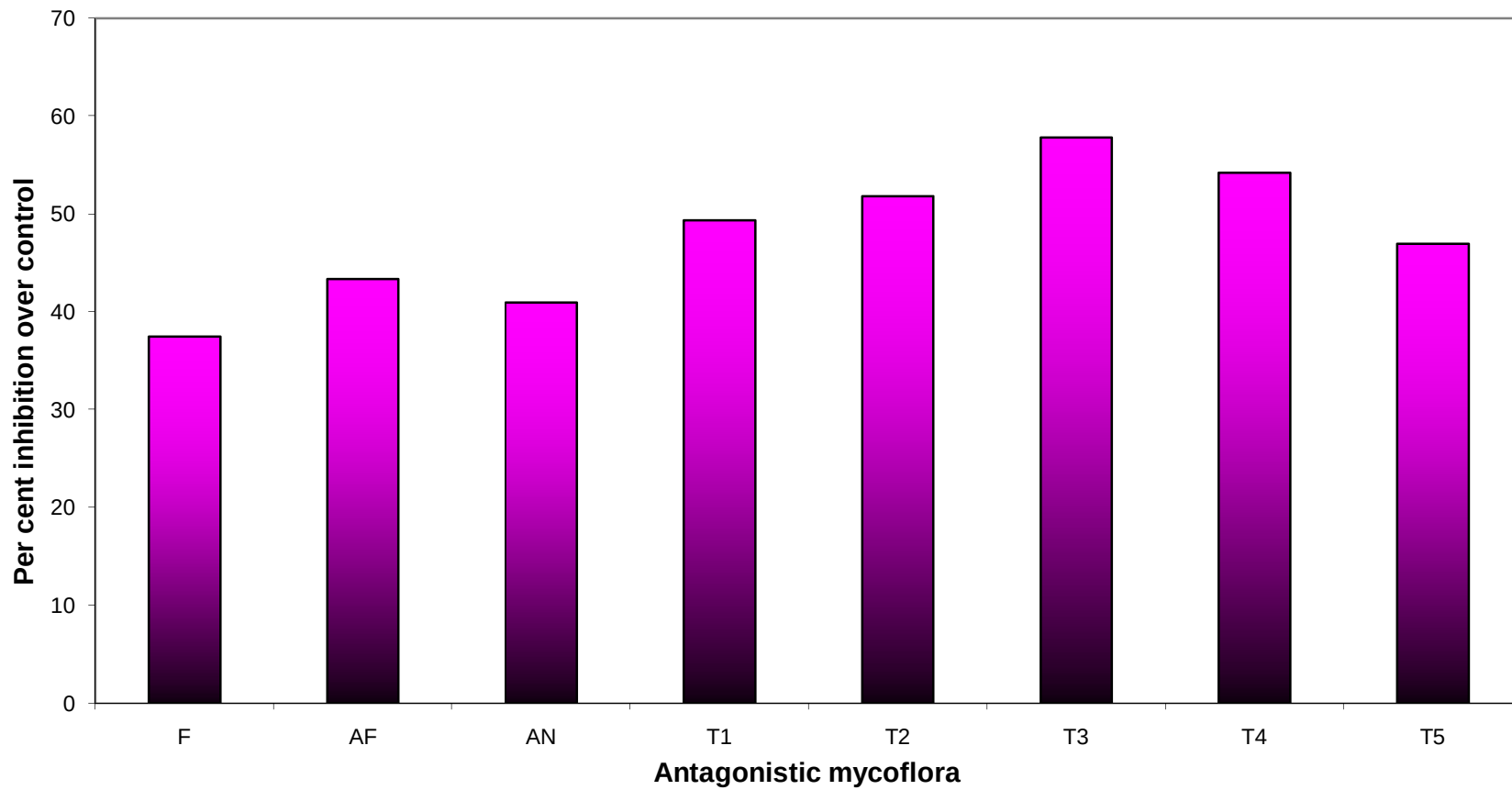


Fig. 5: *In vitro* evaluation of the antagonistic activity of bacteria against *R. bataticola* in dual culture technique

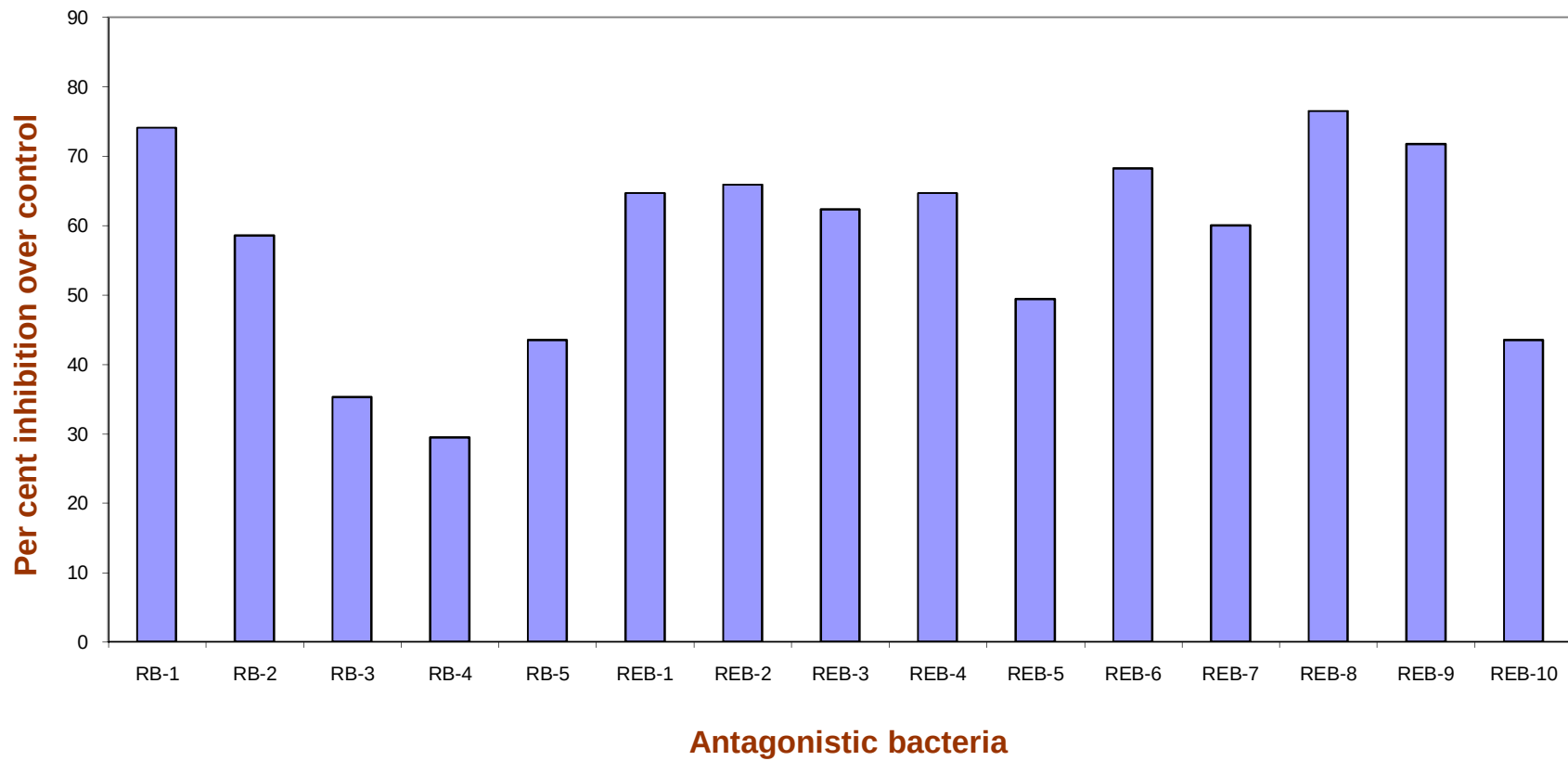


Fig.6: *In vitro* evaluation of the compatibility of the potential antagonistic bacterial isolate REB-8 with different fungicides

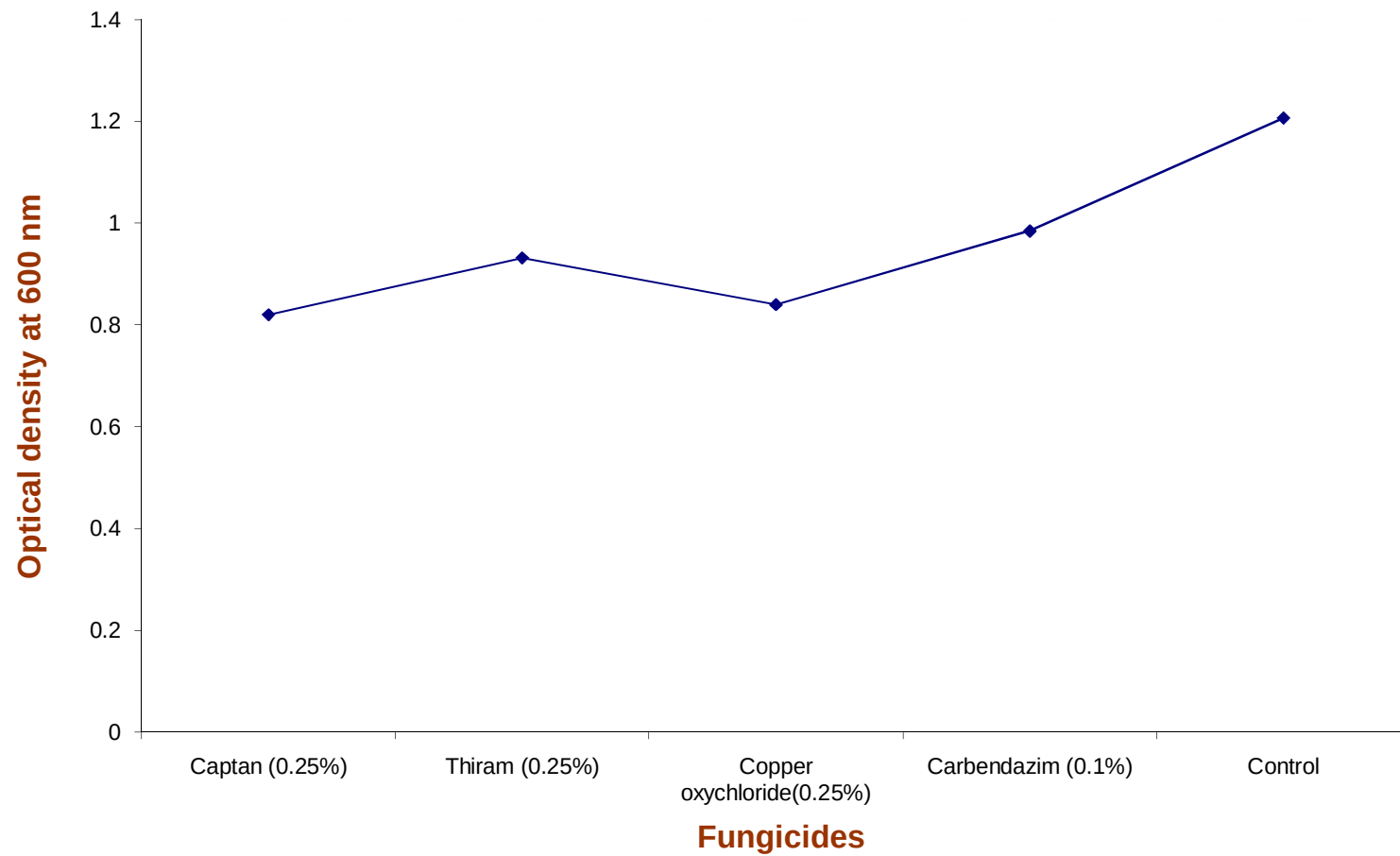


Fig. 7: Efficacy of antagonist and fungicide on per cent incidence of dry root rot chickpea in pot culture

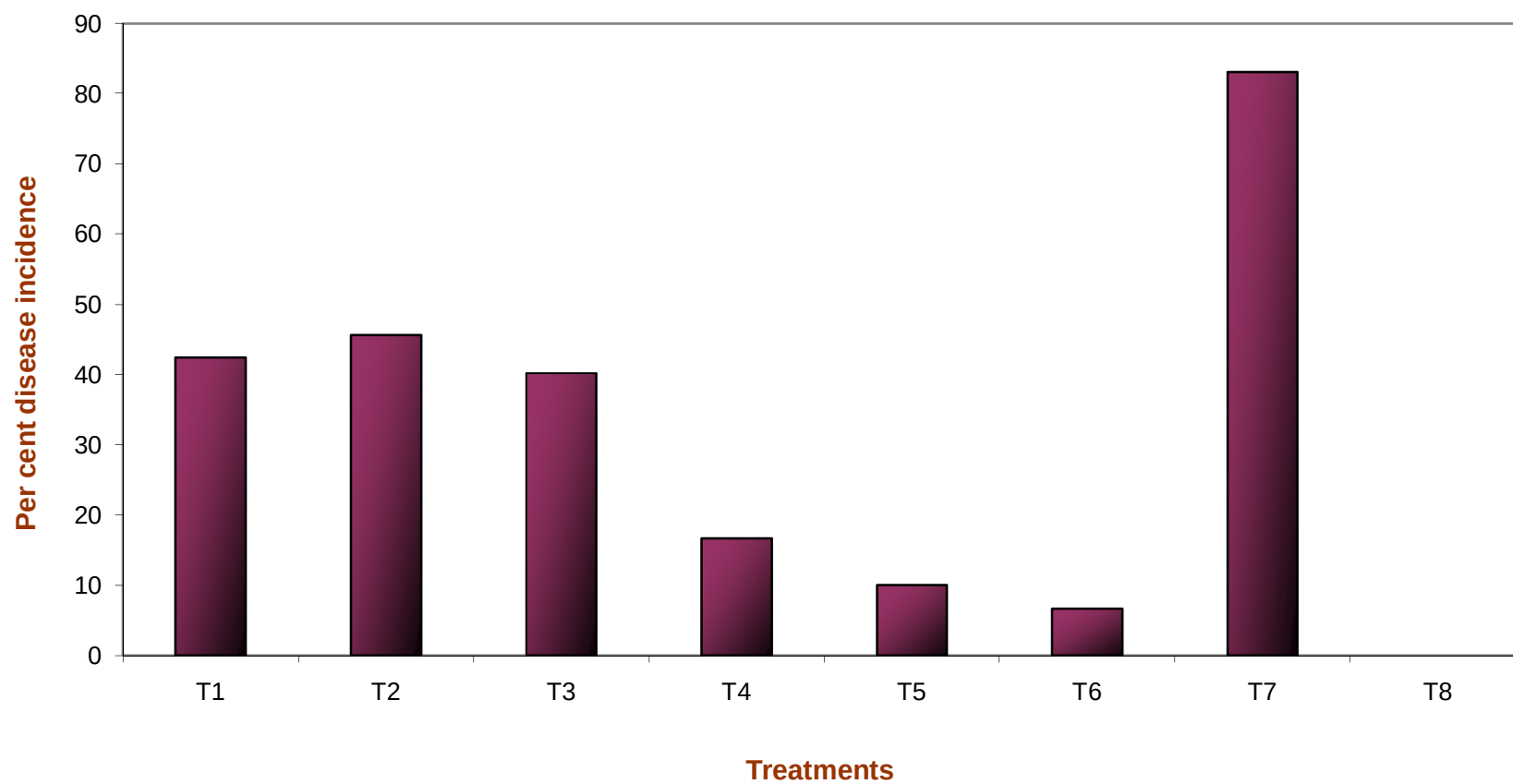


Fig. 8: Effect of potential biocontrol agent (REB-8) and fungicide on plant height of chickpea in pot culture

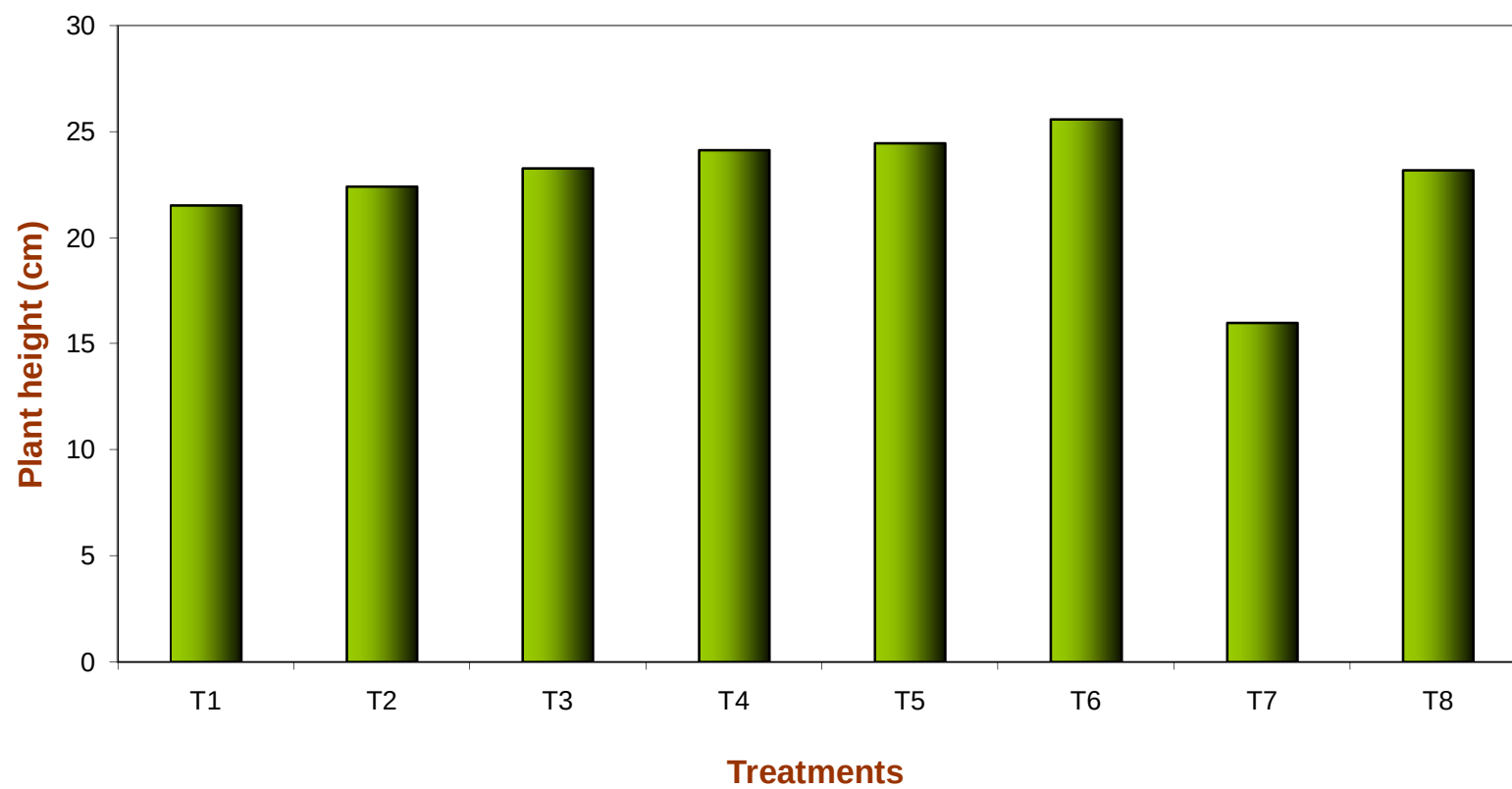


Fig. 9: Effect of potential biocontrol agent (REB-8) and fungicide on root length of chickpea in pot culture

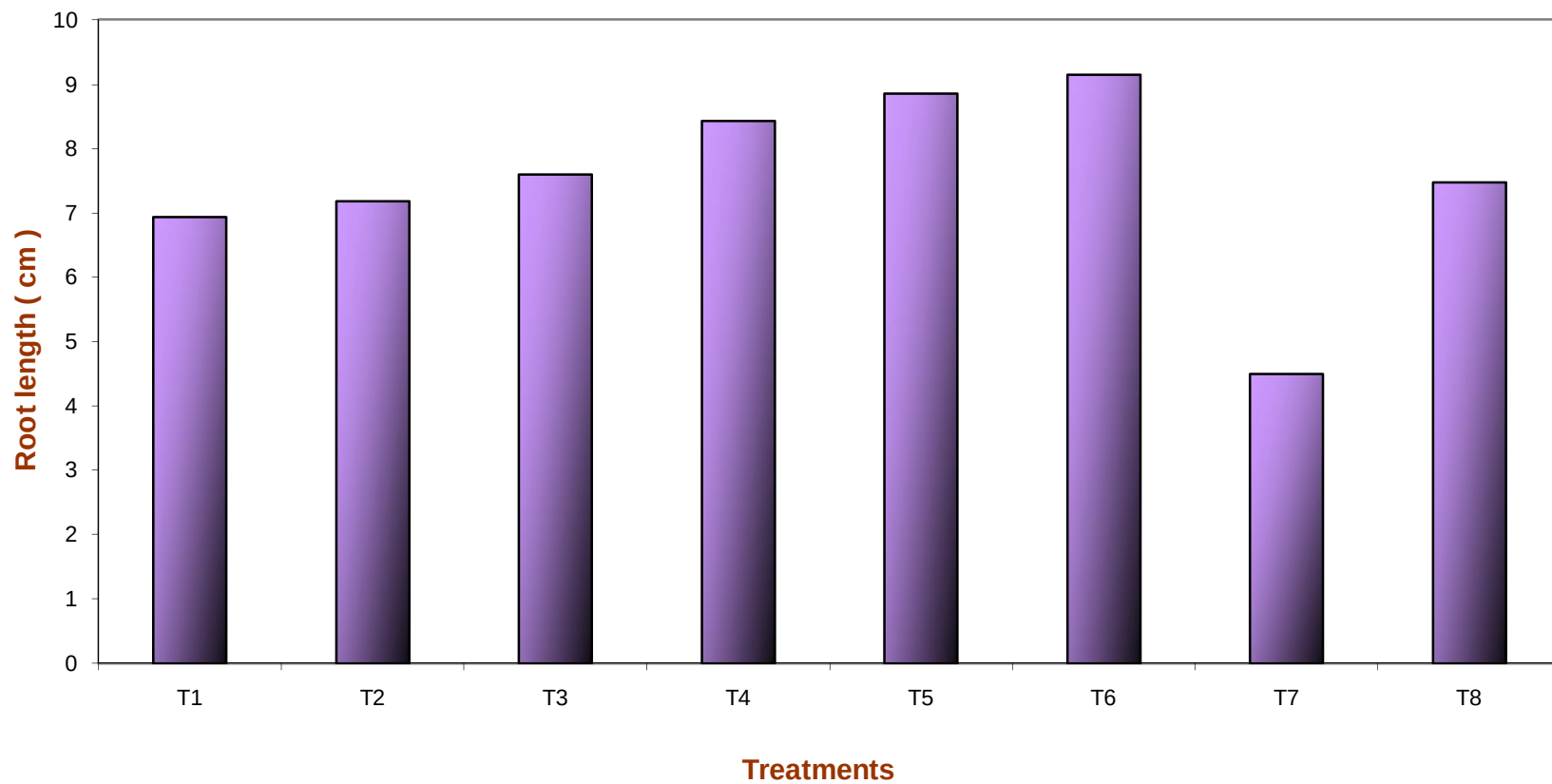


Fig. 10: Effect of potential biocontrol agent (REB-8) and fungicides on dry weight (g) of shoot and root of chickpea in pot culture

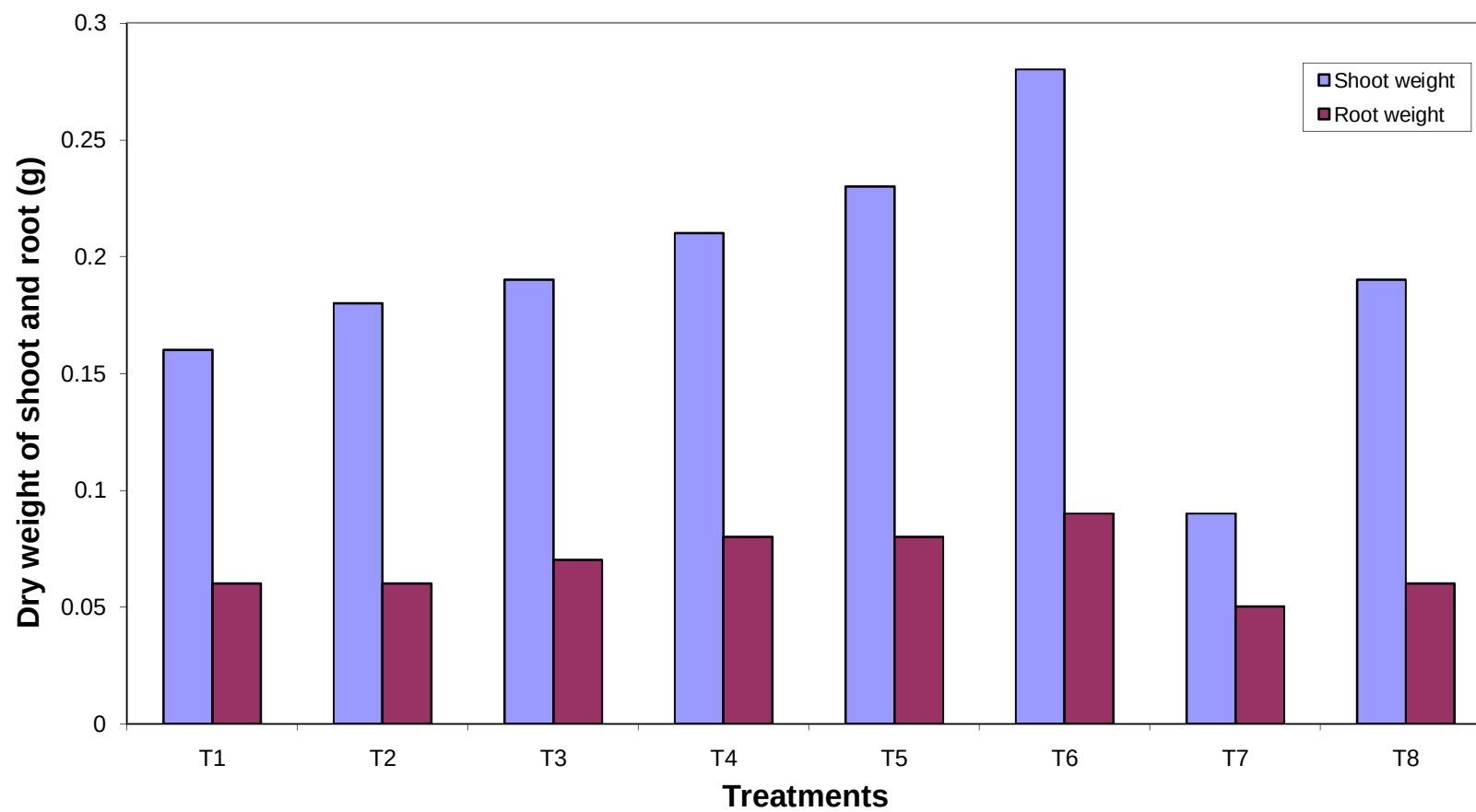


Fig 1a: Map showing the areas selected for survey on the incidence of dry root rot of chickpea in Kurnool district of Andhra Pradesh

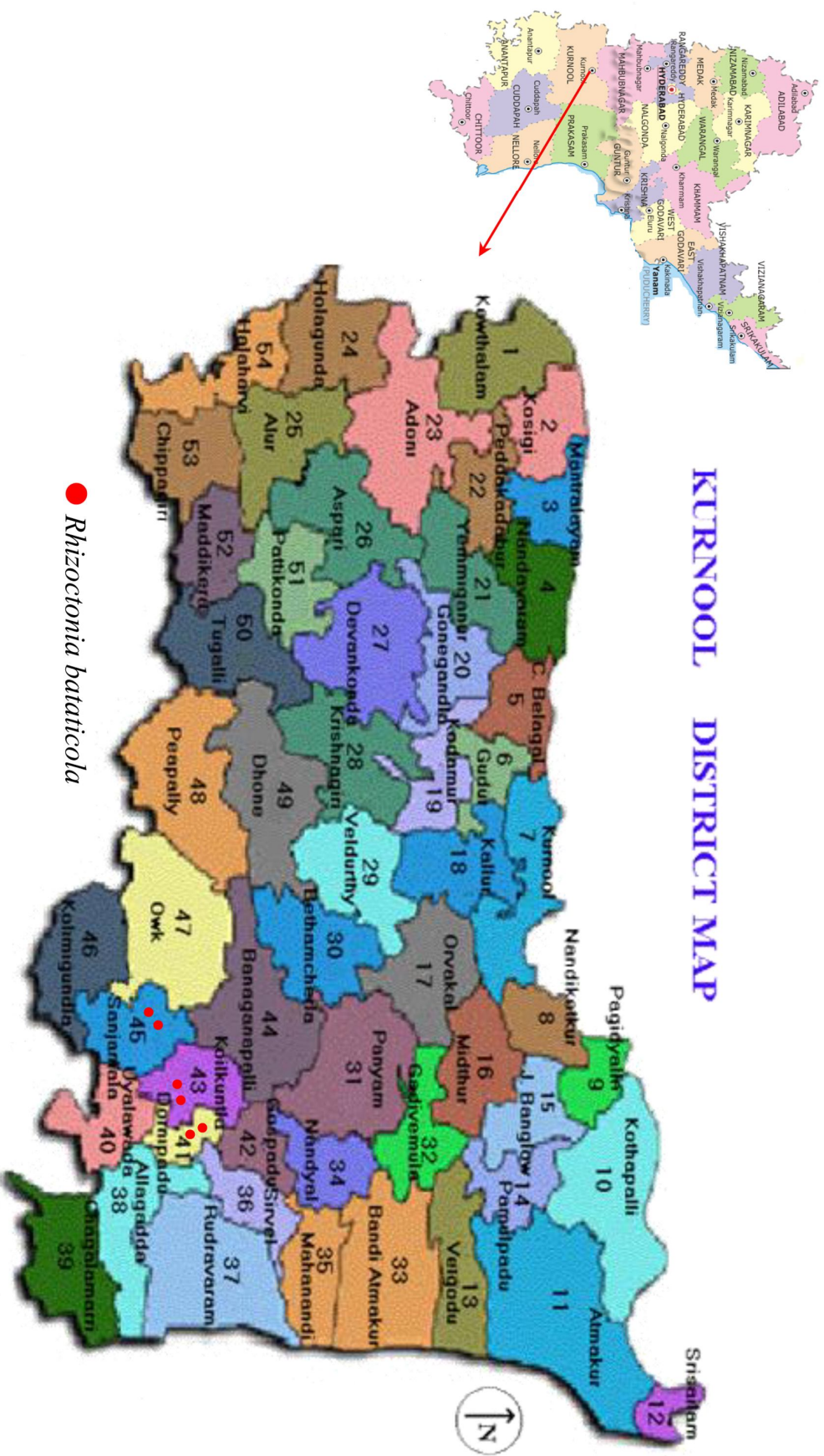
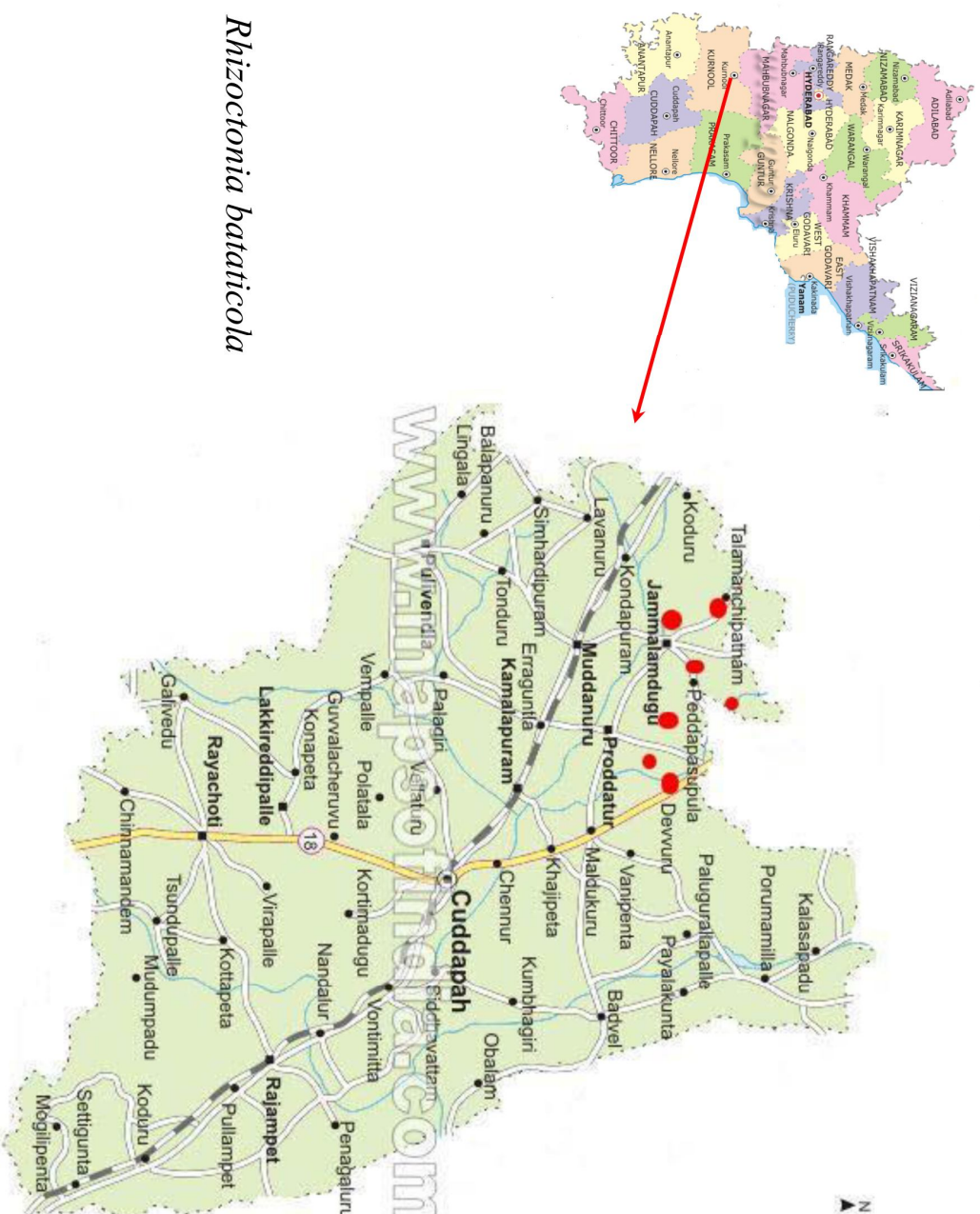


Fig 1b: Map showing the areas selected for survey on the incidence of dry root rot of chickpea in Kadapa district of Andhra Pradesh



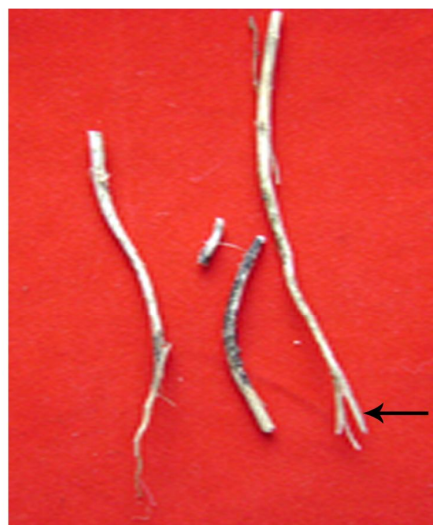
● *Rhizoctonia bataticola*



Plate 1: Field view showing symptoms of dry root rot of chickpea



(a) Darkening of tap root



(b) Shredding of bark

Plate 2: Infected chickpea plants showing typical symptoms of dry root rot

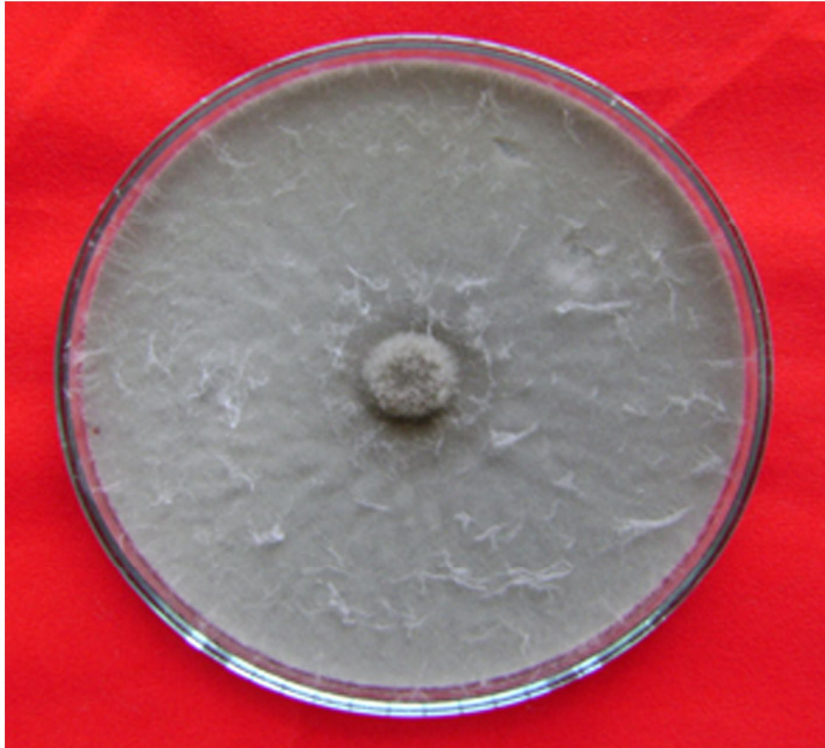


Plate 3: Pure culture of *Rhizoctonia bataticola* (Mycelial stage)

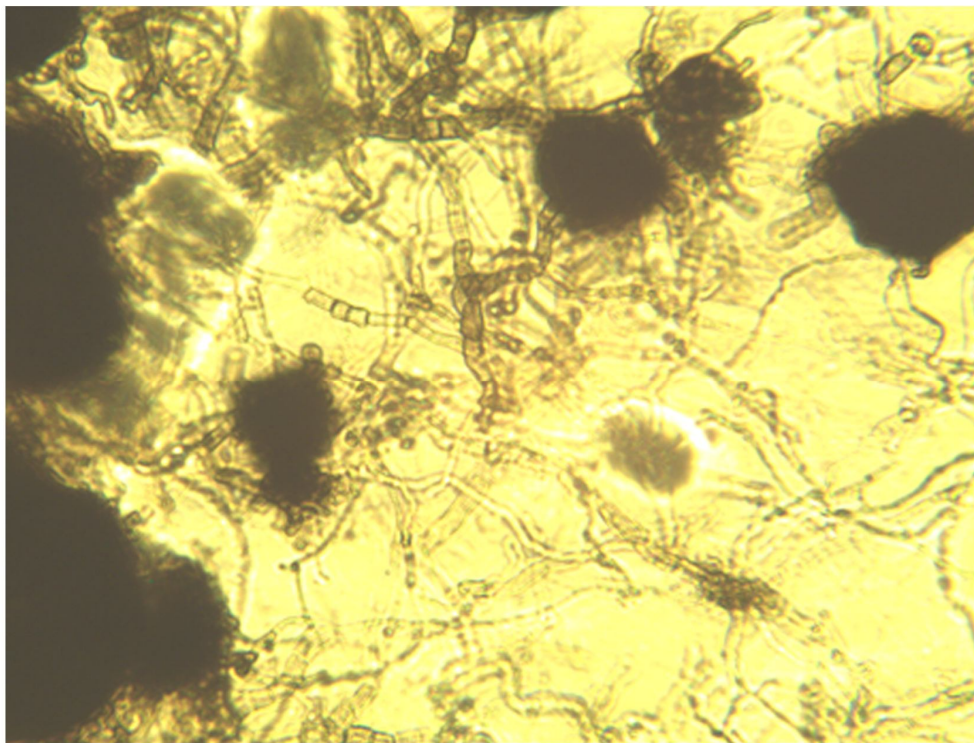


Plate 4: Photomicrograph of *R. bataticola* showing septate and branched hyphae(40x)

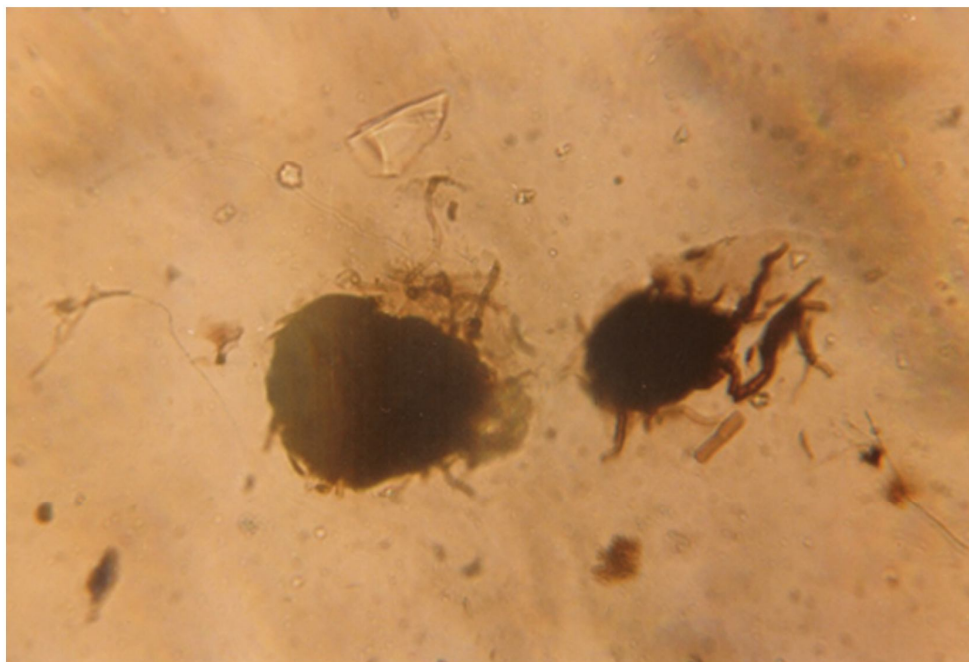


Plate 5: Photomicrograph of sclerotia of *R. bataticola* (40x)



Healthy plants Infected plants



Infected roots

Plate 6: Root rot affected chickpea plants following soil inoculation

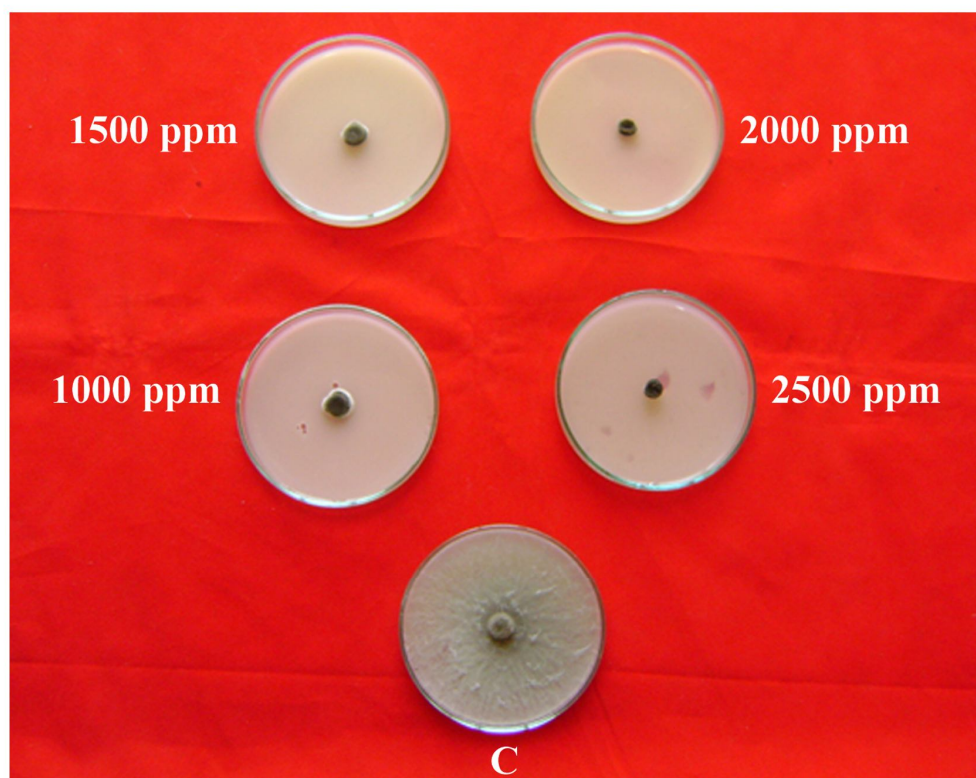


Plate 7: *In vitro* efficacy of captan on mycelial growth of *R. bataticola* by poisoned food technique

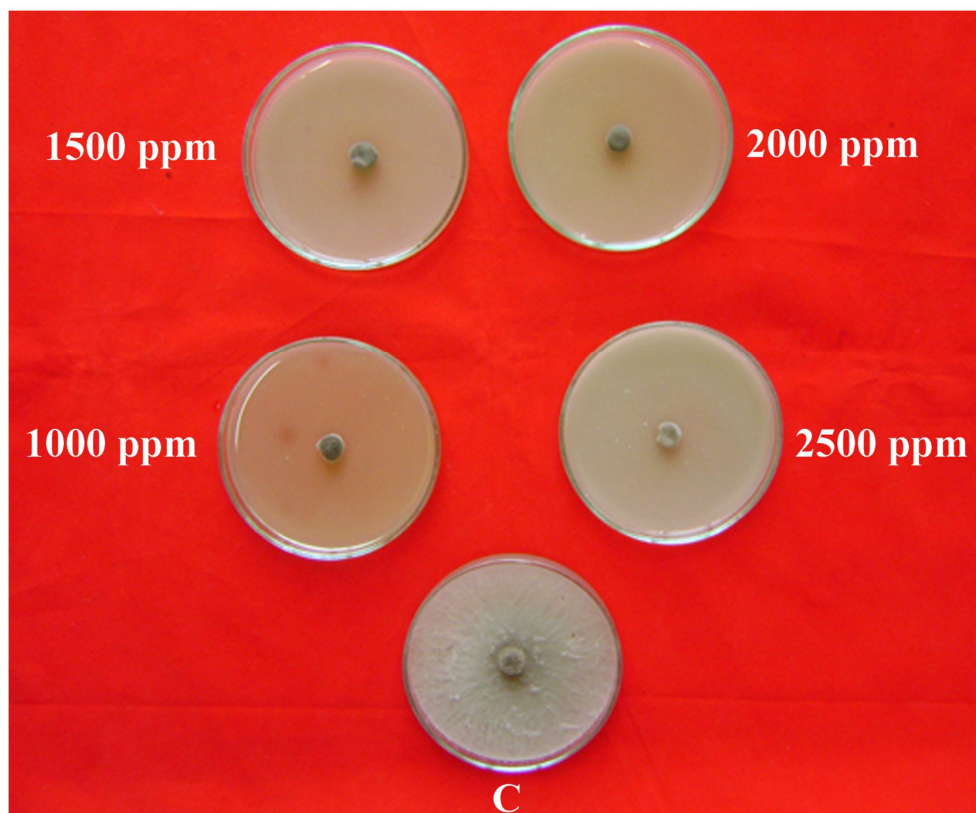


Plate 8: *In vitro* efficacy of thiram on mycelial growth of *R. bataticola* by poisoned food technique

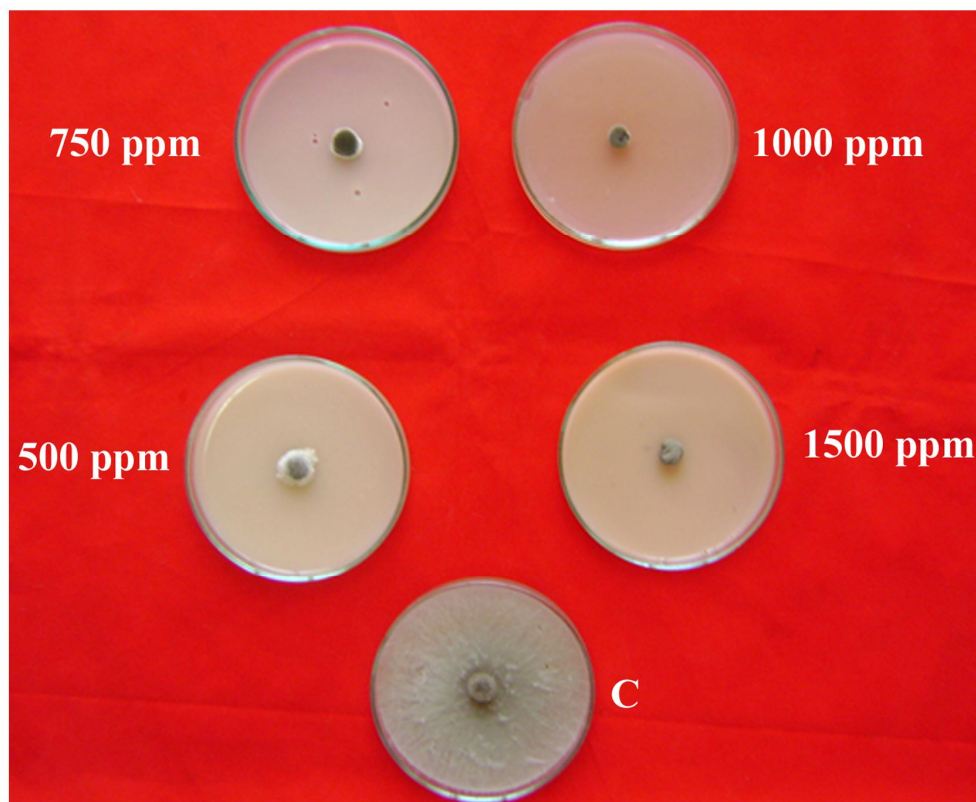


Plate 9: *In vitro* efficacy of carbendazim on mycelial growth of *R. bataticola* by poisoned food technique

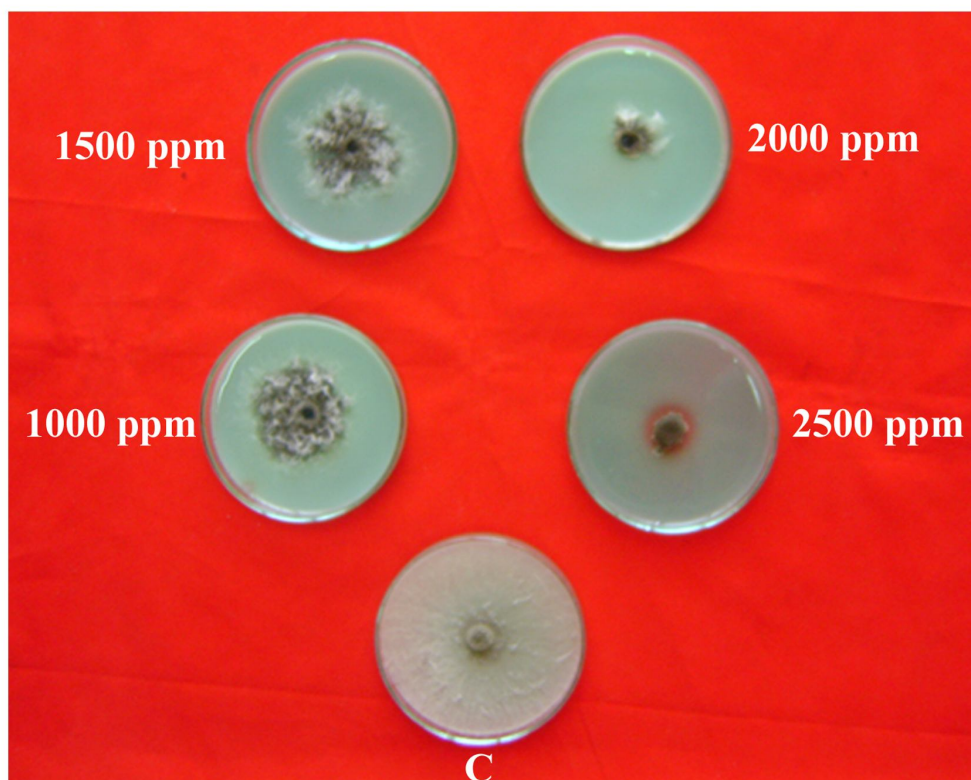


Plate 10: *In vitro* efficacy of copper oxychloride on mycelial growth of *R. bataticola* by poisoned food technique

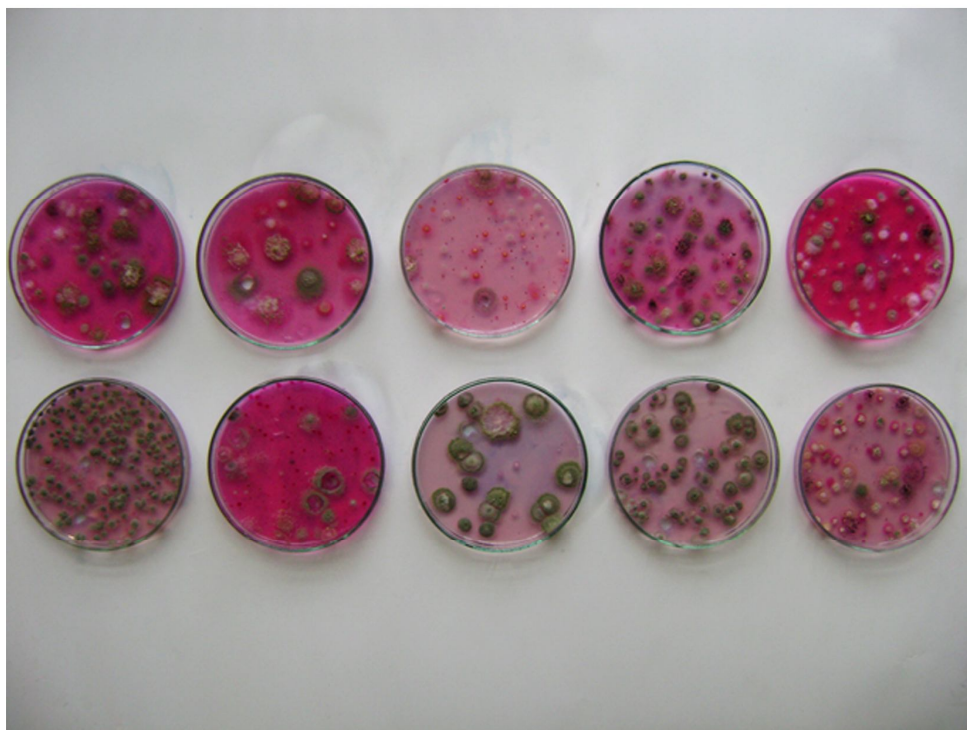


Plate 11: Colonies of antagonistic mycoflora isolated from rhizosphere and root endophytes of chickpea



Plate 12: Colonies of antagonistic bacteria isolated from rhizosphere and root endophytes of chickpea

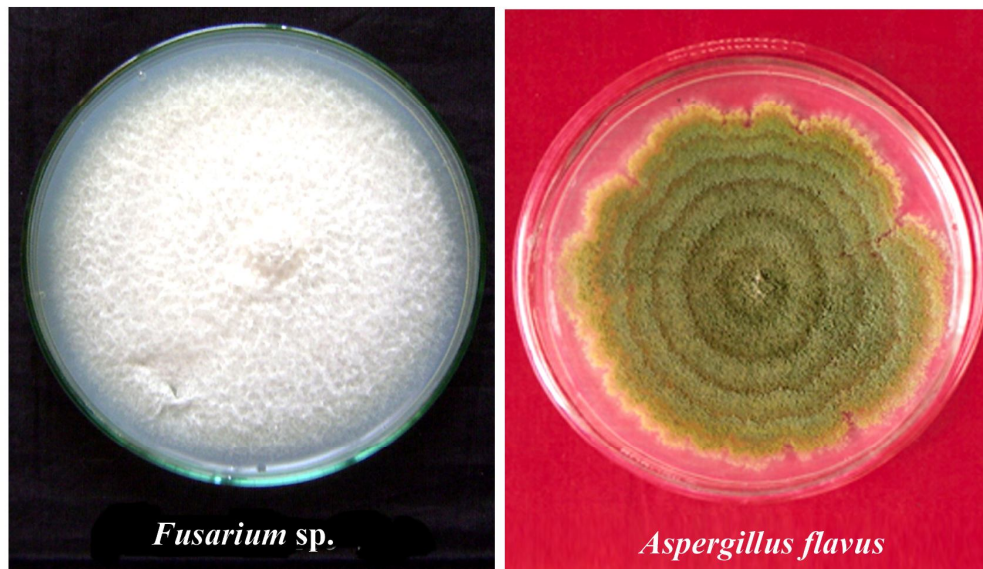


Plate 13: Pure culture of *Fusarium sp.* and *Aspergillus flavus*

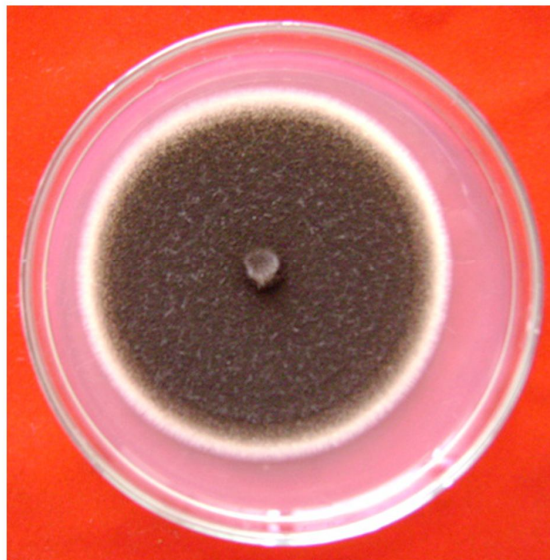


Plate 14: Pure culture of *Aspergillus niger*

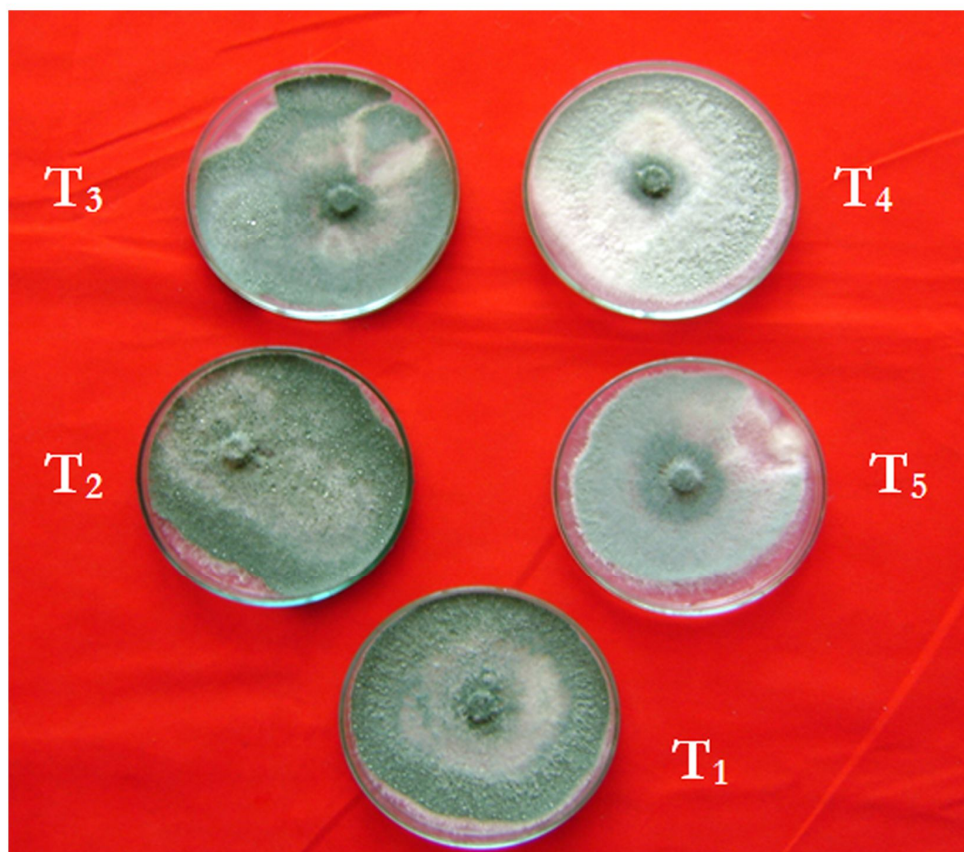


Plate 15: Pure culture of *Trichoderma* isolates-1, 2, 3, 4 and 5 (T_1 to T_5)

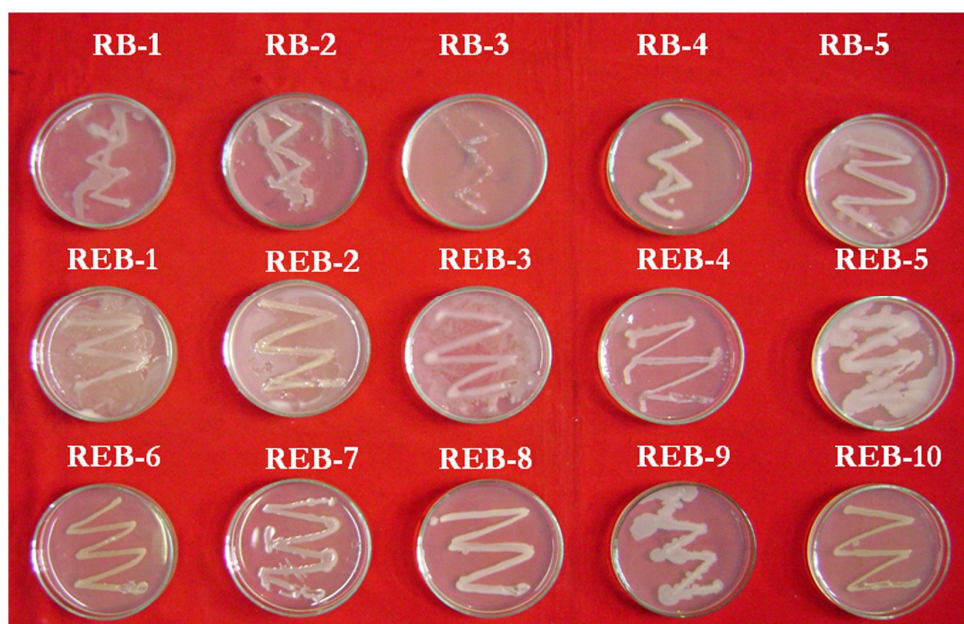


Plate 16: Pure cultures of bacterial isolate rhizosphere bacteria (RB-1 to 5) and root endophytic bacteria (REB 1 to 10)

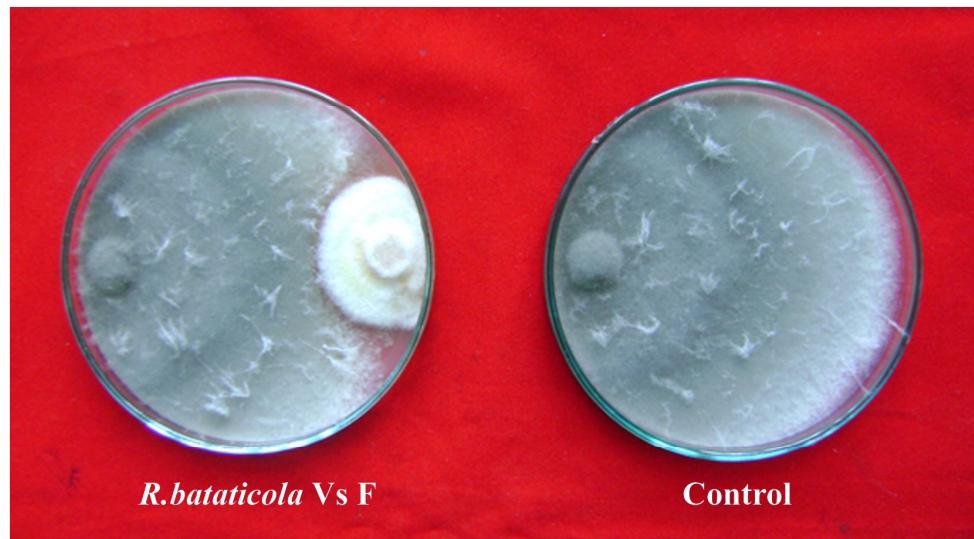


Plate 17: *In vitro* efficacy of *Fusarium* on mycelial growth of *R. bataticola*

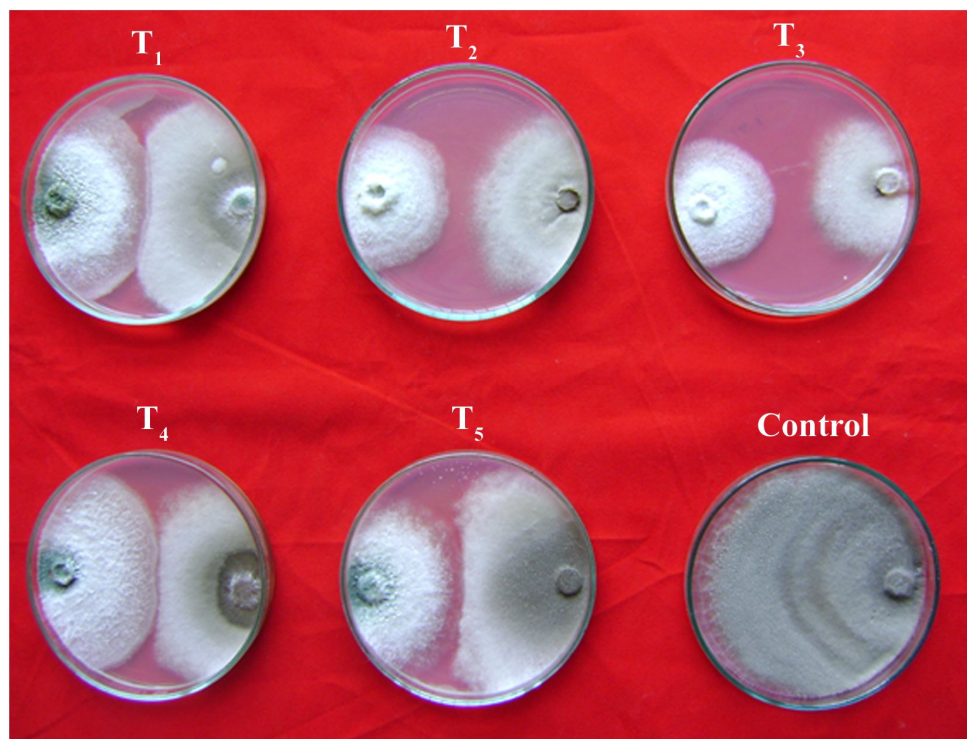


Plate 18: *In vitro* efficacy of *Trichoderma* isolates on mycelial growth of *R. bataticola*

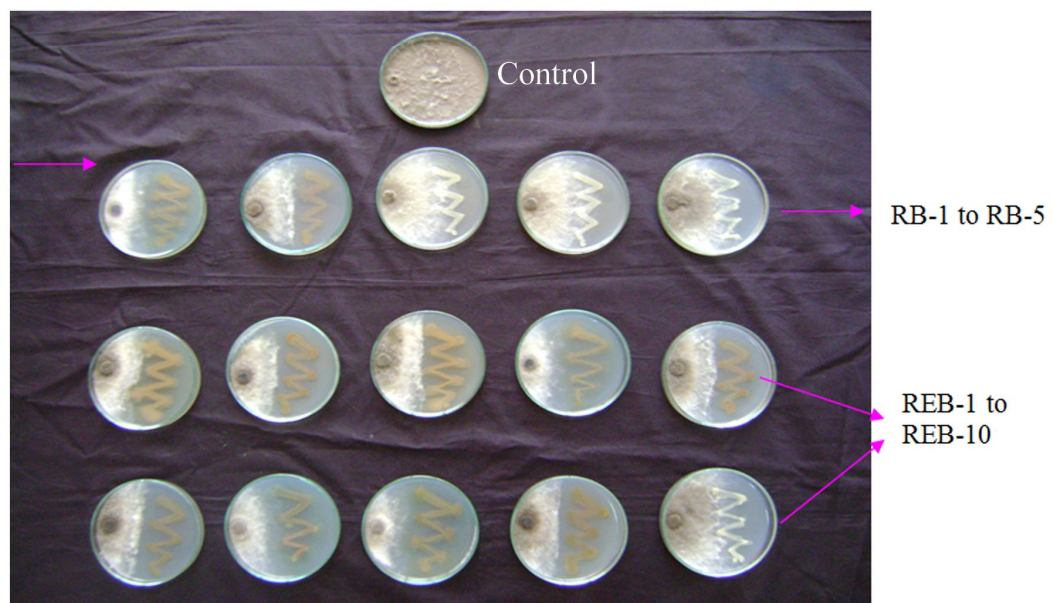


Plate 19: *In vitro* efficacy of bacterial isolates on mycelial growth of *R. bataticola* in dual culture technique



Plate 20: Mass multiplication of *R. bataticola* on sorghum grains



Plate 21: Mass multiplication of potential antagonistic bacteria REB-8 on Nutrient broth



Plate 22: Talc based formulation of potential biocontrol agent (REB-8)



Plate 23: Population of potential antagonistic isolate (REB-8) at the time of application (cfu/g)



Plate 24: Integrated management of *R. bataticola* in chickpea (variety JG-11) under pot culture experiment

Treatments imposed

S.No.	Treatment No.	Treatment
1.	T ₁	Seed treatment with potential biocontrol agent @ 4g/Kg.
2.	T ₂	Seed treatment with fungicide @ 2g/Kg
3.	T ₃	Soil application with potential biocontrol agent
4.	T ₄	Soil drenching with fungicide @ 2.5g/l
5.	T ₅	T ₁ + T ₂
6.	T ₆	T ₃ + T ₄
7.	T ₇	Inoculated control
8.	T ₈	Uninoculated control

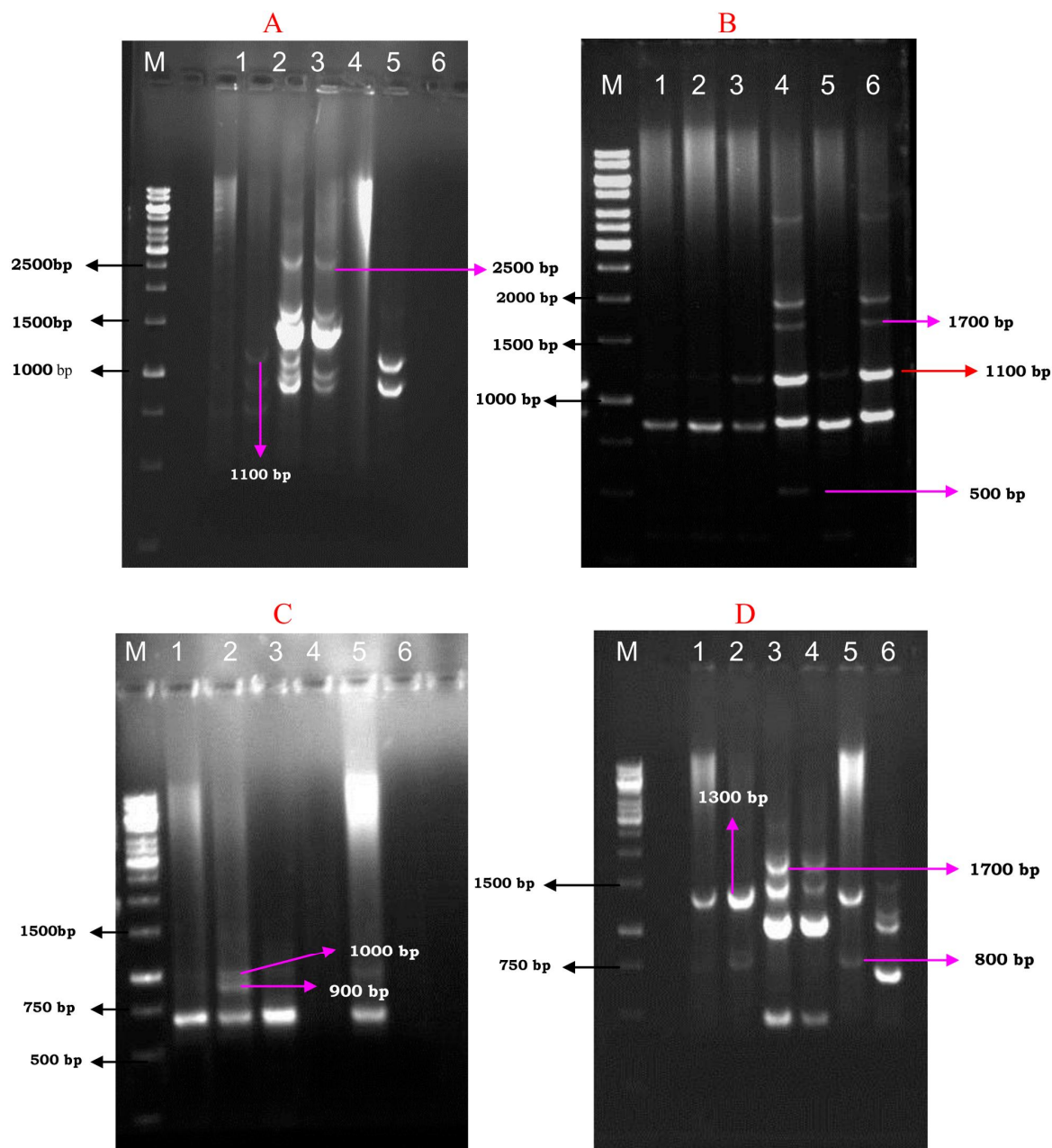


Plate 25a: Random Amplified Polymorphic DNA (RAPD) profiles of antagonistic bacterial isolates with random primers.
M: 1Kb DNA ladder. A: OPA-11, B: OPA-12, C: OPA-14, D: OPA-18, E : OPD-3, Lanes 1-6 are bacterial isolates RB-1, RB-2, RB-3, RB-4, REB-8, and REB-9 respectively

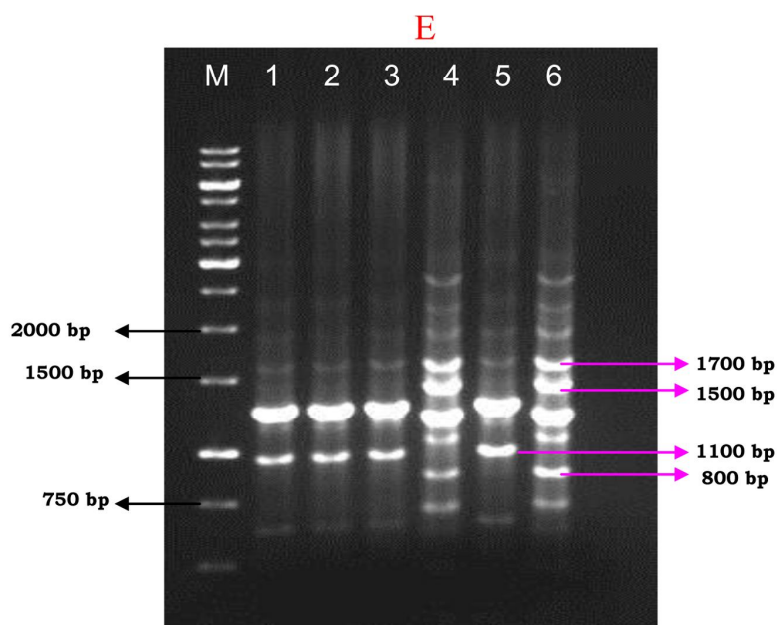


Plate 25b: Random Amplified Polymorphic DNA (RAPD) profiles of antagonistic bacterial isolates with random primer (OPD-3)

M: 1Kb DNA ladder, Lanes 1-6 are bacterial isolates RB-1, RB-2, RB-3, RB-4, REB-8, and REB-9 respectively

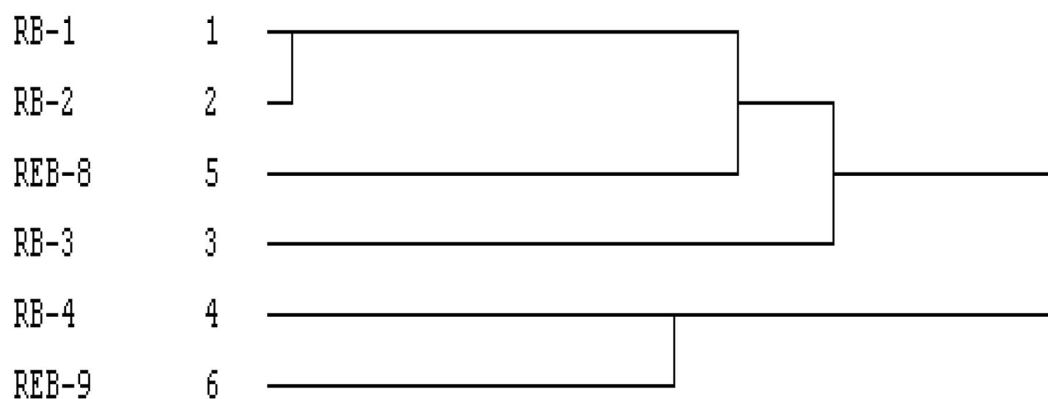


Fig. 11: Dendrogram generated using UPGMA analysis showing polymorphism between antagonistic bacterial isolates using RAPD markers.

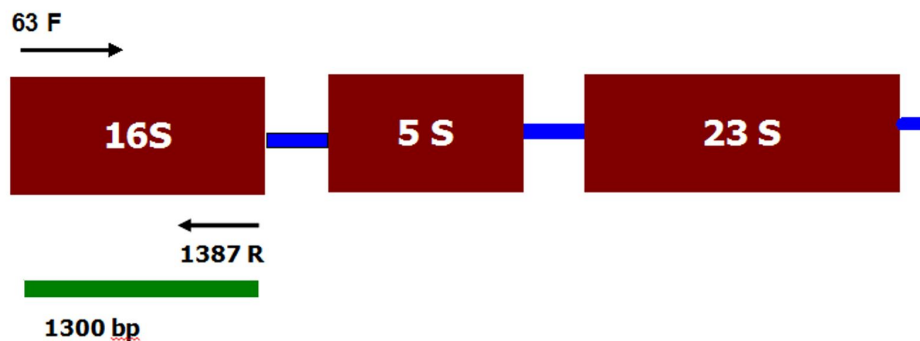


Fig. 12: Structure of rDNA cluster and the position of primers used in the PCR Amplification of 16S rDNA

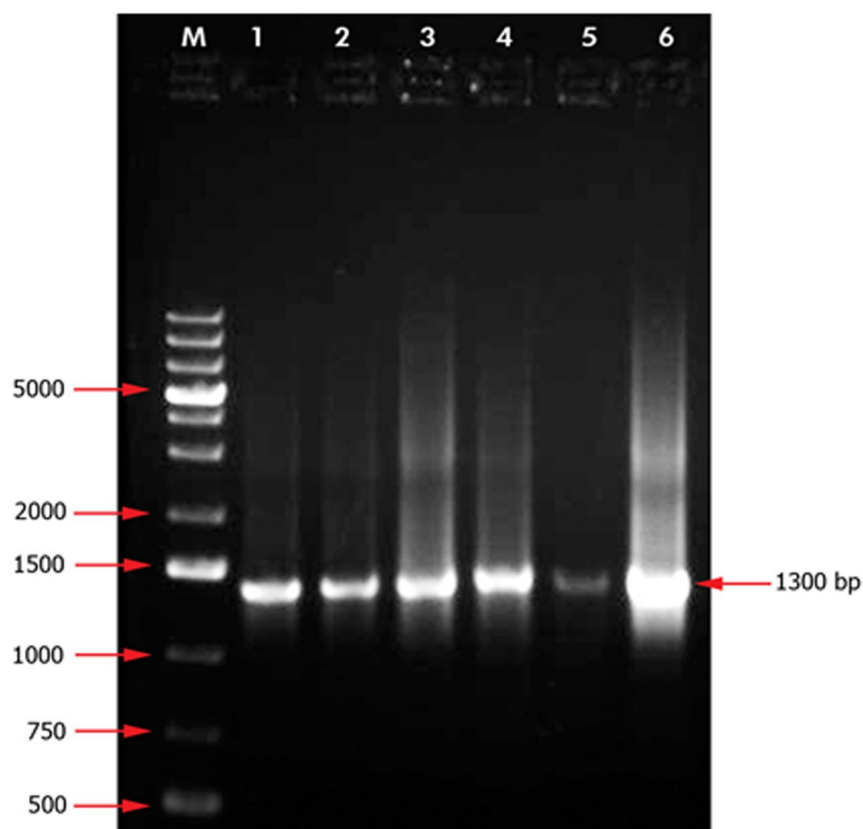


Plate26: Amplification product of 16S rDNA with 63F and 1387R ribosomal DNA primers. Marker = 1 kb DNA ladder: lanes from 1 to 6 represent bacterial isolates RB-1, RB-2, RB-3, RB-4, REB-8, and REB-9 respectively

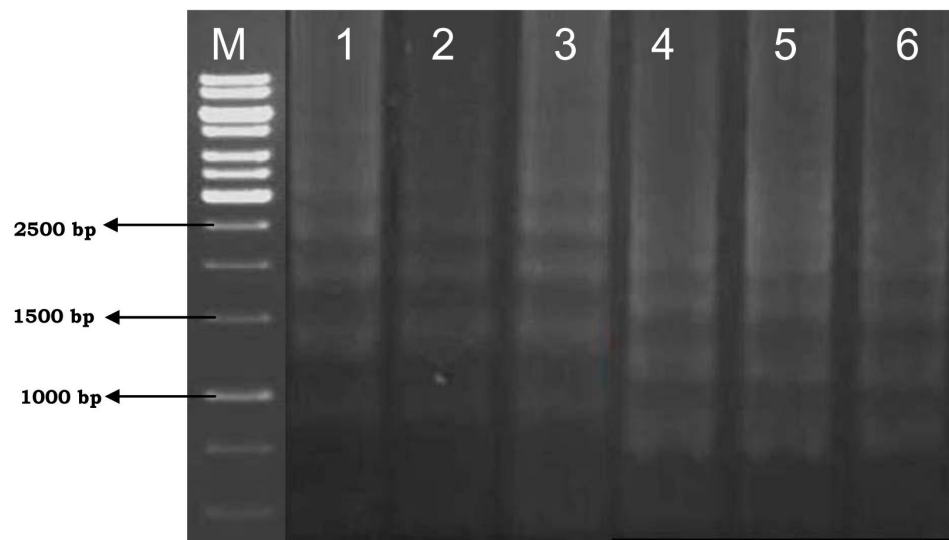


Plate 27: Restriction enzyme digestion of polymerase chain reaction – amplified ribosomal DNA of 16S region from bacterial isolates. Marker = 1 kb DNA ladder, lanes from 1 to 6 represents bacterial isolates RB-1, RB-2, RB-3, RB-4, REB-8, and REB-9 respectively

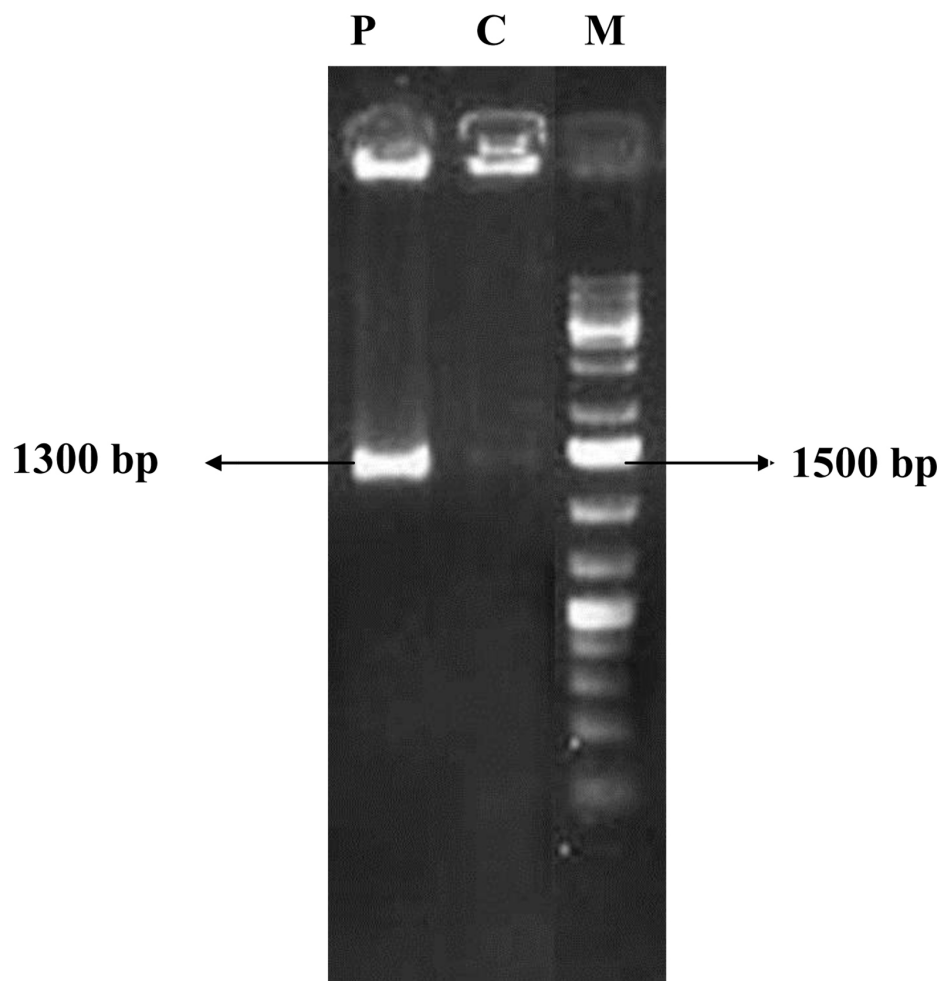


Plate 28: Confirmation of the presence of 1300 bp fragment in recombinant plasmid PCR
M: Marker, C: Control (only plasmid),
P = Recombinant plasmid

CHAPTER –V

DISCUSSION

Chickpea is major grain legume pulse crop of India covering 40 per cent of the area under pulse crops. In chickpea, dry root rot disease caused by *Rhizoctonia bataticola* (Taub) Butler. is a major problem with typical symptoms including withering and drying of the plants, presence of dark tap root showing signs of rotting and devoid of its lateral and finer roots. The most important diagnostic symptoms is shredding of bark and which comes out in the form of flakes (Haware, 1990).

Rhizoctonia bataticola is a non- specialized soil borne fungal pathogen of world–wide importance and has a wide host range of 500 plant species (Sinclair, 1982). The fungus induces a variety of symptoms such as root rots, seedling blights, stem rots, wilts etc., in different host plants.

Biological control has been considered as a potential control strategy against soil borne plant pathogens. In recent years, considerable success has been achieved by introducing antagonists to soil (or) infection court (Papavizas and Lewis 1981; Mukhopadhy and Kaur, 1990). Biocontrol agents also act indirectly by inducing systemic resistance in plants by increased nutrient uptake and make them unavailable to plant pathogens and by inactivating the pathogen enzymes (Chaube *et al.* 2001).

An integrated approach by including the fungicidal tolerant native antagonists isolated from rhizosphere soil and root endophytes appears to be possible solution for effective management of dry root rot.

Molecular characterization of the potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD) and 16S rDNA analysis are important to distinguish different isolates at genetic level.

The present investigation was undertaken to explore the feasibility of integrating biocontrol agents as a part of management of dry root rot of chickpea caused by *Rhizoctonia bataticola*. The results of the present investigation are discussed here under.

5.1 SURVEY

A preliminary roving survey was carried out in rabi, 2009-10 on the occurrence of dry root rot in major chickpea growing mandals of Kadapa and Kurnool district of Andhra Pradesh. During survey, three mandals of Kurnool districts viz., Kovelakuntla, Sanjamala, Dornipadu and three mandals of Kadapa district viz., Jammalamadugu, Peddamandium, Rajupalem were selected. In the present study, survey was conducted in mandals of Kurnool district Kovelakuntla , Sanjamala and Dornipadu and the average percentage of disease incidence as 9.53, 13.50 and 7.71 per cent respectively, whereas in Kadapa district, three mandals viz., Jammalamadugu, Peddamandium and Rajupalem were surveyed and

average per cent disease incidence as 7.55, 7.68 and 6.22 respectively. The highest average per cent disease incidence was recorded in Sanjamal mandal (13.50%) and least average per cent disease incidence was noticed in Rajupalem mandal (6.22%) Kadapa district.

Similarly, Singh and Sirohi (2003) also reported that the incidence of dry root rot of chickpea was highest in Una (4.86 %) and Sirmour (3.04 %) districts of Himachal Pradesh.

5.2 THE PATHOGEN

5.2.1 Isolation of the pathogen

The pathogen associated with dry root rot of chickpea collected from Reddipalli village of Sanjamala mandal of Kurnool district was isolated using tissue segment method. The dry root rot affected plants were identified based on symptoms like drying and withering of plants and when these plants were pulled out showed blackening of tap root and devoid of lateral and finer roots and shredding of bark which comes out in the form of flakes. These symptoms were similar to the reports of Singh and Agarwal (2002).

5.2.2 Identification of the pathogen and its maintenance

The pathogen isolated from root region was purified by single hyphal tip method and maintained on PDA for further studies. Based on mycological characters (4.2.2) the pathogen was identified as *Rhizoctonia bataticola* (Taub) Butler. Similar reports were observed by Barnett and

Hunter ((1972) and Sajeena *et al.* (2004). Based on these, the isolated organism from such infected chickpea plants were identified as *Rhizoctonia bataticola*.

5.2.3 Pathogenicity test

Pathogenicity of the fungus was established by inoculation and re-isolation from artificially infected plants. In the present investigation, soil inoculation method was followed to establish the disease.

Several workers found soil inoculation method as most suitable in establishing the disease caused by *Rhizoctonia bataticola* (Vishwadar and Sarabhay (1993) in soybean; Prameela Devi and Singh (1998) in greengram; Kateria *et al.* (2007) in chickpea).

5.3 *IN VITRO* EVALUATION OF EFFICACY OF FUNGICIDES AGAINST *R. bataticola*

One of the objectives of present investigation is to include an effective fungicide in combination with potential biocontrol agent as an integrated approach to manage *Rhizoctonia bataticola* under glasshouse conditions.

Keeping this in view, an attempt was made to find out a promising fungicide against *Rhizoctonia bataticola* under *in vitro* conditions. The four fungicides viz., captan, thiram, copper oxychloride (COC) and carbendazim

were evaluated against *Rhizoctonia bataticola* at different concentrations by using poison food technique.

Among the fungicides evaluated, overall per cent inhibition of *R. bataticola* mycelial growth was maximum with carbendazim followed by thiram and captan. Copper oxychloride (COC) was found to be least effective on *Rhizoctonia bataticola* which inhibited mycelial growth by 63.95 per cent.

The results were in agreement with Khan and Gangopadhyay (2008) who reported that the carbendazim was highly inhibitory (85%) to *R. bataticola* incitant of dry root rot of chickpea under *in vitro*.

Konde *et al.* (2008) revealed that the combination of carbendazim + thiram (0.1 + 0.2 % concentration) was significantly effective in inhibiting (100%) the radial growth of *R. bataticola*.

Paul *et al.* (2008) evaluated the fungicides against the root rot of pea caused by *F. solani*, *Rhizoctonia solani* and *S. sclerotiorum* and the results revealed that carbendazim was most effective fungicide against all the pathogens.

5.4 ANTAGONISTS

In the present investigation, among the 8 fungal antagonists isolated from rhizosphere soil, the *Trichoderma* isolate-3(T₃) showed maximum

inhibition (57.83%) of mycelial growth of *R. bataticola* (Table 4 and Plate 18). Among 15 rhizosphere and root endophytic bacteria tested under *in vitro*, the isolate REB-8 showed maximum inhibition of 76.47 per cent followed by RB-1 (74.11%) and REB-9(71.76%)(Table 5; Plate 19).

These results were in agreement with Khan and Gangopadhyay (2008) who reported the maximum reduction of mycelial growth of *R. bataticola* incitant of dry root rot of chickpea in dual culture technique by *Pseudomonas fluorescens* strain PFBC-25.

Vinod Kumar *et al.* (2007) tested the efficacy of *Pseudomonas fluorescens* isolates against *M. phaseolina* incitant of charcoal rot of chickpea under *in vitro* and reported that the isolate pf 4-99 was found to be effective in inhibiting the mycelial growth of pathogen.

Konde *et al.* (2008) tested the five antagonists against dry root rot of soybean and observed that *Trichoderma viride* (96.39%) was superior in inhibiting the growth of *R. bataticola*.

5.4.1 *In vitro* compatibility of potential native antagonists with fungicides

A biocontrol agent must be effective and compatible with latest crop production practices so that its use can be integrated into the production system. In such an approach, biocontrol agents have been used without any toxic effect on antagonists (Papavizas and Lumsden, 1980). Integrated seed treatment with chemicals and compatible antagonists not only protect the

seed and seedlings from soil borne pathogens but also provide protection from seed borne inoculum. Compatible fungicides are therefore essential for integrated management (Dubey and Patel, 2001).

5.4.1.1 *In vitro* compatibility of potential bacterial isolate (REB-8) with fungicides

Compatibility of native potential antagonistic bacterial isolate (REB-8) with fungicides was evaluated using spectrophotometric method. The results revealed that the isolate REB-8 was found to be more compatible with carbendazim (0.1%) followed by thiram (0.25%) and copper oxychloride (0.25%). The least compatibility was observed with captan (0.25%) (Table 6 and Fig. 6)

Similar observations were made by Khan and Gangopadhyay (2008) who reported that carbendazim and carboxin were least toxic to *Pseudomonas fluorescens* strain PFBC-25 whereas captan was most inhibitory to this strain.

Vidhyasekharan *et al.* (1995) reported that thiram and carbendazim was not inhibitory to *Pseudomonas fluorescens*.

5.5 INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHICKPEA

As the management of soil borne disease is not possible through only one approach, in recent years effort was made to reduce environmental

hazardous and rationalize the use of pesticides and manage diseases more effectively economically, which lead to the emergence of the new discipline called Integrated Disease Management (IDM). For sustainable crop production the components involved should be eco-friendly. So that, beneficial organism would be safe and IDM practices would go a long way helping stabilized crop production (Anahosur, 2001). In this contest the biological control, integration with fungicidal treatment was found to be a more reliable approach to manage soil borne plant pathogens (Mukhopadhyay, 1987).

Keeping in view the importance of integrated disease management and based on the results obtained in the present investigation, a study was under taken for the management of *Rhizoctonia bataticola* of chickpea by combining potential biocontrol agent (REB-8) and a compatible fungicide carbendazim (0.1%) in pot culture.

5.5.1 Mass multiplication of *Rhizoctonia bataticola*

Sterilized sorghum seeds were found to be suitable for mass multiplication of *R. bataticola* as reported by several workers viz., Rajeswari *et al.*, 1999; Durai, 2004. Hence, during the present study the pathogen was mass multiplied on sterilized sorghum seeds and used for soil inoculation.

5.5.2 Preparation of talc based formulation of potential biocontrol agent REB-8

In the present investigation, talc based formulation of potential root endophytic bacteria (REB-8) was prepared and applied to seed @ 4 g Kg⁻¹ and soil by multiplying on FYM and applied @ 100 g /pot before sowing.

Similarly Rangeshwaran *et al.* (2001) also prepared talc based formulations of *Pseudomonas fluorescens* (PDBCA-2) and *P. putida* (PDBCAB 19) which were effective in inhibiting the wilt and root rot of chickpea under *in vitro*.

5.5.3 Observations

The efficacy of integrated approach that included potential antagonist (REB-8) and effective fungicide was tested in pot culture against dry root rot pathogen and observations on per cent disease incidence (PDI) and growth parameters viz., plant height, root length, dry weight of shoot and root in different treatments imposed on chickpea were recorded (Table 7).

5.5.3.1 Per cent disease incidence

In the present investigation, maximum disease control was observed in integrated treatment (T₆) that included soil application with potential BCA (REB-8) + soil drenching with fungicide (carbendazim) (6.67%).

It may be due to the synergetic effect of carbendazim (0.1%) and antagonist on PDI. In integrated control the fungicide might have weakened the pathogen and making them more susceptible to antagonists.

Similar results have been reported by several workers in management of several soil born diseases by integration of biocontrol agents with chemicals.

Survival of 100 per cent of plants infected with *Phytophthora capsici* of Black Pepper was obtained when *Pseudomonas fluorescens* strains were applied in combination with metalaxyl (Diby paul *et al.*, 2005).

Vyas (1994) reported that integration of carbendazim with *T. harzianum* (or) *T. viride* was effective in controlling root rot of soybean caused by *Macrophomina phaseolina*.

Sindhan *et al.* (2002) reported satisfactory disease control when *Pseudomonas fluorescens* was used as seed treatment along with carbendazim against *Macrophomina phaseolina* in chickpea.

Soil application of 2.5 Kg ha⁻¹ of *Trichoderma viride* and *Pseudomonas fluorescens* in talc based formulations to sunflower reduced per cent disease incidence of charcoal rot significantly than in control (Suriachandraselvan, 1997).

Plant growth parameters

In the present investigation, an attempt was made to observe whether the treatments imposed have any stimulatory (or) inhibitory effect on mean plant height, root length and dry weight of shoot and root of chickpea plants.

a) Plant height

In the present investigation the maximum plant height (25.60 cm) was recorded in integrated soil application with potential biocontrol agent + soil application with fungicide and this treatment was found stimulatory on the growth of the plant. Least plant height (16.0 cm) was recorded in inoculated control. It is attributed that carbendazim could arrest the pathogen and antagonists could parasitize the pathogen and promote growth by secreting growth promoting metabolites.

The results were in agreement with Vinod Kumar *et al.* (2007) who reported that *Pseudomonas fluorescens* isolate pf 4-99 reduced the incidence of charcoal rot of chickpea and enhanced the plant height by 29.4 per cent against the control.

Lynch and Whipps (1991) proved that plant growth promotion by rhizobacteria is due to chemical and physical stimulation of plant roots resulting in more rapid emergence, higher chlorophyll level and increased stature. Liu *et al.* (2003) also observed higher shoot per root ratio in plants treated with strains of plant growth promoting rhizobacteria.

b) Root length

In the present investigation, maximum root length (9.15 cm) was recorded in integrated treatment T₆ (soil application with potential BCA + soil drenching with fungicide).

Bharati *et al.* (2004) reported that soil application of *Trichoderma harzianum* and *B. subtilis* reduced the incidence of damping-off in tomato caused by *Pythium aphanidermatum* and also increased shoot and root length and biomass production.

Sendhilvel *et al.* (2005) used *Pseudomonas fluorescens* isolate SVPF₂ in combination of seed treatment and soil application against cowpea root rot and recorded less disease incidence and maximum root length (22.38 cm) under green house conditions.

Jayasree *et al.* (2000) reported that maximum root length and shoot length of sesame and black gram were recorded in combined seed treatment and soil application of *T. harzianum* and *P. fluorescens*.

c) Dry weight of shoot and root

The results of present investigation are in agreement with earlier reports showing that the integrated treatment with biocontrol agents increased weight of shoot and root of chickpea. Maximum shoot and root

dry weights were recorded in integrated treatment T₆ (soil application with potential biocontrol agent + soil drenching with effective fungicide).

Among all the treatments imposed, integrated soil application with potential BCA + soil drenching with fungicide carbendazim i.e. T₆ is highly effective followed by treatment involving seed treatment with BCA + seed treatment with fungicide i.e. (T₅) were found to be superior as they recorded minimum PDI, maximum root length, dry shoot and root weight. The present findings are supported by other workers according to whom the integration of biocontrol agent with compatible fungicide gave significantly higher disease control in several crops than obtained by either biocontrol agent (or) fungicide. (Henis *et al.* 1978, Sawant and Mukhopadhyay, 1990).

The efficacy of native potential biocontrol agent REB-8 will be evaluated under field conditions during *rabi* 2010-11 in Kadapa and Kurnool districts of Andhra Pradesh.

5.6 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS.

Recently, molecular techniques have gained importance in characterization and diagnosis of microbial population. Moreover, the techniques are not influenced by environment, growth independent and are reproducible when compared to conventional methods. Hence, molecular characterization of potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD) and analysis of 16S rDNA has been carried,

which helps in identification of antagonists are of immense use. These molecular techniques can also be used to develop markers in diagnosis of potential biocontrol agents in future.

RAPD is one of the methods of molecular analysis of natural microbial communities lack resolving power when it comes to the identification at the species level and they also fail to give sufficient information to infer the functions of newly isolated organisms. However, these methods have a role in surveying the composition of microbial communities and in the characterization of bacterial isolates.

During the present investigation, the genetic diversity among isolates of potential biocontrol agents which have different degrees of antagonism against *Rhizoctonia bataticola* under *in vitro* was studied using RAPD. A total of 48 reproducible and scorable bands were obtained using five random primers viz., OPA-11, OPA-12, OPA-14, OPA-18 and OPD-3.

The primer OPD-3 amplified unique band of approximately 800 bp and 1100 bp incase of isolate RB-4 and REB-9. The primer OPA-18 amplified unique band of approximately 1700 bp incase of isolate RB-3 and RB-4. Whereas specific band of approximately 800 bp has been amplified in case of isolate RB-2 and REB-8.

The dendrogram formed using scorable bands of all the six isolates using UPGMA based on wards squared equalidean distance method, gave

two clusters in which overall similarity range from 0 to 100%. The cluster I contains three isolates viz., RB-1, RB-2 and REB-8 and cluster – II contains 3 isolates viz, RB-3, RB-4 and REB-9.

These results are in agreement with Ramesh Kumar *et al.* (2002) who studied the genetic variability among the isolates of *Pseudomonas* by RAPD with random primers, and the primer pgs3 produced several bands, including a unique band with size of 800 bp.

Megha *et al.* (2007) studied the diversity of fifteen isolates of Fluorescent pseudomonads using RAPD - PCR with eight random primers viz., OPC-9, OPD-2, OPD-3, OPO-6, OPO-09, OPO-13, 15 and 16. The PCR amplicons of fluorescent pseudomonads obtained from eight random primers produced 127 polymorphic bands. The minimum bands (9) were produced by the primer OPD-02 and maximum number of bands (25) was produced by OPO-16.

RAPD markers could be used for developing Sequence Characterized Amplified Region (SCAR) markers linked to potential antagonists. SCAR marker is a specific and sensitive tool for monitoring the biocontrol in environmental samples. Biological controls of micro organisms have long been promoted as an alternative to conventional fungicides. Before registration of a microbial biocontrol product for commercial scale, it must

be evaluated for its efficacy with regards to potential spread and persistence after release.

The RAPD analysis results further indicated that the potential bacterial isolate REB-8 formed a special group cluster Ib. This indicates that the REB-8 isolate is different from the rest of the isolates under the study. Further a specific band of 1100 was amplified in REB-8 with OPD-3 primer. This band can be used for the development of SCAR marker after cloning and sequencing. This SCAR marker may be validated further for confirmation.

16S region of rDNA was amplified for all the bacterial isolates under the study using specific 63F and 1387R primers. These primers yielded a amplicon of 1300 bp as expected indicating all the identified potential antagonistic microflora are bacteria and belongs to a kingdom of prokaryotes.

The amplified products of 1300 bp fragment of 16S region of rDNA from all the antagonists was digested with *Taq I* enzyme. The results shows that the there was no polymorphism in the restriction banding pattern among the isolate under the study. However, it is worthwhile to digest the 1300 bp fragment with different restriction enzymes for more insight about the polymorphism among the isolates. The amplified product of 1300 bp from all the isolates may be subjected to nucleotide sequence which will provide

information about restriction map of the amplified product from all the isolates.

These results were in agreement with Ayako kawai (2006) who studied the molecular characterization of all isolated strains of *Bacillus* spp. These results suggest that KB-1 and strains of isolates from KB-1 were identical based on 16S rDNA and 16S rDNA - RFLP.

Present study on molecular characterization of potential biocontrol agents by RAPD and analysis of 16S rDNA reveals the existence of polymorphism among the isolates. The potential antagonistic isolate REB-8 will be identified after obtaining sequence of the 1300 bp amplified product of 16S rDNA and blasting with the available sequences in the NCBI data bank.

CHAPTER - VI

SUMMARY

Chickpea (*Cicer arietinum* L.) is one of the major grain legume pulse crops of India. Many diseases occur at different growth stages in chickpea, among these dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler. is one of the important diseases with yield loss over 10 to 100 per cent.

Rhizoctonia bataticola is a serious soil borne plant pathogen of world wide occurrence causing huge losses in about 500 plant species. In the present study the following objectives were carried: (i) Survey for incidence of dry root rot of chickpea in Kadapa and Kurnool districts of Rayalaseema region, Andhra Pradesh (ii) Isolation and identification of pathogen from infected plants (iii) To evaluate the efficacy of fungicides against causal agent of dry root rot under *in vitro* (iv) To isolate potential fungicidal compatible biocontrol agents from rhizosphere and root endophytes under *in vitro* (v) Integrated disease management of dry root rot of chickpea under greenhouse conditions. In addition, molecular characterization of different isolates of potential antagonists was also studied by RAPD and analysis of 16S rDNA. The results obtained in the present investigation are summarized here.

Survey conducted for dry root rot incidence in six major chickpea growing mandals of Kadapa and Kurnool district the indicated, then the disease incidence varied from 6.22 to 13.50 per cent with lowest and highest incidence in Rajupalem and Sanjamala mandal respectively.

The pathogen associated with dry root rot of chickpea was identified as *Rhizoctonia bataticola* based on standard mycological keys (Barnett and Hunter, 1972).

A total of 23 antagonistic microflora (8 fungi and 15 bacteria) were obtained from rhizosphere soil and root endophytes of chickpea. The potential antagonistic isolates were identified based on their ability to inhibit the growth of *Rhizoctonia bataticola* in dual culture technique. Among eight fungal antagonistic isolates, *Trichoderma* isolate-3 (T₃) was superior with highest per cent (57.83%) inhibition of the growth of *R. bataticola* followed by *Trichoderma* isolate-4(T₄) (54.21%). Regarding root endophytes, the isolate REB-8 showed maximum inhibition (76.47%) of growth of *R. bataticola* followed by RB-1 (74.11%) and REB-9 (71.76%).

In vitro efficacy of four fungicides viz., thiram, copper oxychloride, captan and carbendazim was evaluated against *Rhizoctonia bataticola* using poison food technique at different concentrations. Carbendazim was found to be effective as it completely inhibited the mycelial growth even at lower concentration.

In vitro compatibility of four fungicides used against *R. bataticola* were tested on potential antagonistic root endophytic bacteria viz., REB-8 at different concentrations by using spectrophotometric method. The carbendazim showed high compatibility followed by thiram and copper oxychloride, whereas captan was found to be less compatible with REB-8.

Among the eight treatments imposed, treatment T₆ (soil application of potential biocontrol agent (REB-8) + soil drenching with fungicide (carbendazim) was found to be superior as it recorded the least PDI of 6.67 per cent. This treatment also recorded maximum plant height (25.60 cm), root length (9.15 cm) and maximum shoot and root dry weights i.e. 0.09 g and 0.05 g respectively when compared to other treatments.

RAPD banding profile with five different random primers viz., OPA-11, OPA-12, OPA-14, OPA-18 and OPD-3 revealed the existence of genetic variability among the isolates and were classified into 2 main clusters. Cluster I is sub-divided into cluster Ia (RB-1 and RB-2) and Ib (REB-8). cluster-II contains RB-3, RB-4 and REB-9.

Amplification of 16S rDNA with 63F and 1387R primers which are specific to bacterial 16S rDNA produced approximately 1300 bp fragment. These results show that all the antagonistic isolates are bacteria and belong to prokaryotes.

The amplicon of 1300 bp was subjected to restriction analysis with *TaqI* enzyme in order to see the polymorphism in the 16S rDNA. These 16S rDNA-RFLP results revealed that there was no polymorphism among isolates with the above restriction endonuclease. The 1300bp amplified product of 16S rDNA region was cloned into the vector and sent for sequencing to MWG technologies, Bangalore.

LITERATURE CITED

- Ahmed Q and Mohammad A 1986 Losses in yield due to *Rhizoctonia* root rot of chickpea in Bihar. Indian Phytopathology 39: 590-592.
- Ainsworth G C 1961 Dictionary of fungi Common Wealth Mycological Institute, Kew, Survey, England pp: 547
- Anahosur K H 2001 Integrated Management of potato sclerotium wilt caused by *Sclerotium rolfsii*. Indian Phytopathology 54; 158-166.
- Ananthan P K, B Palanna, E G Ebenezak, and R Bhaskaran 2003 Screening of fungal and bacterial antagonists against *Macrophomina phaseolina* Pub. In National seminar on IPDMSA, Annamalai University.,pp 26.
- Aneja K R 1993 Experiments in Microbiology, plant pathology and tissue culture. Wishwa Prakashan, New Delhi (A division of Wiley Eastern Ltd.,) pp. 471.
- Anonymous 1965 Plant disease survey for the year ending 30th June 1964. 34th Annual Report N S W Department of Agricultural Biology Branch pp 38.
- * Ayako Kawai, Kaori Kusunoki, Daigo Aiuchi, Masanori Koike, Masayuki Tani and Katsuhisa Kuramochi 2006 Biological control of *verticillium* black spot of Japanese radish using *Bacillus* spp. and genotypic differentiation of selected antifungal *Bacillus* strain with antibiotic marker. Res. Bull. Obihiro., 27: 109-119.
- Bagri R K, Jain S C, Jain K L and Pankaj Sharma 2004 Detection and control of seed borne fungi causing root rot disease in chickpea (*Cicer arietinum* L.). Journal of Mycology and Plant Pathology, 34 (2): 658-660.

- Baldev B, Ramanujam S and Jain H K 1988 Pulse crops. Fungal diseases of pulse crops pp.266-267.
- Barnett H L and Barry B Hunter 1972 Illustrated genera of imperfect fungi. Burgess publishing company, Minnesota.
- Bharati S L Singh R Verma A and Upadhyay R S 2004 Biocontrol of damping off of tomato caused by *Pythium aphanidermatum* using *Trichoderma harzianum* and *Bacillus subtilis* (Abstr.). Indian Phytopathology 57: 362-369.
- Bora L C and Deka S N 2008 Bioformulation of *Pseudomonas aeruginosa* in organic substrates and its Role in Management of Bacterial wilt of chilli (*Capsicum annum*). Journal of Mycology and Plant Pathology 38: 80-83.
- * Butler E J 1918 Fungi and Diseases in plants. Thacker Spink and Co. Calcutta pp 547.
- Chattopadhyay C and Kalpana Sastry R 2002 Combining viable disease controls tools for management of sesame stem rot caused by *Macrophomina phaseolina* (Tassi) Goid. Indian Journal of Plant Protection 30 (2): 132-138.
- Chaube H S, Vishwakarma S N, Arvinder Kaur, Jameel Akhatar and Anurag 2001 Biological control of plant diseases. Indian farmers digest October pp.23-29.
- Chauhan M S 1988 Relative efficacy of different methods for the control of seedling disease of cotton caused by *Rhizoctonia bataticola*. Indian Journal of Mycology and Plant Pathology, 18:25-30
- Dhingra O D and Sinclair J B 1978 Biology and pathology of *Macrophomina phaseolina*. Pathology 44: 196-202.

- Dhingra O D and Sinclair J B 1995 Basic plant pathology methods CRC Press, London.
- Diby Paul, Jisha P J, Sharma Y R and Anand Raj M 2005 Rhizospheric *Pseudomonas fluorescens* as rejuvenating and root proliferating agents in black pepper. Journal of Biological Control 19; 173-178.
- Dileep kumar B S and Dubey H C 1992 Seed bacterization with fluorescent pseudomonads for enhanced plant growth, yield and disease control. Soil Biology and Biochemistry 24:539-542.
- Dubey S C and Patel 2001 Determination of tolerance of *Thanatephorus cucumeris*, *Trichoderma viride*, *Glioladium virens* and *Rhizobium* spp. to fungicides. Indian Phytopathology 54: 98-101.
- Durai M 2004 Management of charcoal rot of sesame (*Sesamum indicum* L.) incited by *Macrophomina phaseolina* (Tassi) Goid, M.Sc. (Ag.) Thesis submitted to Acharya N.G. Ranga Agricultural University Hyderbad, Andhra Pradesh.
- Ebenazar E G and Wesely E G 2000 Effect of fungicides on *Macrophomina phaseolina* causing root rot in greengram. Pestology 24(3):12-14.
- Gaikwad S J, Sen B and Meshram S U 2002 Effect of bottle gourd seed coating with antagonists on seedlings quantum of the pathogen inside the seedlings and population of the soil against *Fusarium oxysporum*. Plant and Soil, 101: 205-210.
- Gaur R B, Sharma R N, Sharma R R and Vinod Singh Gautam 2005 Efficacy of *Trichoderma* for *Rhizoctonia* root rot control in chickpea. Journal of Mycology and Plant Pathology, 35(1): 144-150.

- Girija V K and Umamaheswaran 2003 Basal rot of balsam and its management through bioagents. *Plant Disease Research*, 18(1): 52-55.
- Gogoi N K Phookan A K and Narzary B D 2002 Management of collar rot of elephant's foot yam. *Indian Phytopathology* 55: 238-240.
- Gomez K A and Gomez A A 1984 Statistical procedures for agricultural research (second edition). Jhon Wiley and Sons, New York.
- Gupta V 2004 Compatibility of biocontrol agent *Trichoderma harzianum* with pesticides. *Journal of Mycology and Plant Pathology*, 34:504-505.
- Gupta S ,Kalha C S ,Vaid A and Rizvi S E H 2005 Integrated management of anthracnose of french bean caused by *Colletotrichum lindemuthianum*. *Journal of Mycology and Plant Pathology* 35; 432-435.
- Gupta R B L 1995 Management of *Macrophomina* stem blight of cowpea through seed dressing and spray fungicides. *Indian Journal of Mycology and Plant Pathology*, 25:112-118.
- Gurha S N and Shubha Trivedi 2008 Status of soil borne pathogens infecting chickpea in Karnataka state. *Annual Journal of Plant Protection Sciences*, 16(1): 257-258.
- * Haffmaster D E, Mc Laughlin J H, Roy W W and Chester K S 1943 The problem of dry root rot caused by *Macrophomina phaseolina*. *Phytopathology*, 33:1113-1114.
- * Hansford C G 1943 Tomato diseases and their control. Jamaica Department of Agricultural Microbiology, pp 12.

- Haware M P 1990 *Fusarium* wilt and other important diseases of chickpea in the Mediterranean area. Options mediterraneennes-serie seminairies, 9: 61-64.
- Henis Y, Ghaffar A and Baker R 1978 Integrated control of *Rhizoctonia solani* damping - off of radish: Effect of successive plantings, PCNB, and *Trichoderma harzianum* on pathogen and disease. Phytopathology 68: 900-907.
- Hwang S F, Gossen B D, Chang K F, Turnbull G D, Howard R J and Blade S F 2003 Etiology, impact and control of *Rhizoctonia* seedling blight and root rot of chickpea on the Canadian prairies. Canadian Journal of Plant Sciences, 83(4): 959-967.
- Jackson B J and Backman P A 1993 Biological and cultural disease control alternatives and supplements in IPM system. Plant Disease, 77: 311-315.
- Jaiman R K and Jain S C 2008 *Macrophomina phaseolina* in cluster bean (*Cymopsis tetragonaloba*) seeds and its control. Indian Journal of Mycology and Plant Pathology, 38(2): 403-404.
- Jain N K, Khare M N and Sharma H C 1973 Variation among the isolates of *Rhizoctonia bataticola* from Urdbean plant parts and soil. Mysore Journal of Agricultural Sciences, 6: 461-465.
- Jayalakshmi V, Sivaprakasam K and Seetharaman K 2003 Effect of *Pseudomonas fluorescens* on *Rhizoctonia solani* on the growth of rice plant. Journal of Biological Control, 17(2): 189-192.
- Jayasree K, Shanmugham, Raguchander T, Ramanatham A and Samiyappan R 2000 Evaluation of *Pseudomonas fluorescens*-1(pf-1) against blackgram and sesame root rot disease. Journal of Biological Control, 14(2):55-61.

- Jeyarajan R, Ramakrishnan G, Dinakaran D and Sridhar R 1994 Development of product of *Trichoderma viride*, and *Bacillus subtilis* for biocontrol of root rot disease. In: Dwiedi BK (ed.) Biotechnology in India. Bioved Research society, Allahabad, India pp: 25-36
- Jharia H K and Khare M N 1986 Biological control of *Rhizoctonia bataticola* causing disease in soybean. Indian Phytopathology, 39:148-153.
- Johnson L F and Curl E A 1977 Methods for research on the ecology of soil borne plant pathogens. Burgess Publishing Company, Minneapolis, pp.27.
- Kataria H R and Grover R K 1977 Comparison of fungicides for the control of *Rhizoctonia solani* causing damping off of mungbean (*Phaseolus aureus*). Annals of Applied Biology, 83: 79-85.
- Katariya L, Gaur V K and R Sharma 2007 Assessment of genetic variability in *Rhizoctonia bataticola* infecting chickpea isolates using pathogenicity and RAPD markers. Indian Journal of Mycology and Plant Pathology, 37(3): 491-494.
- Kaushal R P 2008 Management of root rot in chickpea. Journal of Food Legumes, 21(3): 178-181.
- Khan M A and Gangopadhyay S 2008 Efficacy of *Pseudomonas fluorescens* in controlling root rot of chickpea caused by *Macrophomina phaseolina*. Indian Journal of Mycology and Plant Pathology, 38(3): 580-587.
- Khalid- Iftikar and Ilyas, M B 2000 Screening of chickpea germplasm against dry root rot disease (*Macrophomina phaseolina*) in pots /glass house. Pakistan Journal of Phytopathology, 12(1): 66-70

- Kheri H K and Chandra A S 1991 Antagonism of *Trichoderma viride* to *Macrophomina phaseolina* and its application in the control of dry root rot of mungbean. *Indian Phytopathology*. 44(11):60-63.
- Khot, G G, Taura, P and Dadarwal, K P (1996) *Rhizoctonia* from chickpea (*Cicer arietinum* L.) rhizosphere effective in wilt control and promote nodulation. *Indian Journal of Microbiology*, 36(4): 217-222.
- Kishore G K Pande S and Podile A R 2005 Biological control of collar rot disease with broad-spectrum antifungal bacteria with groundnut. *Canadian Journal of Microbiology* 51: 123-132.
- Konde S A, Raut B T and Glade M R 2008 Chemical and biological management of root rot (*Rhizoctonia bataticola*) of soybean. *Annals of Plant Physiology*, 22(2): 275-277.
- Kratisharma and Tribhuwan Singh 2000 Seed and seedling infection of *Rhizoctonia bataticola* in *Vigna radiata*. *Journal of Mycology and Plant Pathology*, 1: 15-18.
- Liu J Ovaskim D H Charles T C and Glick B R 2000 An Acc deaminase minus mutant of enterobacter cloacae UW 4 no longer promotes root elongation. *Current Microbiology* 41: 101-105.
- * Lynch J M and Whipps J M 1991 Substrate flow in the rhizosphere. *The rhizosphere and plant growth* pp. 15-24.
- Manoranjitam S K, Prakasham V, Rajappan K and Amutha G 2000 Effect of two antagonists on damping off disease of tomato. *Indian Phytopathology* 53: 441-443.
- * Martin J P 1950 Use of acid rose Bengal and streptomycin in the plate: *Method of estimating soil fungi* 69: 215-233.

- * Mc Rae 1929 New diseases reported during the year 1928 from india. International Bulletin of Plant protection, 3: 21-22.
- Meena B M, Tera dimani S S, Balakrishnan and Siddeswaran K 2003 Effect of plant products and biological agents in the management of tamarind nursery diseases. National seminar on IPDMSA Annamali University, March 20, 21. pp 63.
- Megha Y J Alagawad A R and Krishnaraj P U 2007 Diversity of Fluorescent pseudomonads isolated from the forest soils of the Western Ghats of Uttara Kannada. Current Science, 93(10): 1433-1437.
- Mendes R, Pizzirani-Klenier A A, Araujo W L and Raijmakers J M 2007 Diversity of cultivated endophytic bacteria from sugarcane: genetic and biochemical characterization of *Burkholderia cepacia* complex isolates. Applied and Environmental Microbiology 73: 7259-7267.
- Misbah S, Hassan Yusof M Y, Hanifah YA and Abu Bakar S 2005 Genomic species identification of *Acinetobacter* of clinical isolates by 16S rDNA sequencing. Singapore Med J, 46 (9): 461-466.
- Mondal D K and Bhattacharya P K 2003 Management of gram root rot caused by *Macrophomina phaseolina* (Tossi) Goid with antagonistic bacteria (*Bacillus* sp.). Journal of Mycopathological Research, 41(2): 147-151.
- Morton D J and Stroube W H 1995 Antagonistic and stimulatory effects of soil microorganisms upon *Sclerotium rolfsii*. Phytopathology, 45: 417-420.
- Mukhopadhyay A N 1987 Biological control of soil borne plant pathogens by *Trichoderma* sp. Indian Journal of Mycology and Plant Pathology 17:1-9.

- Murray M and Thompson W 1980 The isolation of high weight plant DNA. Nucleic acid research, 8(3): 4321-4325.
- Muthukumar A and Bhaskaran R 2007 Efficacy of anti-microbial metabolites of *Pseudomonas fluorescens migula* against *Rhizoctonia solani* kuhn and *Pythium* sp. Journal of Biological Control, 21(1): 105-110.
- Muthusamy S and V Mariappan 1991 Disintegration of sclerotia of *Macrophomina phaseolina* (Soybean isolate) by oil cake extract. Indian Phytopathology, 44(2): 271-273.
- * Nene Y L, Haware MP and M V Reddy 1978 Diagnosis of some wilt like disorders of chickpea (*Cicer arietinum* L.) information Bulletin No. 3, ICRISAT pp. 44.
- Nene YL and Thapliyal PN 1993 Fungicides in plant disease control. III ed Oxford and IBH Publishing Company Private Limited, Calcuta pp 531.
- Naseema Beevi S, Ancy Salim M, Priya Mohan Deepti S and Naseema A 2005 Compatability of *Trichoderma harzianum Rifai* with botanicals and synthetic pesticides. Geobios, 32(3): 80-82.
- Nautiyal C S 1997 Selection of Chickpea rhizosphere competent *Pseudomonas fluorescens* NBRI1303 antagonistic to *Fusarium oxysporum* f. sp. *ciceris*, *Rhizoctonia bataticola* and *pythium* sp. Current Microbiology, 35: 52-58.
- Padmalatha K V 2002 Studies on the management of charcoal rot of sunflower (*Helilanthus annuus* L.), M.Sc (Ag.) Thesis submitted to Acharya N.G. Ranga Agricultural University Hyderabad, Andhra Pradesh.

- Pall S, Lakshmi J P and Beohar A B L 1990 Efficacy of fungicides for controlling *Macrophomina phaseolina* (Tassi) Goid in Urd bean. Research and Development Reporter, 2: 213.
- * Pandey G and Singh RB 1990 Survey of root diseases of greengram in Allahabad region. Bioved, 1(1): 93-94.
- Pan S and Bhagat S 2008 Characterization of antagonistic potential of *Trichoderma* sp. against some soil borne plant pathogens. Journal of Biological Control 22(1): 43-49.
- Parmer N and Dadarwal K R 1997 *Rhizobacteria* from the rhizosphere and rhizoplane of chickpea. Indian Journal of Microbiology, 37(4): 205-210.
- Papavizas G C and Lumsden R D 1980 Biological control of soil borne fungal propagules. Annual Review of Phytopathology 18: 389-413.
- * Papavizas G C and Lewis J A 1981 Introduction and augmentation of microbial antagonists for the control of soil born pathogen In: Biological control in crop production (ed Paparzas G C) Osmum Totawa. pp: 305-322.
- Papavizas G C 1985 *Trichoderma* and *Gliocladium*: Biology, Ecology and potential for biocontrol. Annual Review of Phytopathology 23: 23-54.
- Parakhia A M and Vaishnav M U 1986 Biocontrol of *Rhizoctonia bataticola* Indian Phytopathology 39: 439-440.
- Paramasivan M, Mohan S and Muthukrisnan N 2007 Management of coleus dry root rot pathogen *Macrophomina phaseolina* by fungal and bacterial antagonists. Indian Journal of Plant Protection 35(1):133-135.

- Patel S T and Anahosur 2001 Potential antagonism of *T. harzianum* against *Fusarium* spp. *Macrophomina phaseolina* and *Sclerotium rolfsii*. Indian Journal of Mycology and Plant Pathology 31: 365-369.
- Patibanda A K, Upadhyay J P and Mukhopadhyay A N 2002 Efficacy of *Trichoderma harzianum* Rifai alone or in combination with fungicides against *Sclerotium* wilt of groundnut. Journal of Biological Control 16: 57-63.
- Paul Y S, Meena Devi and Kapoor A S 2008 Integrated management of wilt root rot complex of pea. Indian Journal of Mycology and Plant Pathology, 38(3): 571-576.
- Pershney N L, Gade R M and Thakare K G 1992 Sensitivity and adaptability of *Rhizoctonia bataticola* to different fungicides. Journal of Soil and Crops. 2: 35-38.
- Philip C T, Kartha K K, Joshi R K and Nema K G 1969 A new *Rhizoctonia* disease of mungbean in Madhya Pradesh, JNKVV Research Journal, 3: 40-43.
- Prajapati R K, Gangawar R K, Srivastava S S L and Ahmad S 2002 Efficacy of fungicides, non target pesticides and bioagents against the dry root rot of chickpea. Annals of Plant Protection Sciences, 10(1): 154-155.
- Prajapati R K, Srivastava S S L and Chaudhary R G 2003 Incidence of chickpea dry root rot in Uttar pradesh and Uttaranchal and efficacy of seed dressing fungicides on seed germination and seedling infection. Farm science Journal, 12(2): 170-171.
- Prameela Devi and Singh R H 1998 Studies on virulence of *Macrophomina phaseolina* isolates from blackgram and greengram. Journal of Mycology and Plant Pathology, 22(2): 196-198.

- Prasanthi S K, Kulakarni Sikant and Anahosur K H 2000 Management of safflower root rot caused by *Rhizoctonia bataticola* by antagonistic microorganisms. *Plant Disease Research* 15(2): 146-150.
- Poddar R K Singh D V and Dubey S C 2004 Management of chickpea wilt through combination of fungicides and bioagents. *Indian Phytopathology* 57: 39-43.
- Prasad R D, Rangeshwaran R, Anuroop C P and Rashmi HJ 2002 Biological control of wilt and root rot of chickpea under field conditions. *Annals of Plant Protection Sciences* 10: 1, 72-75.
- Raghuchander T, Samiyapan R and Arjunan G 1993 Biocontrol of *Macrophomina* root rot of mungbean. *Indian Phytopathology* 46: 379-382.
- Rajeena B P M and Ahmad R 2007 Control of seed borne fungi of okra with *Pseudomonas fluorescens* and aqueous leaf extract of henna (*Lawsonia inermis* L.). *Journal of Mycology and Plant Pathology*, 37(3): 485-487.
- Rajeevpant and Mukhopadhyay A N 2001 Integrated Management of seed and seedling rot complex of soybean. *Indian Phytopathology* 54: 346-350.
- Rajendar Singh , Sindhan G S , Parashar, R D , Indra- Hooda, Singh R, and Hooda- I 1998 Application of antagonists in relation to dry root rot and biochemical status of chickpea plants. *Plant Disease Research*.13 (1): 35-37.
- Rajpurohit T S 1997 Management of *Macrophomina* stem and root rot of Sesame through fungicidal seed treatment and varietal resistance. *Journal of Mycology and Plant Pathology*, 27(1): 98-104.

- Rajeswari R 1997 Studies on biological control of *Macrophomina phaseolina* (Tassi) and incitant of dry root rot of mungbean. Ph.D Thesis submitted to Acharya N.G. Ranga Agricultural University Hyderabad, Andhra Pradesh.
- Rajeswari B, Chandrasekhara Rao K and Pramod Chandra Kumar C 1999 Efficacy of antagonists and carbendazim against dry root rot of mungbean incited by *Macrophomina phaseolina* (Tassi) Goid under glasshouse conditions. Journal of Biological Control 13: 93-99.
- Ramesh Kumar N, Thirumalai Arasu V and Gunasekharan P 2002 Genotyping of antifungal compounds producing plant growth-promoting rhizobacteria, *Pseudomonas fluorescens*. Current science 82 (12): 1463-1466.
- Ramesh R and V S Korianthimath 2006 Management of groundnut root rot by *Trichoderma viride* and *Pseudomonas fluorescens* under rainfed conditions. Indian Journal of Plant Protection, 34(2): 23-31.
- Rangaswamy G and Mahadevan A 1999 Diseases of crop plants in India.(4th edition) Prentice Hall of India Pvt. Ltd. New Delhi, pp. 607.
- Rangaswamy G 1996. Diseases of crop plants. III rd ed. pp. 257.
- Rangeshwaran R, Prasad R D and Anuroop C P 2001 Field evaluation of two bacterial antagonists *Pseudomonas putida* (PDBCAB 2) against wilt and root rot of chickpea. Journal of Biological Control 15: 165-170.
- Raut B T and Somani R B 1987 Efficacy of different fungicides IV. Field trials on root rot of chickpea. PKV Research Journal 11: 182-184.
- Roopali Sharma, Sekhar Varsheney, Deeksha Joshi and Shalini Shinghal, Singh U S 2002 Farm yard manure: A promising substrate for mass

- multiplication and developing commercial formulation of *Trichoderma harzianum*. Indian Phytopathology, 55: 382-388.
- Sahu A K and Jena N 1997 Seed microflora of greengram (*Phaseolus aureus* Rox b) cultivars of Orissa and their impact on seed germination. Journal of Mycological Research, 35(2): 93-97.
- Sajeena A, Salalrajan F, Seetharaman K and Mohan Babu R 2004 Evaluation of biocontrol agents against dry root rot of blackgram (*Vigna mungo*). Journal of Mycology and Plant Pathology, 34(2): 341-343.
- Sankar S and Jeyarajan R 1995 Biological control of sesame root rot by seed treatment with *Trichoderma* sp. and *Bacillus subtilis*. Indian Journal of Mycology and Plant Pathology, 26: 217-220.
- Sarkar S K and Pradhan S K 1999 Incidence of leaf blight of sunhemp caused by *Macrophomina phaseolina*. Journal of Mycology and Plant Pathology, 29(1): 13-15.
- Sawant I and Mukhopadhyay A N 1990 Integration of metalaxyl with *Trichoderma harzianum* for the control of *Pythium* damping off in sugar beet. Phytopathology 43: 535-541.
- Saxena D R and Saxena M 1995 Evaluation of antagonists against dry root rot pathogen of chickpea. Indian Journal of Mycology and Plant Pathology, 25: 130-134.
- Selvarajan R and Jeyarajan R 1996 Inhibition of chickpea root rot pathogen, *Fusarium solani* and *Macrophomina phaseolina* by antagonists. Indian Journal of Mycology and Plant Pathology, 26(3): 248-251.
- Sendhilvel V, Bhuvaneswari D, Kanmozhi S, Kavitha K and Raguchander 2005 Management of cowpea root rot caused by *Macrophomina*

- phaseolina* (Tassi) Goid. using plant growth promoting Rhizobacteria. Journal of Biological Control, 19(1): 41-46.
- Sharma J and Tripathi H S 2001 Biological and chemical control of web blight disease of Urd bean. Indian Phytopathology, 54(2): 267-269.
- Sharma N, Basat Prabha and Avantina Sharma 2009 Potential of endophytic fungi as bioagent against *Rhizoctonia solani* and *S. rolfsii*. Indian Journal of Mycology and Plant Pathology, 39(33): 266-270.
- * Sinclair J B 1982 Compendium of soybean diseases. American Phytopathological Society, pp 104.
- Sindhan G S, Hooda I and Karwasra S S 2002 Biological control of dry root rot of chickpea caused by *Rhizoctonia bataticola*. Plant Disease Research, 17(1): 68-71.
- Singh P J and Mehrotra R S 1980 Biological control of *Rhizoctonia bataticola* on gram by coating seeds with *Bacillus* and *Streptomyces* sp. and their influence on plant growth. Plant and Soil, 56: 575-583.
- Singh S K, Nene Y L and Reddy M V 1990 Some histopathological observations of chickpea roots infected by *Rhizoctonia bataticola*. International Chickpea News Letter, 23(2): 24-25.
- Singh R D N and Kaiser S A K 1995 Evaluation of some systemic and non systemic fungicides against the charcoal rot of pathogen *Macrophomina phaseolina*(Tassi) Goid. of maize. Journal of Tropical Agriculture, 33: 54-58.
- Singh S D 1999 Soil borne diseases of chickpea. File under plant disease B 10 (field crops). Issued March 1999. 4782000399 M S. S. D Singh @Cgiar.org/http://grep.icrisat.cgiar.org.

- Singh A S and S N Agarwal 2002 Prevalence of chickpea wilt and dry root rot diseases in Vindhyan plateau zone of Madhya Pradesh. Indian Journal of Pulses Research, 15(2): 14-19.
- Singh A and Sirohi A 2003 Status of chickpea diseases in Himachal Pradesh, India. International Chickpea and Pigionpea News letter 10: 29-31.
- Singh R and Sinha A P 2007 Management of sheath blight of rice with *Pseudomonas fluorescens*. Journal of Mycology and Plant Pathology, 37(1): 18-21.
- * Shahzad S and Ghaffer A 1986 *Macrophomina phaseolina* on some new hosts FAO Plant Protection Bulletin, 34: 163.
- Suriachandraselvan M 1997 Studies on the charcoal rot of sunflower (*Helianthus annus* L.) caused by *Macrophomina phaseolina* (Tassi.) Goid. M.Sc(Ag.). Thesis, submitted to Tamilnadu Agriculture University, Coimbatore pp.295.
- Suriachandraselvan M, Salarajan F, Aiyanthan K E A and Seetharaman 2004 Inhibition of sunflower charcoal rot pathogen, *Macrophomina phaseolina* by fungal antagonists. Indian Journal of Mycology and Plant Pathology, 34(2): 364-365.
- Taya R S, Tirpathi and Anwar M S 1988 Influence of soil type, soil moisture and fertilizer on the severity of chickpea dry root rot caused by *Rhizoctonia bataticola*. Indian Journal of Mycology and Plant Pathology 18(2): 133-136.
- Tewari R K S and Mukhopadhyay A N 2003 Management of chickpea root rot and collar rot by integration of biological and chemical seed treatment . Indian Phytopathology 56(1): 39-42.

- Tiwari P K and Thrimurthy V S 2007 Isolation and characterization of the *Pseudomonas fluorescens* from rhizosphere of different crops. Journal of Mycology and Plant Pathology, 37(2): 231-234.
- Tiwari R K S and Ashok Singh 2004 Efficacy of fungicides on *Rhizoctonia solani* and *Sclerotium rolfsii* and their effect on *Trichoderma harzianum* and *Rhizobium leguminosarum*. Journal of Mycology and Plant Pathology, 30(2): 134-139.
- Tuite J 1969 Plant pathology methods: fungi and bacteria. Burgess Publishing Company, Minneapolis, USA pp.239
- Umamaheswari M P, Muthuswamy M and Alice D 2002 Evaluation of antagonists against jasmine wilt caused by *Sclerotium rolfsii* (Sacc.). Journal of Biological Control 16; 135-140
- Upadhyay J P, Lal H C and Roy S 2004 Effect of fungicides cakes and plant by products on *Trichoderma viride*. Journal of Mycology and Plant Pathology 34: 313-315.
- * Valiente C, K Diaz, S Gacitua, M Martinez and E Sanfuentes 2008 Control of charcoal root rot of *Pinus radiata* nurseries with antagonistic bacteria. World Journal of Microbial Biotechnology, 24: 557-568.
- Venkateswar Rao K 1998 An integrated approach for management of sesame stem and root rot caused by *Macrophomina phaseolina* (Tassi) Goid. M.Sc. (Ag.) Thesis submitted to Acharya N.G. Ranga Agricultural University Hyderabad, Andhra Pradesh.
- Vidyasekaran P and Muthamilan 1995 Development of formulations of *Pseudomonas fluorescens* for control of chickpea wilt. Plant Disease 79: 782-786.

- Vijay Mohan, Prasad S M, Barnwal M K, and Kudada N 2006 Fungicidal management of dry root rot disease and yield of chickpea. *Journal of Applied Biology*, 16(1): 42-44.
- Vinod Kumar, Anuj Kumar and R N Khadarwar 2007 Antagonistic potential of fluorescent *Pseudomonads* and control of charcoal rot of chickpea caused by *Macrophomina phaseolina*. *Journal of Environmental Biology*, 28(1): 15-20.
- Vishwadar and Sarbhoy A K 1993 An atypical isolate of *Rhizoctonia bataticola*. *Indian Phytopathology*, pp. 245-246.
- * Vyas S C and Khare M N 1986 Biological control of dry root rot of soybean caused by *Rhizoctonia bataticola* with carbendazim and antagonist. Seminar on management of soil borne diseases of crop plants, January 8-10, TNAU, Coimbatore, India.
- Vyas S C 1987 Effect of seed treatment fungicides thiram and carbendazim on the antagonists of soybean and dry root rot pathogen *Rhizoctonia bataticola*. Abstracts of papers presented in workshop on biological control of plant diseases, TNAU, Coimbatore, India pp.23.
- Vyas S C 1994 Integrated biological and chemical control of dry root rot on soybean. *Indian Journal of Mycology and Plant Pathology*, 24: 132-134.
- Williams J G K, Kubelic A R, Linak K J, Rafalski J A and Tinger S V 1990 DNA polymorphism amplified by arbitrary primers are useful as genetic markers. *Nucleic acid research* 18: 7213-7218.
- Yang C Y 1977 Mungbean disease and control. First international Mung Symposium AVRDC, Taiwan, pp. 141-146.

Zaidi N W and Singh U S 2004 Development of improved technology for mass multiplication and delivery of fungal (*Trichoderma*) and bacterial (*Pseudomonas*) biocontrol agents. Journal of Mycology and Plant Pathology, 34: 732-737.

Ziedan E H C 2006 Manipulating endophytic bacteria for biological control to soil borne diseases of peanut. Journal of Applied Sciences Research, 2: 97-105.

* Originals not seen